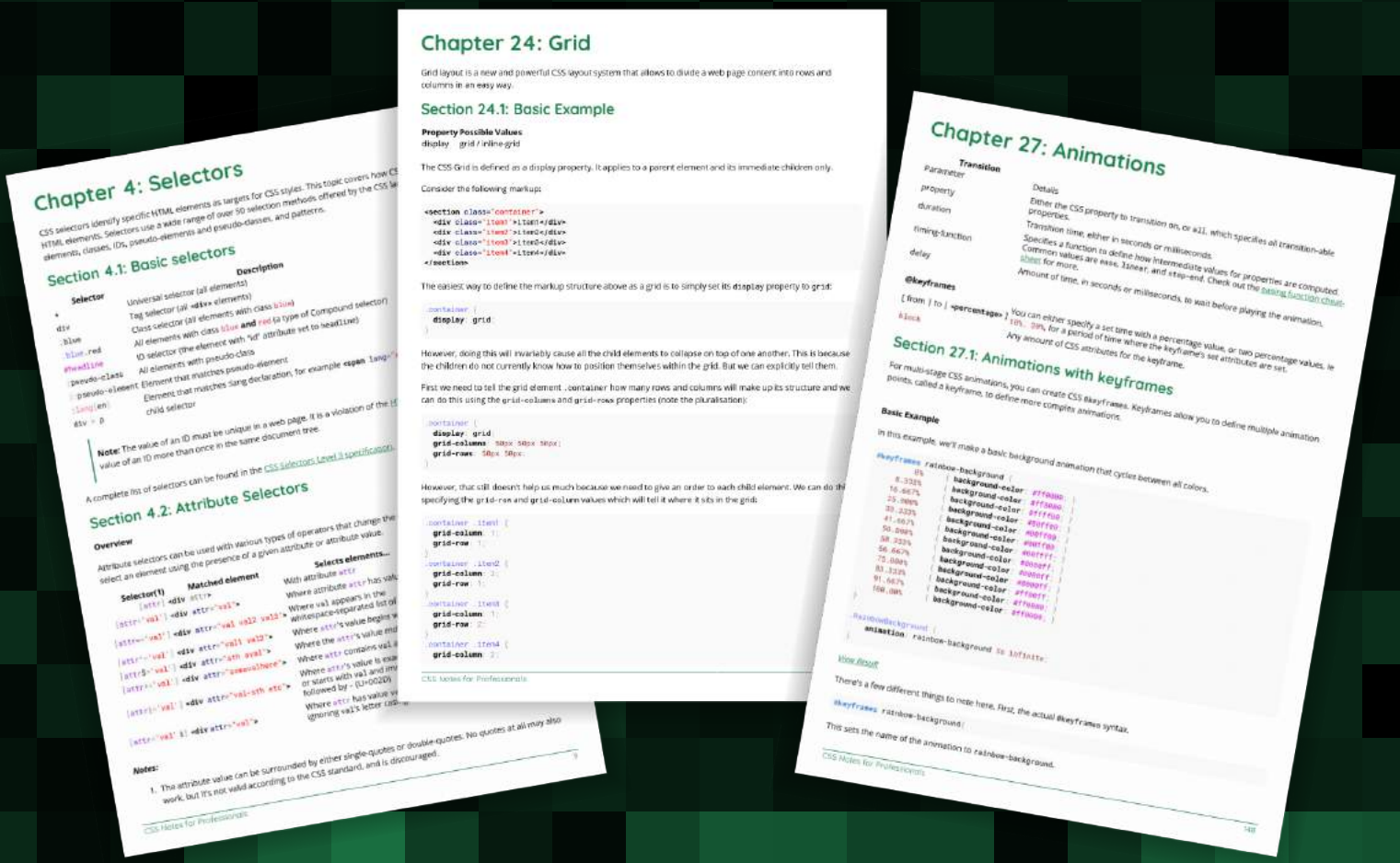
CSS

Notes for Professionals



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Contents

About ... 1
Chapter 1: Getting started with CSS ... 2
Section 1.1: External Stylesheet ... 2
Section 1.2: Internal Styles ... 3
Section 1.3: CSS @import rule (one of CSS at-rule) ... 4
Section 1.4: Inline Styles ... 4
Section 1.5: Changing CSS with JavaScript ... 4
Section 1.6: Styling Lists with CSS ... 5
Chapter 2: Structure and Formatting of a CSS Rule ... 7
Section 2.1: Property Lists ... 7
Section 2.2: Multiple Selectors ... 7
Section 2.3: Rules, Selectors, and Declaration Blocks ... 7
Chapter 3: Comments ... 8
Section 3.1: Single Line ... 8
Section 3.2: Multiple Line ... 8
Chapter 4: Selectors ... 9
Section 4.1: Basic selectors ... 9
Section 4.2: Attribute Selectors ... 9
Section 4.3: Combinators ... 12
Section 4.4: Pseudo-classes ... 13
Section 4.5: Child Pseudo Class ... 15
Section 4.6: Class Name Selectors ... 16
Section 4.7: Select element using its ID without the high specificity of the ID selector ... 17
Section 4.8: The :last-of-type selector ... 17
Section 4.9: CSS3 :in-range selector example ... 17
Section 4.10: A. The :not pseudo-class example & B. :focus-within CSS pseudo-class ... 18
Section 4.11: Global boolean with checkbox:checked and ~ (general sibling combinator) ... 19
Section 4.12: ID selectors ... 20
Section 4.13: How to style a Range input ... 21
Section 4.14: The :only-child pseudo-class selector example ... 21
Chapter 5: Backgrounds ... 22
Section 5.1: Background Color ... 22
Section 5.2: Background Gradients ... 24
Section 5.3: Background Image ... 25
Section 5.4: Background Shorthand ... 26
Section 5.5: Background Size ... 27
Section 5.6: Background Position ... 31
Section 5.7: The background-origin property ... 32
Section 5.8: Multiple Background Image ... 34
Section 5.9: Background Attachment ... 35
Section 5.10: Background Clip ... 36
Section 5.11: Background Repeat ... 37
Section 5.12: background-blend-mode Property ... 37
Section 5.13: Background Color with Opacity ... 38
Chapter 6: Centering ... 39
Section 6.1: Using Flexbox ... 39
Section 6.2: Using CSS transform ... 40

Section 6.3: Using margin: 0 auto; ... 41
Section 6.4: Using text-align ... 42
Section 6.5: Using position: absolute ... 42
Section 6.6: Using calc() ... 43
Section 6.7: Using line-height ... 43
Section 6.8: Vertical align anything with 3 lines of code ... 44
Section 6.9: Centering in relation to another item ... 44
Section 6.10: Ghost element technique (Michał Czernow's hack) ... 45
Section 6.11: Centering vertically and horizontally without worrying about height or width ... 46
Section 6.12: Vertically align an image inside div ... 47
Section 6.13: Centering with fixed size ... 47
Section 6.14: Vertically align dynamic height elements ... 49
Section 6.15: Horizontal and Vertical centering using table layout ... 49

Chapter 7: The Box Model ... 51
Section 7.1: What is the Box Model? ... 51
Section 7.2: box-sizing ... 52

Chapter 8: Margins ... 55
Section 8.1: Margin Collapsing ... 55
Section 8.2: Apply Margin on a Given Side ... 57
Section 8.3: Margin property simplification ... 58
Section 8.4: Horizontally center elements on a page using margin ... 58
Section 8.5: Example 1: ... 59
Section 8.6: Negative margins ... 59

Chapter 9: Padding ... 61
Section 9.1: Padding Shorthand ... 61
Section 9.2: Padding on a given side ... 62

Chapter 10: Border ... 63
Section 10.1: border-radius ... 63
Section 10.2: border-style ... 64
Section 10.3: Multiple Borders ... 65
Section 10.4: border (shorthands) ... 66
Section 10.5: border-collapse ... 66
Section 10.6: border-image ... 67
Section 10.7: Creating a multi-colored border using border-image ... 67
Section 10.8: border-[left|right|top|bottom] ... 68

Chapter 11: Outlines ... 69
Section 11.1: Overview ... 69
Section 11.2: outline-style ... 69

Chapter 12: Overflow ... 71
Section 12.1: overflow-wrap ... 71
Section 12.2: overflow-x and overflow-y ... 72
Section 12.3: overflow: scroll ... 73
Section 12.4: overflow: visible ... 73
Section 12.5: Block Formatting Context Created with Overflow ... 74

Chapter 13: Media Queries ... 76
Section 13.1: Terminology and Structure ... 76
Section 13.2: Basic Example ... 77
Section 13.3: mediatype ... 77
Section 13.4: Media Queries for Retina and Non Retina Screens ... 78

Section 13.5: Width vs Viewport ... 79
Section 13.6: Using Media Queries to Target Different Screen Sizes ... 79
Section 13.7: Use on link tag ... 80
Section 13.8: Media queries and IE8 ... 80
Chapter 14: Floats ... 81
Section 14.1: Float an Image Within Text ... 81
Section 14.2: clear property ... 82
Section 14.3: Clearfix ... 83
Section 14.4: In-line DIV using float ... 84
Section 14.5: Use of overflow property to clear floats ... 86
Section 14.6: Simple Two Fixed-Width Column Layout ... 86
Section 14.7: Simple Three Fixed-Width Column Layout ... 87
Section 14.8: Two-Column Lazy/Greedy Layout ... 88
Chapter 15: Typography ... 89
Section 15.1: The Font Shorthand ... 89
Section 15.2: Quotes ... 90
Section 15.3: Font Size ... 90
Section 15.4: Text Direction ... 90
Section 15.5: Font Stacks ... 91
Section 15.6: Text Overflow ... 91
Section 15.7: Text Shadow ... 91
Section 15.8: Text Transform ... 92
Section 15.9: Letter Spacing ... 92
Section 15.10: Text Indent ... 93
Section 15.11: Text Decoration ... 93
Section 15.12: Word Spacing ... 94
Section 15.13: Font Variant ... 94
Chapter 16: Flexible Box Layout (Flexbox) ... 96
Section 16.1: Dynamic Vertical and Horizontal Centering (align-items, justify-content) ... 96
Section 16.2: Sticky Variable-Height Footer ... 102
Section 16.3: Optimally fit elements to their container ... 103
Section 16.4: Holy Grail Layout using Flexbox ... 104
Section 16.5: Perfectly aligned buttons inside cards with flexbox ... 105
Section 16.6: Same height on nested containers ... 107
Chapter 17: Cascading and Specificity ... 109
Section 17.1: Calculating Selector Specificity ... 109
Section 17.2: The !important declaration ... 111
Section 17.3: Cascading ... 112
Section 17.4: More complex specificity example ... 113
Chapter 18: Colors ... 115
Section 18.1: currentColor ... 115
Section 18.2: Color Keywords ... 116
Section 18.3: Hexadecimal Value ... 122
Section 18.4: rgb() Notation ... 122
Section 18.5: rgba() Notation ... 123
Section 18.6: hsl() Notation ... 123
Section 18.7: hsla() Notation ... 124
Chapter 19: Opacity ... 126
Section 19.1: Opacity Property ... 126
Section 19.2: IE Compatibility for `opacity` ... 126

Chapter 20: Length Units ... 127
Section 20.1: Creating scalable elements using rems and ems ... 127
Section 20.2: Font size with rem ... 128
Section 20.3: vmin and vmax ... 129
Section 20.4: vh and vw ... 129
Section 20.5: using percent % ... 129
Chapter 21: Pseudo-Elements ... 131
Section 21.1: Pseudo-Elements ... 131
Section 21.2: Pseudo-Elements in Lists ... 131
Chapter 22: Positioning ... 133
Section 22.1: Overlapping Elements with z-index ... 133
Section 22.2: Absolute Position ... 134
Section 22.3: Fixed position ... 135
Section 22.4: Relative Position ... 135
Section 22.5: Static positioning ... 135
Chapter 23: Layout Control ... 137
Section 23.1: The display property ... 137
Section 23.2: To get old table structure using div ... 139
Chapter 24: Grid ... 141
Section 24.1: Basic Example ... 141
Chapter 25: Tables ... 143
Section 25.1: table-layout ... 143
Section 25.2: empty-cells ... 143
Section 25.3: border-collapse ... 143
Section 25.4: border-spacing ... 144
Section 25.5: caption-side ... 144
Chapter 26: Transitions ... 145
Section 26.1: Transition shorthand ... 145
Section 26.2: cubic-bezier ... 145
Section 26.3: Transition (longhand) ... 147
Chapter 27: Animations ... 148
Section 27.1: Animations with keyframes ... 148
Section 27.2: Animations with the transition property ... 149
Section 27.3: Syntax Examples ... 150
Section 27.4: Increasing Animation Performance Using the `will-change` Attribute ... 151
Chapter 28: 2D Transforms ... 152
Section 28.1: Rotate ... 152
Section 28.2: Scale ... 153
Section 28.3: Skew ... 153
Section 28.4: Multiple transforms ... 153
Section 28.5: Translate ... 154
Section 28.6: Transform Origin ... 155
Chapter 29: 3D Transforms ... 156
Section 29.1: Compass pointer or needle shape using 3D transforms ... 156
Section 29.2: 3D text effect with shadow ... 157
Section 29.3: backface-visibility ... 158
Section 29.4: 3D cube ... 159
Chapter 30: Filter Property ... 161
Section 30.1: Blur ... 161

Section 30.2: Drop Shadow (use box-shadow instead if possible) ... 161
Section 30.3: Hue Rotate ... 162
Section 30.4: Multiple Filter Values ... 162
Section 30.5: Invert Color ... 163
Chapter 31: Cursor Styling ... 164
Section 31.1: Changing cursor type ... 164
Section 31.2: pointer-events ... 164
Section 31.3: caret-color ... 165
Chapter 32: box-shadow ... 166
Section 32.1: bottom-only drop shadow using a pseudo-element ... 166
Section 32.2: drop shadow ... 167
Section 32.3: inner drop shadow ... 167
Section 32.4: multiple shadows ... 168
Chapter 33: Shapes for Floats ... 170
Section 33.1: Shape Outside with Basic Shape – circle() ... 170
Section 33.2: Shape margin ... 171
Chapter 34: List Styles ... 173
Section 34.1: Bullet Position ... 173
Section 34.2: Removing Bullets / Numbers ... 173
Section 34.3: Type of Bullet or Numbering ... 173
Chapter 35: Counters ... 175
Section 35.1: Applying roman numerals styling to the counter output ... 175
Section 35.2: Number each item using CSS Counter ... 175
Section 35.3: Implementing multi-level numbering using CSS counters ... 176
Chapter 36: Functions ... 178
Section 36.1: calc() function ... 178
Section 36.2: attr() function ... 178
Section 36.3: var() function ... 178
Section 36.4: radial-gradient() function ... 179
Section 36.5: linear-gradient() function ... 179
Chapter 37: Custom Properties (Variables) ... 180
Section 37.1: Variable Color ... 180
Section 37.2: Variable Dimensions ... 180
Section 37.3: Variable Cascading ... 180
Section 37.4: Valid/Invalids ... 181
Section 37.5: With media queries ... 182
Chapter 38: Single Element Shapes ... 184
Section 38.1: Trapezoid ... 184
Section 38.2: Triangles ... 184
Section 38.3: Circles and Ellipses ... 187
Section 38.4: Bursts ... 188
Section 38.5: Square ... 190
Section 38.6: Cube ... 190
Section 38.7: Pyramid ... 191
Chapter 39: Columns ... 193
Section 39.1: Simple Example (column-count) ... 193
Section 39.2: Column Width ... 193
Chapter 40: Multiple columns ... 195
Section 40.1: Create Multiple Columns ... 195

Section 40.2: Basic example ... 195
Chapter 41: Inline-Block Layout ... 196
Section 41.1: Justified navigation bar ... 196
Chapter 42: Inheritance ... 197
Section 42.1: Automatic inheritance ... 197
Section 42.2: Enforced inheritance ... 197
Chapter 43: CSS Image Sprites ... 198
Section 43.1: A Basic Implementation ... 198
Chapter 44: Clipping and Masking ... 199
Section 44.1: Clipping and Masking: Overview and Difference ... 199
Section 44.2: Simple mask that fades an image from solid to transparent ... 201
Section 44.3: Clipping (Circle) ... 201
Section 44.4: Clipping (Polygon) ... 202
Section 44.5: Using masks to cut a hole in the middle of an image ... 203
Section 44.6: Using masks to create images with irregular shapes ... 204
Chapter 45: Fragmentation ... 206
Section 45.1: Media print page-break ... 206
Chapter 46: CSS Object Model (CSSOM) ... 207
Section 46.1: Adding a background-image rule via the CSSOM ... 207
Section 46.2: Introduction ... 207
Chapter 47: Feature Queries ... 208
Section 47.1: Basic @supports usage ... 208
Section 47.2: Chaining feature detections ... 208
Chapter 48: Stacking Context ... 209
Section 48.1: Stacking Context ... 209
Chapter 49: Block Formatting Contexts ... 212
Section 49.1: Using the overflow property with a value different to visible ... 212
Chapter 50: Vertical Centering ... 213
Section 50.1: Centering with display: table ... 213
Section 50.2: Centering with Flexbox ... 213
Section 50.3: Centering with Transform ... 214
Section 50.4: Centering Text with Line Height ... 214
Section 50.5: Centering with Position: absolute ... 214
Section 50.6: Centering with pseudo element ... 215
Chapter 51: Object Fit and Placement ... 217
Section 51.1: object-fit ... 217
Chapter 52: CSS design patterns ... 220
Section 52.1: BEM ... 220
Chapter 53: Browser Support & Prefixes ... 222
Section 53.1: Transitions ... 222
Section 53.2: Transform ... 222
Chapter 54: Normalizing Browser Styles ... 223
Section 54.1: normalize.css ... 223
Section 54.2: Approaches and Examples ... 223
Chapter 55: Internet Explorer Hacks ... 226
Section 55.1: Adding Inline Block support to IE6 and IE7 ... 226
Section 55.2: High Contrast Mode in Internet Explorer 10 and greater ... 226
Section 55.3: Internet Explorer 6 & Internet Explorer 7 only ... 227

Section 55.4: Internet Explorer 8 only 227
Chapter 56: Performance 228
Section 56.1: Use transform and opacity to avoid trigger layout 228
Credits 231
You may also like 236

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Chapter 1: Getting started with CSS

Version	Release Date
1	1996-12-17
2	1998-05-12
3	2015-10-13

Section 1.1: External Stylesheet

An external CSS stylesheet can be applied to any number of HTML documents by placing a `<link>` element in each HTML document.

The attribute `rel` of the `<link>` tag has to be set to `"stylesheet"`, and the `href` attribute to the relative or absolute path to the stylesheet. While using relative URL paths is generally considered good practice, absolute paths can be used, too. In HTML5 the `type` attribute can be omitted.

It is recommended that the `<link>` tag be placed in the HTML file's `<head>` tag so that the styles are loaded before the elements they style. Otherwise, users will see a flash of unstyled content.

Example

hello-world.html

```
<!DOCTYPE html>
<html>
    <head>
        <meta charset="utf-8" />
        <link rel="stylesheet" type="text/css" href="style.css">
    </head>
    <body>
        <h1>Hello world!</h1>
        <p>I ♥ CSS</p>
    </body>
</html>
```

style.css

```
h1 {
    color: green;
    text-decoration: underline;
}
p {
    font-size: 25px;
    font-family: 'Trebuchet MS', sans-serif;
}
```

Make sure you include the correct path to your CSS file in the href. If the CSS file is in the same folder as your HTML file then no path is required (like the example above) but if it's saved in a folder, then specify it like this `href="foldername/style.css"`.

```
<link rel="stylesheet" type="text/css" href="foldername/style.css">
```

External stylesheets are considered the best way to handle your CSS. There's a very simple reason for this: when you're managing a site of, say, 100 pages, all controlled by a single stylesheet, and you want to change your link

colors from blue to green, it's a lot easier to make the change in your CSS file and let the changes "cascade" throughout all 100 pages than it is to go into 100 separate pages and make the same change 100 times. Again, if you want to completely change the look of your website, you only need to update this one file.

You can load as many CSS files in your HTML page as needed.

```
<link rel="stylesheet" type="text/css" href="main.css">
<link rel="stylesheet" type="text/css" href="override.css">
```

CSS rules are applied with some basic rules, and order does matter. For example, if you have a main.css file with some code in it:

```
p.green { color: #00FF00; }
```

All your paragraphs with the 'green' class will be written in light green, but you can override this with another .css file just by including it *after* main.css. You can have override.css with the following code follow main.css, for example:

```
p.green { color: #006600; }
```

Now all your paragraphs with the 'green' class will be written in darker green rather than light green.

Other principles apply, such as the '!important' rule, specificity, and inheritance.

When someone first visits your website, their browser downloads the HTML of the current page plus the linked CSS file. Then when they navigate to another page, their browser only needs to download the HTML of that page; the CSS file is cached, so it does not need to be downloaded again. Since browsers cache the external stylesheet, your pages load faster.

Section 1.2: Internal Styles

CSS enclosed in **`<style></style>`** tags within an HTML document functions like an external stylesheet, except that it lives in the HTML document it styles instead of in a separate file, and therefore can only be applied to the document in which it lives. Note that this element *must* be inside the **`<head>`** element for HTML validation (though it will work in all current browsers if placed in `body`).

```
<head>
    <style>
        h1 {
            color: green;
            text-decoration: underline;
        }
        p {
            font-size: 25px;
            font-family: 'Trebuchet MS', sans-serif;
        }
    </style>
</head>
<body>
    <h1>Hello world!</h1>
    <p>I ♥ CSS</p>
</body>
```

Section 1.3: CSS @import rule (one of CSS at-rule)

The @import CSS at-rule is used to import style rules from other style sheets. These rules must precede all other types of rules, except @charset rules; as it is not a nested statement, @import cannot be used inside conditional group at-rules. `@import`.

How to use @import

You can use @import rule in following ways:

A. With internal style tag

```
<style>
 @import url('/css/styles.css');
</style>
```

B. With external stylesheet

The following line imports a CSS file named `additional-styles.css` in the root directory into the CSS file in which it appears:

```
@import '/additional-styles.css';
```

Importing external CSS is also possible. A common use case are font files.

```
@import 'https://fonts.googleapis.com/css?family=Lato';
```

An optional second argument to `@import` rule is a list of media queries:

```
@import '/print-styles.css' print;
@import url('landscape.css') screen and (orientation:landscape);
```

Section 1.4: Inline Styles

Use inline styles to apply styling to a specific element. Note that this is ***not*** optimal. Placing style rules in a **`<style>`** tag or external CSS file is encouraged in order to maintain a distinction between content and presentation.

Inline styles override any CSS in a **`<style>`** tag or external style sheet. While this can be useful in some circumstances, this fact more often than not reduces a project's maintainability.

The styles in the following example apply directly to the elements to which they are attached.

```
<h1 style="color: green; text-decoration: underline;">Hello world!</h1>
<p style="font-size: 25px; font-family: 'Trebuchet MS';">I ♥ CSS</p>
```

Inline styles are generally the safest way to ensure rendering compatibility across various email clients, programs and devices, but can be time-consuming to write and a bit challenging to manage.

Section 1.5: Changing CSS with JavaScript

Pure JavaScript

It's possible to add, remove or change CSS property values with JavaScript through an element's `style` property.

```
var el = document.getElementById("element");
el.style.opacity = 0.5;
el.style.fontFamily = 'sans-serif';
```

Note that style properties are named in lower camel case style. In the example you see that the css property `font-family` becomes `fontFamily` in javascript.

As an alternative to working directly on elements, you can create a **`<style>`** or **`<link>`** element in JavaScript and append it to the **`<body>`** or **`<head>`** of the HTML document.

jQuery

Modifying CSS properties with jQuery is even simpler.

```
$('#element').css('margin', '5px');
```

If you need to change more than one style rule:

```
$('#element').css({
    margin: "5px",
    padding: "10px",
    color: "black"
});
```

jQuery includes two ways to change css rules that have hyphens in them (i.e. `font-size`). You can put them in quotes or camel-case the style rule name.

```
$('.example-class').css({
    "background-color": "blue",
    fontSize: "10px"
});
```

See also

- JavaScript documentation – Reading and Changing CSS Style.
- jQuery documentation – CSS Manipulation

Section 1.6: Styling Lists with CSS

There are three different properties for styling list-items: `list-style-type`, `list-style-image`, and `list-style-position`, which should be declared in that order. The default values are disc, outside, and none, respectively. Each property can be declared separately, or using the `list-style` shorthand property.

`list-style-type` defines the shape or type of bullet point used for each list-item.

Some of the acceptable values for `list-style-type`:

- disc
- circle
- square
- decimal
- lower-roman
- upper-roman
- none

(For an exhaustive list, see the W3C specification wiki)

To use square bullet points for each list-item, for example, you would use the following property-value pair:

```
li {
    list-style-type: square;
}
```

The **`list-style-image`** property determines whether the list-item icon is set with an image, and accepts a value of `none` or a URL that points to an image.

```
li {
    list-style-image: url(images/bullet.png);
}
```

The **`list-style-position`** property defines where to position the list-item marker, and it accepts one of two values: "inside" or "outside".

```
li {
  list-style-position: inside;
}
```

Chapter 2: Structure and Formatting of a CSS Rule

Section 2.1: Property Lists

Some properties can take multiple values, collectively known as a **property list**.

```
/* Two values in this property list */
span {
    text-shadow: yellow 0 0 3px, green 4px 4px 10px;
}

/* Alternate Formatting */
span {
    text-shadow:
        yellow 0 0 3px,
        green 4px 4px 10px;
}
```

Section 2.2: Multiple Selectors

When you group CSS selectors, you apply the same styles to several different elements without repeating the styles in your style sheet. Use a comma to separate multiple grouped selectors.

```
div, p { color: blue }
```

So the blue color applies to all `<div>` elements and all `<p>` elements. Without the comma only `<p>` elements that are a child of a `<div>` would be red.

This also applies to all types of selectors.

```
p, .blue, #first, div span{ color : blue }
```

This rule applies to:

- `<p>`
- elements of the `blue` class
- element with the ID `first`
- every `<span>` inside of a `<div>`

Section 2.3: Rules, Selectors, and Declaration Blocks

A CSS **rule** consists of a **selector** (e.g. `h1`) and **declaration block** (`{}`).

```
h1 {}
```

Chapter 3: Comments

Section 3.1: Single Line

```
/* This is a CSS comment */
div {
   color: red; /* This is a CSS comment */
}
```

Section 3.2: Multiple Line

```
/*
    This
    is
    a
    CSS
    comment
*/
div {
   color: red;
}
```

Chapter 4: Selectors

CSS selectors identify specific HTML elements as targets for CSS styles. This topic covers how CSS selectors target HTML elements. Selectors use a wide range of over 50 selection methods offered by the CSS language, including elements, classes, IDs, pseudo-elements and pseudo-classes, and patterns.

Section 4.1: Basic selectors

Selector	Description
`*`	Universal selector (all elements)
`div`	Tag selector (all `<div>` elements)
`.blue`	Class selector (all elements with class `blue`)
`.blue.red`	All elements with class `blue` **and** `red` (a type of Compound selector)
`#headline`	ID selector (the element with "id" attribute set to `headline`)
`:pseudo-class`	All elements with pseudo-class
`::pseudo-element`	Element that matches pseudo-element
`:lang(en)`	Element that matches :lang declaration, for example `<span lang="en">`
`div > p`	child selector

> **Note:** The value of an ID must be unique in a web page. It is a violation of the HTML standard to use the value of an ID more than once in the same document tree.

A complete list of selectors can be found in the CSS Selectors Level 3 specification.

Section 4.2: Attribute Selectors

Overview

Attribute selectors can be used with various types of operators that change the selection criteria accordingly. They select an element using the presence of a given attribute or attribute value.

Selector(1)	Matched element	Selects elements...	CSS Version
`[attr]`	`<div attr>`	With attribute `attr`	2
`[attr='val']`	`<div attr="val">`	Where attribute `attr` has value `val`	2
`[attr~='val']`	`<div attr="val val2 val3">`	Where `val` appears in the whitespace-separated list of `attr`	2
`[attr^='val']`	`<div attr="val1 val2">`	Where `attr`'s value *begins* with `val`	3
`[attr$='val']`	`<div attr="sth aval">`	Where the `attr`'s value *ends* with `val`	3
`[attr*='val']`	`<div attr="somevalhere">`	Where `attr` contains `val` anywhere	3
`[attr|='val']`	`<div attr="val-sth etc">`	Where `attr`'s value is exactly `val`, or starts with `val` and immediately followed by `-` (U+002D)	2
`[attr='val' i]`	`<div attr="val">`	Where `attr` has value `val`, ignoring `val`'s letter casing.	4(2)

Notes:

1. The attribute value can be surrounded by either single-quotes or double-quotes. No quotes at all may also work, but it's not valid according to the CSS standard, and is discouraged.

2. There is no single, integrated CSS4 specification, because it is split into separate modules. However, there are "level 4" modules. See browser support.

Details

`[attribute]`

Selects elements with the given attribute.

```
div[data-color] {
  color: red;
}
```

```
<div data-color="red">This will be red</div>
<div data-color="green">This will be red</div>
<div data-background="red">This will NOT be red</div>
```

Live Demo on JSBin

`[attribute="value"]`

Selects elements with the given attribute and value.

```
div[data-color="red"] {
  color: red;
}
```

```
<div data-color="red">This will be red</div>
<div data-color="green">This will NOT be red</div>
<div data-color="blue">This will NOT be red</div>
```

Live Demo on JSBin

`[attribute*="value"]`

Selects elements with the given attribute and value where the given attribute contains the given value anywhere (as a substring).

```
[class*="foo"] {
  color: red;
}
```

```
<div class="foo-123">This will be red</div>
<div class="foo123">This will be red</div>
<div class="bar123foo">This will be red</div>
<div class="barfooo123">This will be red</div>
<div class="barfo0">This will NOT be red</div>
```

Live Demo on JSBin

`[attribute~="value"]`

Selects elements with the given attribute and value where the given value appears in a whitespace-separated list.

```
[class~="color-red"] {
  color: red;
}
```

```
<div class="color-red foo-bar the-div">This will be red</div>
<div class="color-blue foo-bar the-div">This will NOT be red</div>
```

Live Demo on JSBin

```
[attribute^="value"]
```

Selects elements with the given attribute and value where the given attribute begins with the value.

```
[class^="foo-"] {
  color: red;
}
```

```
<div class="foo-123">This will be red</div>
<div class="foo-234">This will be red</div>
<div class="bar-123">This will NOT be red</div>
```

Live Demo on JSBin

```
[attribute$="value"]
```

Selects elements with the given attribute and value where the given attribute ends with the given value.

```
[class$="file"] {
  color: red;
}
```

```
<div class="foobar-file">This will be red</div>
<div class="foobar-file">This will be red</div>
<div class="foobar-input">This will NOT be red</div>
```

Live Demo on JSBin

```
[attribute|="value"]
```

Selects elements with a given attribute and value where the attribute's value is exactly the given value or is exactly the given value followed by `-` (U+002D)

```
[lang|="EN"] {
  color: red;
}
```

```
<div lang="EN-us">This will be red</div>
<div lang="EN-gb">This will be red</div>
<div lang="PT-pt">This will NOT be red</div>
```

Live Demo on JSBin

```
[attribute="value" i]
```

Selects elements with a given attribute and value where the attribute's value can be represented as `Value`, `VALUE`, `vAlUe` or any other case-insensitive possibility.

```
[lang="EN" i] {
  color: red;
}
```

```
<div lang="EN">This will be red</div>
<div lang="en">This will be red</div>
<div lang="PT">This will NOT be red</div>
```

Live Demo on JSBin

Specificity of attribute selectors
`0-1-0`

Same as class selector and pseudoclass.

```
*[type=checkbox] // 0-1-0
```

Note that this means an attribute selector can be used to select an element by its ID at a lower level of specificity than if it was selected with an ID selector: `[id="my-ID"]` targets the same element as `#my-ID` but with lower specificity.

See the Syntax Section for more details.

Section 4.3: Combinators

Overview

Selector	Description
`div span`	Descendant selector (all `<span>`s that are descendants of a `<div>`)
`div > span`	Child selector (all `<span>`s that are a direct child of a `<div>`)
`a ~ span`	General Sibling selector (all `<span>`s that are siblings after an `<a>`)
`a + span`	Adjacent Sibling selector (all `<span>`s that are immediately after an `<a>`)

> **Note:** Sibling selectors target elements that come after them in the source document. CSS, by its nature (it cascades), cannot target *previous* or *parent* elements. However, using the flex `order` property, a previous sibling selector can be simulated on visual media.

Descendant Combinator: `selector selector`

A descendant combinator, represented by at least one space character (), selects elements that are a descendant of the defined element. This combinator selects **all** descendants of the element (from child elements on down).

```
div p {
  color:red;
}
```

```
<div>
  <p>My text is red</p>
  <section>
    <p>My text is red</p>
  </section>
</div>

<p>My text is not red</p>
```

Live Demo on JSBin

In the above example, the first two `<p>` elements are selected since they are both descendants of the `<div>`.

Child Combinator: `selector > selector`

The child (`>`) combinator is used to select elements that are **children**, or **direct descendants**, of the specified element.

```
div > p {
  color:red;
}
```

```
<div>
  <p>My text is red</p>
  <section>
    <p>My text is not red</p>
  </section>
</div>
```

Live Demo on JSBin

The above CSS selects only the first `<p>` element, as it is the only paragraph directly descended from a `<div>`.

The second `<p>` element is not selected because it is not a direct child of the `<div>`.

Adjacent Sibling Combinator: `selector + selector`

The adjacent sibling (+) combinator selects a sibling element that immediate follows a specified element.

```
p + p {
  color:red;
}
```

```
<p>My text is not red</p>
<p>My text is red</p>
<p>My text is red</p>
<hr>
<p>My text is not red</p>
```

Live Demo on JSBin

The above example selects only those `<p>` elements which are *directly preceded* by another `<p>` element.

General Sibling Combinator: `selector ~ selector`

The general sibling (~) combinator selects *all* siblings that follow the specified element.

```
p ~ p {
  color:red;
}
```

```
<p>My text is not red</p>
<p>My text is red</p>
<hr>
<h1>And now a title</h1>
<p>My text is red</p>
```

Live Demo on JSBin

The above example selects all `<p>` elements that are *preceded* by another `<p>` element, whether or not they are immediately adjacent.

Section 4.4: Pseudo-classes

Pseudo-classes are **keywords** which allow selection based on information that lies outside of the document tree or

that cannot be expressed by other selectors or combinators. This information can be associated to a certain state (state and dynamic pseudo-classes), to locations (structural and target pseudo-classes), to negations of the former (negation pseudo-class) or to languages (lang pseudo-class). Examples include whether or not a link has been followed (`:visited`), the mouse is over an element (`:hover`), a checkbox is checked (`:checked`), etc.

Syntax

```
selector:pseudo-class {
    property: VALUE;
}
```

List of pseudo-classes:

Name	Description
`:active`	Applies to any element being activated (i.e. clicked) by the user.
`:any`	Allows you to build sets of related selectors by creating groups that the included items will match. This is an alternative to repeating an entire selector.
`:target`	Selects the current active #news element (clicked on a URL containing that anchor name)
`:checked`	Applies to radio, checkbox, or option elements that are checked or toggled into an "on" state.
`:default`	Represents any user interface element that is the default among a group of similar elements.
`:disabled`	Applies to any UI element which is in a disabled state.
`:empty`	Applies to any element which has no children.
`:enabled`	Applies to any UI element which is in an enabled state.
`:first`	Used in conjunction with the `@page` rule, this selects the first page in a printed document.
`:first-child`	Represents any element that is the first child element of its parent.
`:first-of-type`	Applies when an element is the first of the selected element type inside its parent. This may or may not be the first-child.
`:focus`	Applies to any element which has the user's focus. This can be given by the user's keyboard, mouse events, or other forms of input.
`:focus-within`	Can be used to highlight a whole section when one element inside it is focused. It matches any element that the :focus pseudo-class matches or that has a descendant focused.
`:full-screen`	Applies to any element displayed in full-screen mode. It selects the whole stack of elements and not just the top level element.
`:hover`	Applies to any element being hovered by the user's pointing device, but not activated.
`:indeterminate`	Applies radio or checkbox UI elements which are neither checked nor unchecked, but are in an indeterminate state. This can be due to an element's attribute or DOM manipulation.
`:in-range`	The `:in-range` CSS pseudo-class matches when an element has its value attribute inside the specified range limitations for this element. It allows the page to give a feedback that the value currently defined using the element is inside the range limits.
`:invalid`	Applies to **`<input>`** elements whose values are invalid according to the type specified in the `type=` attribute.
`:lang`	Applies to any element who's wrapping **`<body>`** element has a properly designated `lang=` attribute. For the pseudo-class to be valid, it must contain a valid two or three letter language code.
`:last-child`	Represents any element that is the last child element of its parent.
`:last-of-type`	Applies when an element is the last of the selected element type inside its parent. This may or may not be the last-child.

`:left`	Used in conjunction with the `@page` rule, this selects all the left pages in a printed document.
`:link`	Applies to any links which haven't been visited by the user.
`:not()`	Applies to all elements which **do not** match the value passed to (`:not(p)` or `:not(.class-name)` for example. It must have a value to be valid and it can only contain one selector. However, you can chain multiple `:not` selectors together.
`:nth-child`	Applies when an element is the `n`-th element of its parent, where `n` can be an integer, a mathematical expression (e.g `n+3`) or the keywords `odd` or `even`.
`:nth-of-type`	Applies when an element is the `n`-th element of its parent of the same element type, where `n` can be an integer, a mathematical expression (e.g `n+3`) or the keywords `odd` or `even`.
`:only-child`	The `:only-child` CSS pseudo-class represents any element which is the only child of its parent. This is the same as `:first-child:last-child` or `:nth-child(1):nth-last-child(1)`, but with a lower specificity.
`:optional`	The `:optional` CSS pseudo-class represents any element that does not have the required attribute set on it. This allows forms to easily indicate optional fields and to style them accordingly.
`:out-of-range`	The `:out-of-range` CSS pseudo-class matches when an element has its value attribute outside the specified range limitations for this element. It allows the page to give a feedback that the value currently defined using the element is outside the range limits. A value can be outside of a range if it is either smaller or larger than maximum and minimum set values.
`:placeholder-shown`	**Experimental.** Applies to any form element currently displaying placeholder text.
`:read-only`	Applies to any element which is not editable by the user.
`:read-write`	Applies to any element that is editable by a user, such as **`<input>`** elements.
`:right`	Used in conjunction with the `@page` rule, this selects all the right pages in a printed document.
`:root`	matches the root element of a tree representing the document.
`:scope`	CSS pseudo-class matches the elements that are a reference point for selectors to match against.
`:target`	Selects the current active #news element (clicked on a URL containing that anchor name)
`:visited`	Applies to any links which have has been visited by the user.

> The `:visited` pseudoclass can't be used for most styling in a lot of modern browsers anymore because it's a security hole. See this link for reference.

Section 4.5: Child Pseudo Class

> "The :nth-child(an+b) CSS pseudo-class matches an element that has an+b-1 siblings before it in the document tree, for a given positive **or zero value** for n" - MDN :nth-child

pseudo-selector	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
`:first-child`	✓									
`:nth-child(3)`			✓							
`:nth-child(n+3)`			✓	✓	✓	✓	✓	✓	✓	✓
`:nth-child(3n)`			✓			✓			✓	

`:nth-child(3n+1)`	✓		✓		✓		✓
`:nth-child(-n+3)`	✓	✓	✓				
`:nth-child(odd)`	✓		✓		✓		✓		✓
`:nth-child(even)`		✓		✓		✓		✓		✓
`:last-child`										✓
`:nth-last-child(3)`							✓			

Section 4.6: Class Name Selectors

The class name selector select all elements with the targeted class name. For example, the class name `.warning` would select the following `<div>` element:

```
<div class="warning">
    <p>This would be some warning copy.</p>
</div>
```

You can also combine class names to target elements more specifically. Let's build on the example above to showcase a more complicated class selection.

CSS

```
.important {
    color: orange;
}
.warning {
    color: blue;
}
.warning.important {
    color: red;
}
```

HTML

```
<div class="warning">
    <p>This would be some warning copy.</p>
</div>

<div class="important warning">
    <p class="important">This is some really important warning copy.</p>
</div>
```

In this example, all elements with the `.warning` class will have a blue text color, elements with the `.important` class with have an orange text color, and all elements that have *both* the `.important` and `.warning` class name will have a red text color.

Notice that within the CSS, the `.warning.important` declaration did not have any spaces between the two class names. This means it will only find elements which contain both class names `warning` and `important` in their `class` attribute. Those class names could be in any order on the element.

If a space was included between the two classes in the CSS declaration, it would only select elements that have parent elements with a `.warning` class names and child elements with `.important` class names.

Section 4.7: Select element using its ID without the high specificity of the ID selector

This trick helps you select an element using the ID as a value for an attribute selector to avoid the high specificity of the ID selector.

HTML:

```
<div id="element">...</div>
```

CSS

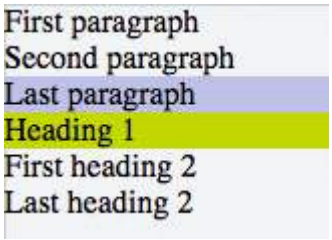
```
#element { ... } /* High specificity will override many selectors */

[id="element"] { ... } /* Low specificity, can be overridden easily */
```

Section 4.8: The :last-of-type selector

The `:last-of-type` selects the element that is the last child, of a particular type, of its parent. In the example below, the css selects the last paragraph and the last heading `h1`.

```
p:last-of-type {
  background: #C5CAE9;
}
h1:last-of-type {
  background: #CDDC39;
}
```

```
<div class="container">
    <p>First paragraph</p>
    <p>Second paragraph</p>
    <p>Last paragraph</p>
    <h1>Heading 1</h1>
    <h2>First heading 2</h2>
    <h2>Last heading 2</h2>
</div>
```



jsFiddle

Section 4.9: CSS3 :in-range selector example

```
<style>
input:in-range {
    border: 1px solid blue;
}
</style>



<input type="number" min="10" max="20" value="15">
```

```
<p>The border for this value will be blue</p>
```

The `:in-range` CSS pseudo-class matches when an element has its value attribute inside the specified range limitations for this element. It allows the page to give a feedback that the value currently defined using the element is inside the range limits.[1]

Section 4.10: A. The :not pseudo-class example & B. :focus-within CSS pseudo-class

A. The syntax is presented above.

The following selector matches all `<input>` elements in an HTML document that are not disabled and don't have the class `.example`:

HTML:

```
<form>
    Phone: <input type="tel" class="example">
    E-mail: <input type="email" disabled="disabled">
    Password: <input type="password">
</form>
```

CSS:

```
input:not([disabled]):not(.example){
   background-color: #ccc;
}
```

The `:not()` pseudo-class will also support comma-separated selectors in Selectors Level 4:

CSS:

```
input:not([disabled], .example){
   background-color: #ccc;
}
```

Live Demo on JSBin

See background syntax here.

B. The :focus-within CSS pseudo-class

HTML:

```
  <h3>Background is blue if the input is focused .</p>
  <div>
    <input type="text">
  </div>
```

CSS:

```
div {
  height: 80px;
}
input{
  margin:30px;
```

```
}
div:focus-within {
  background-color: #1565C0;
}
```

```
div {
  height: 80px;
}
input{
  margin:30px;
}
div:focus-within {
  background-color: #1565C0;
}
```

Background is blue if the input is focused .

:focus-within CSS pseudo-class - UNOFF

Global 13.62%

The `:focus-within` pseudo-class matches elements that either themselves match `:focus` or that have descendants which match `:focus`.

Current aligned | Usage relative | Date relative | Show all

IE	Edge *	Firefox	Chrome	Safari	Opera	iOS Safari *	Opera Mini *	Android Browser *	Chrome for Android
		52	49			9.3		4.4	
	14	53	58		45	10.2		4.4.4	
11	15	54	[1] 59	10.1	[1] 46	10.3	all	56	59
	16	55	60	11	[1] 47	11			
		56	61	TP	[1] 48				
		57	62						

Notes | Known issues (0) | Resources (11) | Feedback

[1] Can be enabled via the "Experimental Web Platform Features" flag

Section 4.11: Global boolean with checkbox:checked and ~ (general sibling combinator)

With the ~ selector, you can easily implement a global accessible boolean without using JavaScript.

Add boolean as a checkbox

To the very beginning of your document, add as much booleans as you want with a unique `id` and the `hidden` attribute set:

```
<input type="checkbox" id="sidebarShown" hidden />
<input type="checkbox" id="darkThemeUsed" hidden />

<!-- here begins actual content, for example: -->
<div id="container">
    <div id="sidebar">
        <!-- Menu, Search, ... -->
    </div>

    <!-- Some more content ... -->
</div>

<div id="footer">
    <!-- ... -->
</div>
```

Change the boolean's value

You can toggle the boolean by adding a `label` with the `for` attribute set:

```
<label for="sidebarShown">Show/Hide the sidebar!</label>
```

Accessing boolean value with CSS

The normal selector (like `.color-red`) specifies the default properties. They can be overridden by following `true` / `false` selectors:

```
/* true: */
<checkbox>:checked ~ [sibling of checkbox & parent of target] <target>

/* false: */
<checkbox>:not(:checked) ~ [sibling of checkbox & parent of target] <target>
```

Note that **`<checkbox>`**, `[sibling ...]` and **`<target>`** should be replaced by the proper selectors. `[sibling ...]` can be a specific selector, (often if you're lazy) simply `*` or nothing if the target is already a sibling of the checkbox.

Examples for the above HTML structure would be:

```
#sidebarShown:checked ~ #container #sidebar {
    margin-left: 300px;
}

#darkThemeUsed:checked ~ #container,
#darkThemeUsed:checked ~ #footer {
    background: #333;
}
```

In action

See this fiddle for a implementation of these global booleans.

Section 4.12: ID selectors

ID selectors select DOM elements with the targeted ID. To select an element by a specific ID in CSS, the `#` prefix is used.

For example, the following HTML `div` element…

```
<div id="exampleID">
    <p>Example</p>
</div>
```

…can be selected by `#exampleID` in CSS as shown below:

```
#exampleID {
    width: 20px;
}
```

> **Note**: The HTML specs do not allow multiple elements with the same ID

Section 4.13: How to style a Range input

HTML

```
<input type="range"></input>
```

CSS

Effect	Pseudo Selector
Thumb	`input[type=range]::-webkit-slider-thumb`, `input[type=range]::-moz-range-thumb`, `input[type=range]::-ms-thumb`
Track	`input[type=range]::-webkit-slider-runnable-track`, `input[type=range]::-moz-range-track`, `input[type=range]::-ms-track`
OnFocus	`input[type=range]:focus`
Lower part of the track	`input[type=range]::-moz-range-progress`, `input[type=range]::-ms-fill-lower` (not possible in WebKit browsers currently - JS needed)

Section 4.14: The :only-child pseudo-class selector example

The `:only-child` CSS pseudo-class represents any element which is the only child of its parent.

HTML:

```
<div>
  <p>This paragraph is the only child of the div, it will have the color blue</p>
</div>

<div>
  <p>This paragraph is one of the two children of the div</p>
  <p>This paragraph is one of the two children of its parent</p>
</div>
```

CSS:

```
p:only-child {
  color: blue;
}
```

The above example selects the **`<p>`** element that is the unique child from its parent, in this case a **`<div>`**.

Live Demo on JSBin

Chapter 5: Backgrounds

With CSS you can set colors, gradients, and images as the background of an element.

It is possible to specify various combinations of images, colors, and gradients, and adjust the size, positioning, and repetition (among others) of these.

Section 5.1: Background Color

The `background-color` property sets the background color of an element using a color value or through keywords, such as `transparent`, `inherit` or `initial`.

- **transparent**, specifies that the background color should be transparent. This is default.
- **inherit**, inherits this property from its parent element.
- **initial**, sets this property to its default value.

This can be applied to all elements, and `::first-letter`/`::first-line` pseudo-elements.

Colors in CSS can be specified by different methods.

Color names

CSS

```
div {
  background-color: red;  /* red */
}
```

HTML

```
<div>This will have a red background</div>
```

- The example used above is one of several ways that CSS has to represent a single color.

Hex color codes

Hex code is used to denote RGB components of a color in base-16 hexadecimal notation. #ff0000, for example, is bright red, where the red component of the color is 256 bits (ff) and the corresponding green and blue portions of the color is 0 (00).

If both values in each of the three RGB pairings (R, G, and B) are the same, then the color code can be shortened into three characters (the first digit of each pairing). `#ff0000` can be shortened to `#f00`, and `#ffffff` can be shortened to `#fff`.

Hex notation is case-insensitive.

```
body {
  background-color: #de1205; /* red */
}

.main {
```

```
  background-color: #00f; /* blue */
}
```

RGB / RGBa

Another way to declare a color is to use RGB or RGBa.

RGB stands for Red, Green and Blue, and requires of three separate values between 0 and 255, put between brackets, that correspond with the decimal color values for respectively red, green and blue.

RGBa allows you to add an additional alpha parameter between 0.0 and 1.0 to define opacity.

```
header {
  background-color: rgb(0, 0, 0); /* black */
}

footer {
  background-color: rgba(0, 0, 0, 0.5); /* black with 50% opacity */
}
```

HSL / HSLa

Another way to declare a color is to use HSL or HSLa and is similar to RGB and RGBa.

HSL stands for hue, saturation, and lightness, and is also often called HLS:

- Hue is a degree on the color wheel (from 0 to 360).
- Saturation is a percentage between 0% and 100%.
- Lightness is also a percentage between 0% and 100%.

HSLa allows you to add an additional alpha parameter between 0.0 and 1.0 to define opacity.

```
li a {
  background-color: hsl(120, 100%, 50%); /* green */
}

#p1 {
  background-color: hsla(120, 100%, 50%, .3); /* green with 30% opacity */
}
```

Interaction with background-image

The following statements are all equivalent:

```
body {
  background: red;
  background-image: url(partiallytransparentimage.png);
}

body {
  background-color: red;
  background-image: url(partiallytransparentimage.png);
}

body {
  background-image: url(partiallytransparentimage.png);
  background-color: red;
```

```
}

body {
  background: red url(partiallytransparentimage.png);
}
```

They will all lead to the red color being shown underneath the image, where the parts of the image are transparent, or the image is not showing (perhaps as a result of `background-repeat`).

Note that the following is not equivalent:

```
body {
  background-image: url(partiallytransparentimage.png);
  background: red;
}
```

Here, the value of `background` overrides your `background-image`.

For more info on the `background` property, see Background Shorthand

Section 5.2: Background Gradients

Gradients are new image types, added in CSS3. As an image, gradients are set with the `background-image` property, or the `background` shorthand.

There are two types of gradient functions, linear and radial. Each type has a non-repeating variant and a repeating variant:

- `linear-gradient()`
- `repeating-linear-gradient()`
- `radial-gradient()`
- `repeating-radial-gradient()`

linear-gradient()

A `linear-gradient` has the following syntax

```
background: linear-gradient( <direction>?, <color-stop-1>, <color-stop-2>, ...);
```

Value	Meaning
`<direction>`	Could be an argument like `to top`, `to bottom`, `to right` or `to left`; or an angle as `0deg`, `90deg`... . The angle starts from to top and rotates clockwise. Can be specified in deg, grad, rad, or turn. If omitted, the gradient flows from top to bottom
`<color-stop-list>`	List of colors, optionally followed each one by a percentage or length to display it at. For example, `yellow 10%`, `rgba(0,0,0,.5) 40px`, `#fff 100%`...

For example, this creates a linear gradient that starts from the right and transitions from red to blue

```
.linear-gradient {
  background: linear-gradient(to left, red, blue); /* you can also use 270deg */
}
```

You can create a `diagonal` gradient by declaring both a horizontal and vertical starting position.

```
.diagonal-linear-gradient {
  background: linear-gradient(to left top, red, yellow 10%);
```

```
}
```

It is possible to specify any number of color stops in a gradient by separating them with commas. The following examples will create a gradient with 8 color stops

```
.linear-gradient-rainbow {
  background: linear-gradient(to left, red, orange, yellow, green, blue, indigo, violet)
}
```

radial-gradient()

```
.radial-gradient-simple {
  background: radial-gradient(red, blue);
}

.radial-gradient {
  background: radial-gradient(circle farthest-corner at top left, red, blue);
}
```

Value	Meaning
`circle`	Shape of gradient. Values are `circle` or `ellipse`, default is `ellipse`.
`farthest-corner`	Keywords describing how big the ending shape must be. Values are `closest-side`, `farthest-side`, `closest-corner`, `farthest-corner`
`top left`	Sets the position of the gradient center, in the same way as `background-position`.

Repeating gradients

Repeating gradient functions take the same arguments as the above examples, but tile the gradient across the background of the element.

```
.bullseye {
  background: repeating-radial-gradient(red, red 10%, white 10%, white 20%);
}
.warning {
  background: repeating-linear-gradient(-45deg, yellow, yellow 10%, black 10%, black 20% );
}
```

Value	Meaning
`-45deg`	Angle unit. The angle starts from to top and rotates clockwise. Can be specified in deg, grad, rad, or turn.
`to left`	Direction of gradient, default is `to bottom`. Syntax: `to [y-axis(top OR bottom)] [x-axis(left OR right)]` ie `to top right`
`yellow 10%`	Color, optionally followed by a percentage or length to display it at. Repeated two or more times.

Note that HEX, RGB, RGBa, HSL, and HSLa color codes may be used instead of color names. Color names were used for the sake of illustration. Also note that the radial-gradient syntax is much more complex than linear-gradient, and a simplified version is shown here. For a full explanation and specs, see the MDN Docs

Section 5.3: Background Image

The `background-image` property is used to specify a background image to be applied to all matched elements. By default, this image is tiled to cover the entire element, excluding margin.

```
.myClass {
  background-image: url('/path/to/image.jpg');
}
```

To use multiple images as `background-image`, define comma separated `url()`

```
.myClass {
  background-image: url('/path/to/image.jpg'),
                    url('/path/to/image2.jpg');
}
```

The images will stack according to their order with the first declared image on top of the others and so on.

Value	Result
`url('/path/to/image.jpg')`	Specify background image's path(s) or an image resource specified with data URI schema (apostrophes can be omitted), separate multiples by comma
`none`	No background image
`initial`	Default value
`inherit`	Inherit parent's value

More CSS for Background Image

This following attributes are very useful and almost essential too.

```
background-size:     xpx ypx | x% y%;
background-repeat:   no-repeat | repeat | repeat-x | repeat-y;
background-position: left offset (px/%) right offset (px/%) | center center | left top | right
bottom;
```

Section 5.4: Background Shorthand

The `background` property can be used to set one or more background related properties:

Value	Description	CSS Ver.
`background-image`	Background image to use	1+
`background-color`	Background color to apply	1+
`background-position`	Background image's position	1+
`background-size`	Background image's size	3+
`background-repeat`	How to repeat background image	1+
`background-origin`	How the background is positioned (ignored when `background-attachment` is `fixed`)	3+
`background-clip`	How the background is painted relative to the `content-box`, `border-box`, or the `padding-box`	3+
`background-attachment`	How the background image behaves, whether it scrolls along with its containing block or has a fixed position within the viewport	1+
`initial`	Sets the property to value to default	3+
`inherit`	Inherits property value from parent	2+

The order of the values does not matter and every value is optional

Syntax

The syntax of the background shorthand declaration is:

```
background: [<background-image>] [<background-color>]  [<background-position>]/[<background-size>]
[<background-repeat>] [<background-origin>] [<background-clip>] [<background-attachment>]
[<initial|inherit>];
```

Examples

```
background: red;
```

Simply setting a `background-color` with the `red`value.

```
background: border-box red;
```

Setting a `background-clip` to border-box and a `background-color` to red.

```
background: no-repeat center url("somepng.jpg");
```

Sets a `background-repeat` to no-repeat, `background-origin` to center and a `background-image` to an image.

```
background: url('pattern.png') green;
```

In this example, the `background-color` of the element would be set to `green` with `pattern.png`, if it is available, overlayed on the colour, repeating as often as necessary to fill the element. If `pattern.png` includes any transparency then the `green` colour will be visible behind it.

```
background: #000000 url("picture.png") top left / 600px auto no-repeat;
```

In this example we have a black background with an image 'picture.png' on top, the image does not repeat in either axis and is positioned in the top left corner. The `/` after the position is to be able to include the size of the background image which in this case is set as `600px` width and auto for the height. This example could work well with a feature image that can fade into a solid colour.

> **NOTE:** Use of the shorthand background property resets all previously set background property values, even if a value is not given. If you wish only to modify a background property value previously set, use a longhand property instead.

Section 5.5: Background Size

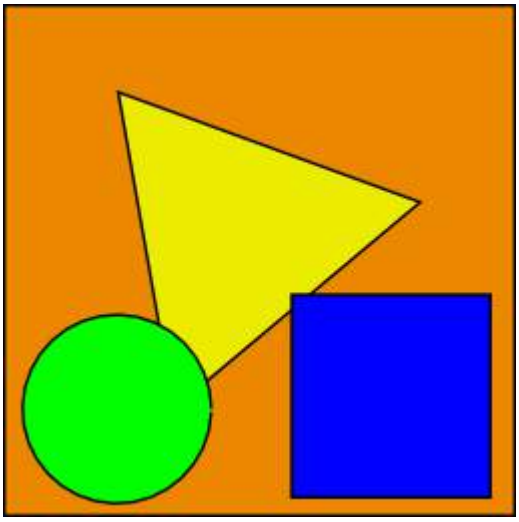
General overview

The `background-size` property enables one to control the scaling of the `background-image`. It takes up to two values, which determine the scale/size of the resulting image in vertical and and horizontal direction. If the property is missing, its deemed `auto` in both `width` and `height`.

`auto` will keep the image's aspect ratio, if it can be determined. The height is optional and can be considered `auto`. Therefore, on a 256 px × 256 px image, all the following `background-size` settings would yield an image with height and width of 50 px:

```
background-size: 50px;
background-size: 50px auto; /* same as above */
background-size: auto 50px;
background-size: 50px 50px;
```

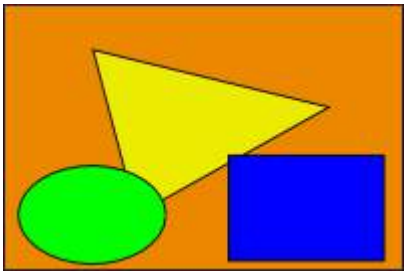
So if we started with the following picture (which has the mentioned size of 256 px × 256 px),



we'll end up with a 50 px × 50 px on the user's screen, contained in the background of our element:

One can also use percentage values to scale the image with respect of the element. The following example would yield a 200 px × 133 px drawn image:

```
#withbackground {
    background-image: url(to/some/background.png);

    background-size: 100% 66%;

    width: 200px;
    height: 200px;

    padding: 0;
    margin: 0;
}
```



The behaviour depends on the `background-origin`.

Keeping the aspect ratio

The last example in the previos section lost its original aspect ratio. The circle got into an ellipse, the square into a rectangle, the triangle into another triangle.

The length or percentage approach isn't flexible enough to keep the aspect ratio at all times. `auto` doesn't help, since you might not know which dimension of your element will be larger. However, to cover certain areas with an

image (and correct aspect ratio) completely or to contain an image with correct aspect ratio completely in a background area, the values, `contain` and `cover` provide the additional functionality.

Eggsplanation for `contain` and `cover`

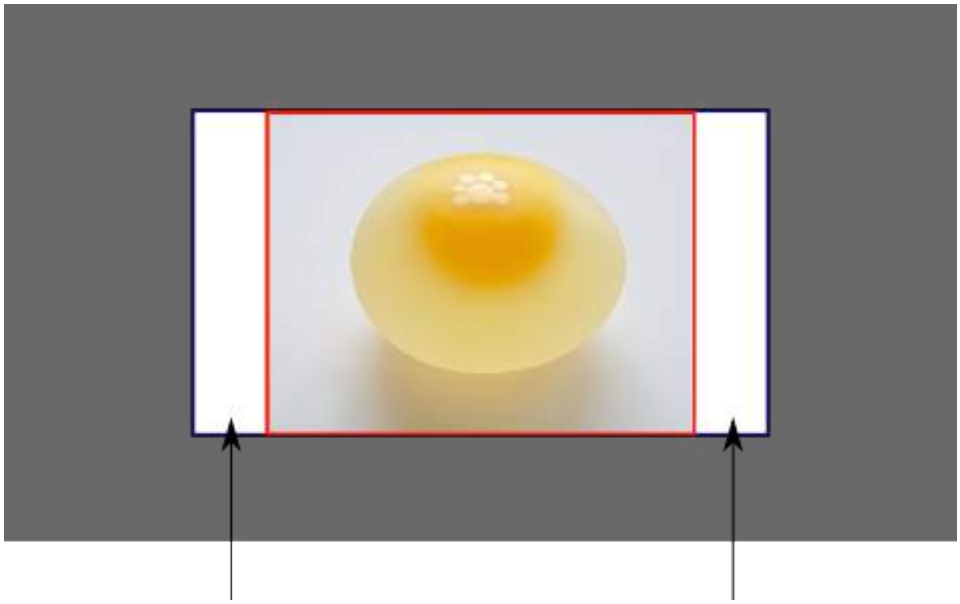
Sorry for the bad pun, but we're going to use a picture of the day by Biswarup Ganguly for demonstration. Lets say that this is your screen, and the gray area is outside of your visible screen. For demonstration, We're going to assume a 16 × 9 ratio.

We want to use the aforementioned picture of the day as a background. However, we cropped the image to 4x3 for some reason. We could set the `background-size` property to some fixed length, but we will focus on `contain` and `cover`. Note that I also assume that we didn't mangle the width and/or height of `body`.

`contain`

> `contain`
>
> Scale the image, while preserving its intrinsic aspect ratio (if any), to the largest size such that both its width and its height can fit inside the background positioning area.

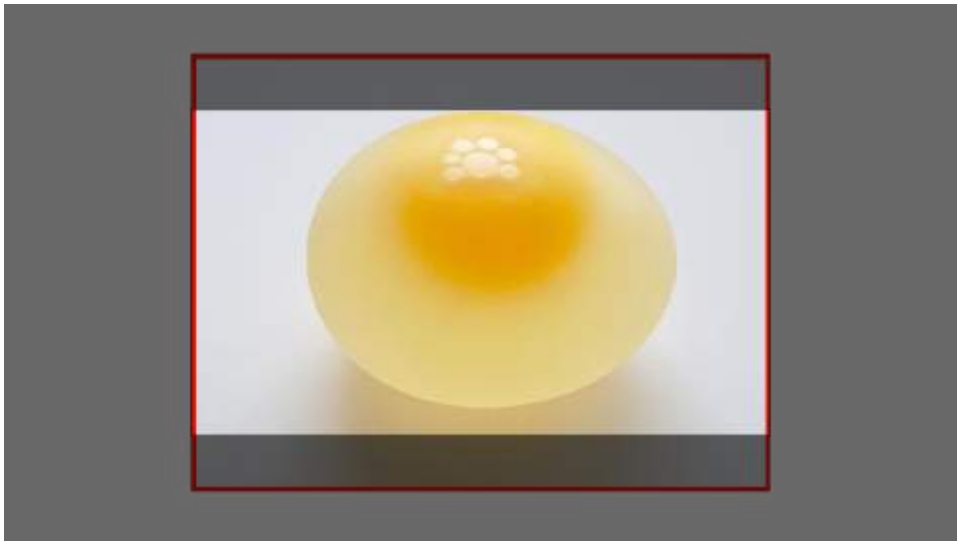
This makes sure that the background image is always completely contained in the background positioning area, however, there could be some empty space filled with your `background-color` in this case:



`cover`

> `cover`
>
> Scale the image, while preserving its intrinsic aspect ratio (if any), to the smallest size such that both its width and its height can completely cover the background positioning area.

This makes sure that the background image is covering everything. There will be no visible `background-color`, however depending on the screen's ratio a great part of your image could be cut off:



Demonstration with actual code

```
div > div {
  background-image: url(http://i.stack.imgur.com/r5CAq.jpg);
  background-repeat: no-repeat;
  background-position: center center;
  background-color: #ccc;
  border: 1px solid;
  width: 20em;
  height: 10em;
}
div.contain {
  background-size: contain;
}
div.cover {
  background-size: cover;
}
/*******************************************
 Additional styles for the explanation boxes
*******************************************/

div > div {
  margin: 0 1ex 1ex 0;
  float: left;
}
div + div {
  clear: both;
  border-top: 1px dashed silver;
  padding-top:1ex;
}
div > div::after {
  background-color: #000;
  color: #fefefe;
```

```
  margin: 1ex;
  padding: 1ex;
  opacity: 0.8;
  display: block;
  width: 10ex;
  font-size: 0.7em;
  content: attr(class);
}
```

```
<div>
  <div class="contain"></div>
  <p>Note the grey background. The image does not cover the whole region, but it's fully
<em>contained</em>.
  </p>
</div>
<div>
  <div class="cover"></div>
  <p>Note the ducks/geese at the bottom of the image. Most of the water is cut, as well as a part
of the sky. You don't see the complete image anymore, but neither do you see any background color;
the image <em>covers</em> all of the <code>&lt;div&gt;</code>.</p>
</div>
```

Note the grey background. The image does not cover the whole region, but it's fully *contained*.

Note the ducks/geese at the bottom of the image. Most of the water is cut, as well as a part of the sky. You don't see the complete image anymore, but neither do you see any background color; the image *covers* all of the `<div>`.

Section 5.6: Background Position

The `background-position` property is used to specify the starting position for a background image or gradient

```
.myClass {
  background-image: url('path/to/image.jpg');
  background-position: 50% 50%;
}
```

The position is set using an **X** and **Y** co-ordinate and be set using any of the units used within CSS.

Unit	Description

value% *value*%	A percentage for the horizontal offset is relative to *(width of background positioning area - width of background image)*. A percentage for the vertical offset is relative to *(height of background positioning area - height of background image)* The size of the image is the size given by `background-size`.
*value*px *value*px	Offsets background image by a length given in pixels relative to the top left of the background positioning area

Units in CSS can be specified by different methods (see here).

Longhand Background Position Properties

In addition to the shorthand property above, one can also use the longhand background properties `background-position-x` and `background-position-y`. These allow you to control the x or y positions separately.

> **NOTE:** This is supported in all browsers except Firefox (versions 31-48) 2. Firefox 49, to be released September 2016, *will* support these properties. Until then, there is a Firefox hack within this Stack Overflow answer.

Section 5.7: The background-origin property

The background-origin property specifies where the background image is positioned.

Note: If the `background-attachment` property is set to `fixed`, this property has no effect.

Default value: `padding-box`

Possible values:

- `padding-box` - The position is relative to the padding box
- `border-box` - The position is relative to the border box
- `content-box` - The position is relative to the content box
- `initial`
- `inherit`

CSS

```
.example {
  width: 300px;
  border: 20px solid black;
  padding: 50px;
  background: url(https://static.pexels.com/photos/6440/magazines-desk-work-workspace-medium.jpg);
  background-repeat: no-repeat;
}

.example1 {}

.example2 { background-origin: border-box; }

.example3 { background-origin: content-box; }
```

HTML

```
<p>No background-origin (padding-box is default):</p>
```

```
<div class="example example1">
  <h2>Lorem Ipsum Dolor</h2>
  <p>Lorem ipsum dolor sit amet, consectetuer adipiscing elit, sed diam nonummy nibh euismod
tincidunt ut laoreet dolore magna aliquam erat volutpat.</p>
  <p>Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut
aliquip ex ea commodo consequat.</p>
</div>

<p>background-origin: border-box:</p>
<div class="example example2">
  <h2>Lorem Ipsum Dolor</h2>
  <p>Lorem ipsum dolor sit amet, consectetuer adipiscing elit, sed diam nonummy nibh euismod
tincidunt ut laoreet dolore magna aliquam erat volutpat.</p>
  <p>Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut
aliquip ex ea commodo consequat.</p>
</div>

<p>background-origin: content-box:</p>
<div class="example example3">
  <h2>Lorem Ipsum Dolor</h2>
  <p>Lorem ipsum dolor sit amet, consectetuer adipiscing elit, sed diam nonummy nibh euismod
tincidunt ut laoreet dolore magna aliquam erat volutpat.</p>
  <p>Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut
aliquip ex ea commodo consequat.</p>
</div>
```

Result:

No background-origin (padding-box is default):

background-origin: border-box:

background-origin: content-box:

More:

https://www.w3.org/TR/css3-background/#the-background-origin

https://developer.mozilla.org/en-US/docs/Web/CSS/background-origin

Section 5.8: Multiple Background Image

In CSS3, we can stack multiple background in the same element.

```
#mydiv {
  background-image: url(img_1.png), /* top image */
                    url(img_2.png), /* middle image */
                    url(img_3.png); /* bottom image */
  background-position: right bottom,
                       left top,
                       right top;
  background-repeat: no-repeat,
                     repeat,
                     no-repeat;
}
```

Images will be stacked atop one another with the first background on top and the last background in the back. `img_1` will be on top, the `img_2` and `img_3` is on bottom.

We can also use background shorthand property for this:

```
#mydiv {
  background: url(img_1.png) right bottom no-repeat,
              url(img_2.png) left top repeat,
              url(img_3.png) right top no-repeat;
}
```

We can also stack images and gradients:

```
#mydiv {
  background: url(image.png) right bottom no-repeat,
              linear-gradient(to bottom, #fff 0%,#000 100%);
}
```

- Demo

Section 5.9: Background Attachment

The background-attachment property sets whether a background image is fixed or scrolls with the rest of the page.

```
body {
  background-image: url('img.jpg');
  background-attachment: fixed;
}
```

Value	Description
scroll	The background scrolls along with the element. This is default.
fixed	The background is fixed with regard to the viewport.
local	The background scrolls along with the element's contents.
initial	Sets this property to its default value.
inherit	Inherits this property from its parent element.

Examples

background-attachment: scroll

The default behaviour, when the body is scrolled the background scrolls with it:

```
body {
  background-image: url('image.jpg');
  background-attachment: scroll;
}
```

background-attachment: fixed

The background image will be fixed and will not move when the body is scrolled:

```
body {
  background-image: url('image.jpg');
  background-attachment: fixed;
}
```

background-attachment: local

The background image of the div will scroll when the contents of the div is scrolled.

```
div {
```

```
  background-image: url('image.jpg');
  background-attachment: local;
}
```

Section 5.10: Background Clip

Definition and Usage: The `background-clip` property specifies the painting area of the background.

Default value: `border-box`

Values

- `border-box` is the default value. This allows the background to extend all the way to the outside edge of the element's border.
- `padding-box` clips the background at the outside edge of the element's padding and does not let it extend into the border;
- `content-box` clips the background at the edge of the content box.
- `inherit` applies the setting of the parent to the selected element.

CSS

```
.example {
  width: 300px;
  border: 20px solid black;
  padding: 50px;
  background: url(https://static.pexels.com/photos/6440/magazines-desk-work-workspace-medium.jpg);
  background-repeat: no-repeat;
}

.example1 {}

.example2 { background-origin: border-box; }

.example3 { background-origin: content-box; }
```

HTML

```
<p>No background-origin (padding-box is default):</p>

<div class="example example1">
  <h2>Lorem Ipsum Dolor</h2>
  <p>Lorem ipsum dolor sit amet, consectetuer adipiscing elit, sed diam nonummy nibh euismod
tincidunt ut laoreet dolore magna aliquam erat volutpat.</p>
  <p>Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut
aliquip ex ea commodo consequat.</p>
</div>

<p>background-origin: border-box:</p>
<div class="example example2">
  <h2>Lorem Ipsum Dolor</h2>
  <p>Lorem ipsum dolor sit amet, consectetuer adipiscing elit, sed diam nonummy nibh euismod
tincidunt ut laoreet dolore magna aliquam erat volutpat.</p>
  <p>Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut
aliquip ex ea commodo consequat.</p>
</div>

<p>background-origin: content-box:</p>
<div class="example example3">
  <h2>Lorem Ipsum Dolor</h2>
```

```
  <p>Lorem ipsum dolor sit amet, consectetuer adipiscing elit, sed diam nonummy nibh euismod
tincidunt ut laoreet dolore magna aliquam erat volutpat.</p>
  <p>Ut wisi enim ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl ut
aliquip ex ea commodo consequat.</p>
</div>
```

Section 5.11: Background Repeat

The background-repeat property sets if/how a background image will be repeated.

By default, a background-image is repeated both vertically and horizontally.

```
div {
  background-image: url("img.jpg");
  background-repeat: repeat-y;
}
```

Here's how a `background-repeat: repeat-y` looks like:

Section 5.12: background-blend-mode Property

```
.my-div {
    width: 300px;
    height: 200px;
    background-size: 100%;
    background-repeat: no-repeat;
    background-image: linear-gradient(to right, black 0%,white 100%),
url('https://static.pexels.com/photos/54624/strawberry-fruit-red-sweet-54624-medium.jpeg');
    background-blend-mode:saturation;
}
```

```
<div class="my-div">Lorem ipsum</div>
```

See result here: https://jsfiddle.net/MadalinaTn/y69d28Lb/

CSS Syntax: background-blend-mode: normal | multiply | screen | overlay | darken | lighten | color-dodge | saturation | color | luminosity;

Section 5.13: Background Color with Opacity

If you set `opacity` on an element it will affect all its child elements. To set an opacity just on the background of an element you will have to use RGBA colors. Following example will have a black background with 0.6 opacity.

```
/* Fallback for web browsers that don't support RGBa */
background-color: rgb(0, 0, 0);

/* RGBa with 0.6 opacity */
background-color: rgba(0, 0, 0, 0.6);

/* For IE 5.5 - 7*/
filter: progid:DXImageTransform.Microsoft.gradient(startColorstr=#99000000, endColorstr=#99000000);

/* For IE 8*/
-ms-filter: "progid:DXImageTransform.Microsoft.gradient(startColorstr=#99000000,
endColorstr=#99000000)";
```

Chapter 6: Centering

Section 6.1: Using Flexbox

HTML:

```
<div class="container">
  <img src="http://lorempixel.com/400/200" />
</div>
```

CSS:

```
html, body, .container {
  height: 100%;
}
.container {
  display: flex;
  justify-content: center; /* horizontal center */
}
img {
  align-self: center; /* vertical center */
}
```

View Result

HTML:

```
<img src="http://lorempixel.com/400/200" />
```

CSS:

```
html, body {
  height: 100%;
}
body {
  display: flex;
  justify-content: center; /* horizontal center */
  align-items: center;     /* vertical center */
}
```

View Result

See Dynamic Vertical and Horizontal Centering under the Flexbox documentation for more details on flexbox and what the styles mean.

Browser Support

Flexbox is supported by all major browsers, except IE versions before 10.

Some recent browser versions, such as Safari 8 and IE10, require vendor prefixes.

For a quick way to generate prefixes there is Autoprefixer, a third-party tool.

For older browsers (like IE 8 & 9) a Polyfill is available.

For a more detailed look at flexbox browser support, see this answer.

Section 6.2: Using CSS transform

CSS transforms are based on the size of the elements so if you don't know how tall or wide your element is, you can position it absolutely 50% from the top and left of a relative container and translate it by 50% left and upwards to center it vertically and horizontally.

Keep in mind that with this technique, the element could end being rendered at a non-integer pixel boundary, making it look blurry. See this answer in SO for a workaround.

HTML

```
<div class="container">
  <div class="element"></div>
</div>
```

CSS

```
.container {
  position: relative;
}

.element {
  position: absolute;
  top: 50%;
  left: 50%;
  transform: translate(-50%, -50%);
}
```

View example in JSFiddle

CROSS BROWSER COMPATIBILITY

The transform property needs prefixes to be supported by older browsers. Prefixes are needed for Chrome<=35, Safari<=8, Opera<=22, Android Browser<=4.4.4, and IE9. CSS transforms are not supported by IE8 and older versions.

Here is a common transform declaration for the previous example:

```
-webkit-transform: translate(-50%, -50%); /* Chrome, Safari, Opera, Android */
    -ms-transform: translate(-50%, -50%); /* IE 9 */
        transform: translate(-50%, -50%);
```

For more information see canluse.

MORE INFORMATION

- The element is being positioned according to the first non-static parent (`position: relative`, `absolute`, or `fixed`). Explore more in this fiddle and this documentation topic.
- For horizontal-only centering, use `left: 50%` and `transform: translateX(-50%)`. The same goes for vertical-only centering: center with `top: 50%` and `transform: translateY(-50%)`.
- Using a non-static width/height elements with this method of centering can cause the centered element to appear squished. This mostly happens with elements containing text, and can be fixed by adding: `margin-right: -50%;` and `margin-bottom: -50%;`. View this fiddle for more information.

Section 6.3: Using margin: 0 auto;

Objects can be centered by using `margin: 0 auto;` if they are block elements and have a defined width.

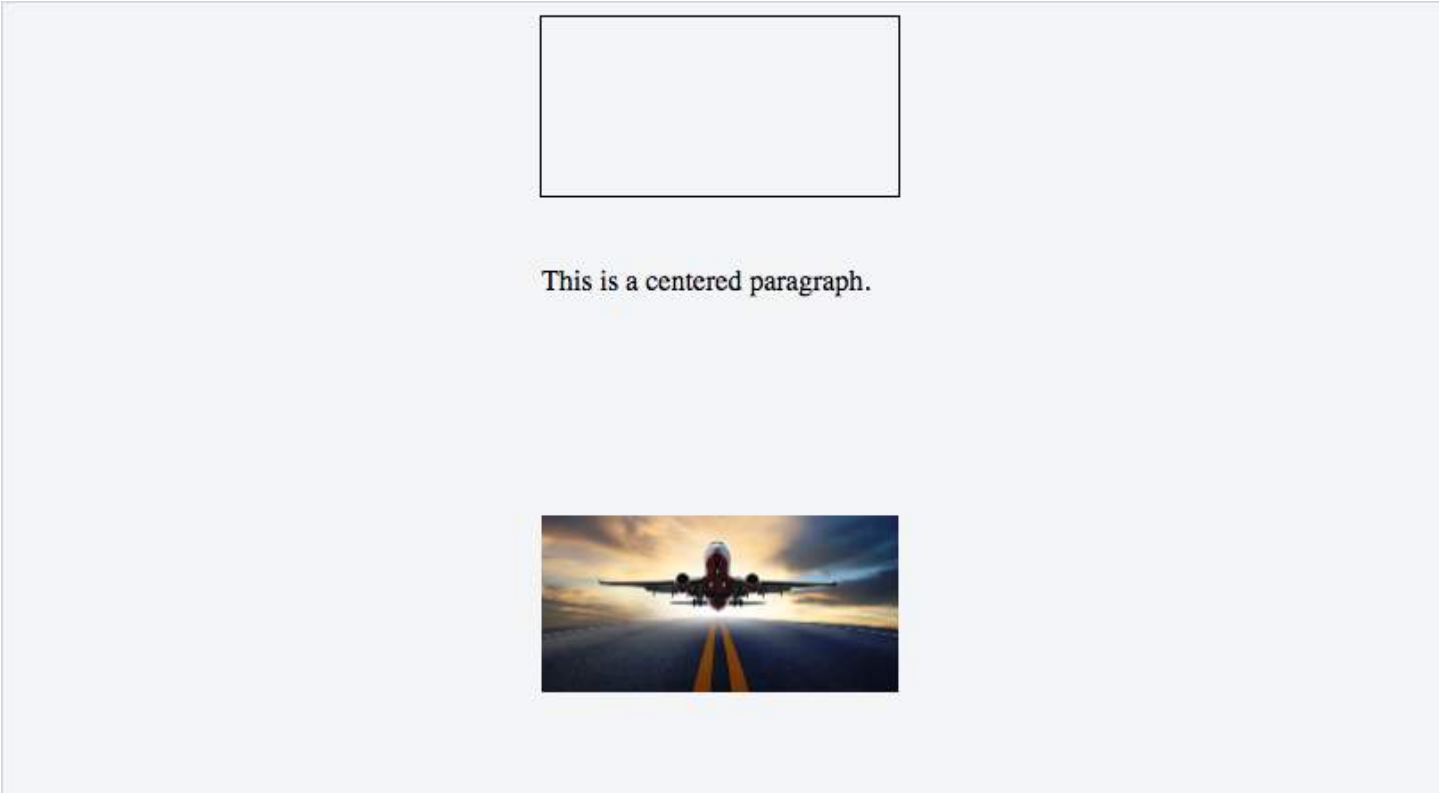
HTML

```
<div class="containerDiv">
    <div id="centeredDiv"></div>
</div>

<div class="containerDiv">
    <p id="centeredParagraph">This is a centered paragraph.</p>
</div>

<div class="containerDiv">
    <img id="centeredImage"
src="https://i.kinja-img.com/gawker-media/image/upload/s--c7Q9b4Eh--/c_scale,fl_progressive,q_80,w_
800/qqyvc3bkpyl3mfhr8all.jpg" />
</div>
```

CSS

```
.containerDiv {
    width: 100%;
    height: 100px;
    padding-bottom: 40px;
}

#centeredDiv {
    margin: 0 auto;
    width: 200px;
    height: 100px;
    border: 1px solid #000;
}

#centeredParagraph {
    width: 200px;
    margin: 0 auto;
}

#centeredImage {
    display: block;
    width: 200px;
    margin: 0 auto;
}
```

Result:



JSFiddle example: Centering objects with margin: 0 auto;

Section 6.4: Using text-align

The most common and easiest type of centering is that of lines of text in an element. CSS has the rule `text-align: center` for this purpose:

HTML

```
<p>Lorem ipsum</p>
```

CSS

```
p {
    text-align: center;
}
```

This does not work for centering entire block elements. `text-align` controls only alignment of inline content like text in its parent block element.

See more about `text-align` in Typography section.

Section 6.5: Using position: absolute

Working in old browsers (IE >= 8)

Automatic margins, paired with values of zero for the `left` and `right` or `top` and `bottom` offsets, will center an absolutely positioned elements within its parent.

View Result

HTML

```
<div class="parent">
  <img class="center" src="http://lorempixel.com/400/200/" />
</div>
```

CSS

```
.parent {
  position: relative;
  height: 500px;
}

.center {
  position: absolute;
  margin: auto;
  top: 0;
  right: 0;
  bottom: 0;
  left: 0;
}
```

Elements that don't have their own implicit width and height like images do, will need those values defined.

Other resources: Absolute Centering in CSS

Section 6.6: Using calc()

The calc() function is the part of a new syntax in CSS3 in which you can calculate (mathematically) what size/position your element occupies by using a variety of values like pixels, percentages, etc. Note: Whenever you use this function, always take care of the space between two values `calc(100% - 80px)`.

CSS

```
.center {
    position: absolute;
    height: 50px;
    width: 50px;
    background: red;
    top: calc(50% - 50px / 2); /* height divided by 2*/
    left: calc(50% - 50px / 2); /* width divided by 2*/
}
```

HTML

```
<div class="center"></div>
```

Section 6.7: Using line-height

You can also use `line-height` to center vertically a single line of text inside a container :

CSS

```
div {
    height: 200px;
    line-height: 200px;
}
```

That's quite ugly, but can be useful inside an `<input />` element. The `line-height` property works only when the text to be centered spans a single line. If the text wraps into multiple lines, the resulting output won't be centered.

Section 6.8: Vertical align anything with 3 lines of code

Supported by IE11+

View Result

Use these 3 lines to vertical align practically everything. Just make sure the div/image you apply the code to has a parent with a height.

CSS

```
div.vertical {
  position: relative;
  top: 50%;
  transform: translateY(-50%);
}
```

HTML

```
<div class="vertical">Vertical aligned text!</div>
```

Section 6.9: Centering in relation to another item

We will see how to center content based on the height of a near element.

Compatibility: IE8+, all other modern browsers.

HTML

```
<div class="content">
  <div class="position-container">
    <div class="thumb">
      <img src="http://lorempixel.com/400/200/">
    </div>
    <div class="details">
      <p class="banner-title">text 1</p>
      <p class="banner-text">content content content content content content content content
content content content content content content</p>
      <button class="btn">button</button>
    </div>
  </div>
</div>
```

CSS

```
.content * {
  box-sizing: border-box;
}
.content .position-container {
  display: table;
}
.content .details {
  display: table-cell;
  vertical-align: middle;
```

```
  width: 33.333333%;
  padding: 30px;
  font-size: 17px;
  text-align: center;
}
.content .thumb {
  width: 100%;
}
.content .thumb img {
  width: 100%;
}
```

Link to JSFiddle

The main points are the 3 `.thumb`, `.details` and `.position-container` containers:

- The `.position-container` must have `display: table`.
- The `.details` must have the real width set `width: ....` and `display: table-cell`, `vertical-align: middle`.
- The `.thumb` must have `width: 100%` if you want that it will take all the remaining space and it will be influenced by the `.details` width.
- The image (if you have an image) inside `.thumb` should have `width: 100%`, but it is not necessary if you have correct proportions.

Section 6.10: Ghost element technique (Michał Czernow's hack)

This technique works even when the container's dimensions are unknown.

Set up a "ghost" element inside the container to be centered that is 100% height, then use `vertical-align: middle` on both that and the element to be centered.

CSS

```
/* This parent can be any width and height */
.block {
  text-align: center;

  /* May want to do this if there is risk the container may be narrower than the element inside */
  white-space: nowrap;
}

/* The ghost element */
.block:before {
  content: '';
  display: inline-block;
  height: 100%;
  vertical-align: middle;

  /* There is a gap between ghost element and .centered,
  caused by space character rendered. Could be eliminated by
  nudging .centered (nudge distance depends on font family),
  or by zeroing font-size in .parent and resetting it back
  (probably to 1rem) in .centered. */
  margin-right: -0.25em;
}
```

```
/* The element to be centered, can also be of any width and height */
.centered {
  display: inline-block;
  vertical-align: middle;
  width: 300px;
  white-space: normal; /* Resetting inherited nowrap behavior */
}
```

HTML

```
<div class="block">
  <div class="centered"></div>
</div>
```

Section 6.11: Centering vertically and horizontally without worrying about height or width

The following technique allows you to add your content to an HTML element and center it both horizontally and vertically **without worrying about its height or width**.

The outer container

- should have `display: table;`

The inner container

- should have `display: table-cell;`
- should have `vertical-align: middle;`
- should have `text-align: center;`

The content box

- should have `display: inline-block;`
- should re-adjust the horizontal text-alignment to eg. `text-align: left;` or `text-align: right;`, unless you want text to be centered

Demo

HTML

```
<div class="outer-container">
   <div class="inner-container">
     <div class="centered-content">
        You can put anything here!
     </div>
   </div>
</div>
```

CSS

```
body {
    margin : 0;
}

.outer-container {
    position : absolute;
    display: table;
    width: 100%; /* This could be ANY width */
```

```
    height: 100%; /* This could be ANY height */
    background: #ccc;
}

.inner-container {
    display: table-cell;
    vertical-align: middle;
    text-align: center;
}

.centered-content {
    display: inline-block;
    text-align: left;
    background: #fff;
    padding: 20px;
    border: 1px solid #000;
}
```

See also **this Fiddle**!

Section 6.12: Vertically align an image inside div

HTML

```
<div class="wrap">
    <img src="http://lorempixel.com/400/200/" />
</div>
```

CSS

```
.wrap {
    height: 50px;/* max image height */
    width: 100px;
    border: 1px solid blue;
    text-align: center;
}
.wrap:before {
  content:"";
  display: inline-block;
  height: 100%;
  vertical-align: middle;
  width: 1px;
}

img {
    vertical-align: middle;
}
```

Section 6.13: Centering with fixed size

If the size of your content is fixed, you can use absolute positioning to 50% with `margin` that reduces half of your content's width and height:

HTML

```
<div class="center">
    Center vertically and horizontally
</div>
```

CSS

```
.center {
    position: absolute;
    background: #ccc;

    left: 50%;
    width: 150px;
    margin-left: -75px;  /* width * -0.5 */

    top: 50%;
    height: 200px;
    margin-top: -100px;  /* height * -0.5 */
}
```

Horizontal centering with only fixed width

You can center the element horizontally even if you don't know the height of the content:

HTML

```
<div class="center">
    Center only horizontally
</div>
```

CSS

```
.center {
    position: absolute;
    background: #ccc;

    left: 50%;
    width: 150px;
    margin-left: -75px;  /* width * -0.5 */
}
```

Vertical centering with fixed height

You can center the element vertically if you know the element's height:

HTML

```
<div class="center">
    Center only vertically
</div>
```

CSS

```
.center {
    position: absolute;
    background: #ccc;

    top: 50%;
    height: 200px;
    margin-top: -100px;  /* width * -0.5 */
}
```

Section 6.14: Vertically align dynamic height elements

Applying css intuitively doesn't produce the desired results because

- `vertical-align:middle` ***isn't*** applicable to block-level elements
- `margin-top:auto` and `margin-bottom:auto` used values would compute as ***zero***
- `margin-top:-50%` percentage-based margin values are calculated relative to the ***width*** of containing block

For widest browser support, a workaround with helper elements:

HTML

```
<div class="vcenter--container">
  <div class="vcenter--helper">
    <div class="vcenter--content">
      <!--stuff-->
    </div>
  </div>
</div>
```

CSS

```
.vcenter--container {
  display: table;
  height: 100%;
  position: absolute;
  overflow: hidden;
  width: 100%;
}
.vcenter--helper {
  display: table-cell;
  vertical-align: middle;
}
.vcenter--content {
  margin: 0 auto;
  width: 200px;
}
```

jsfiddle from original question. This approach

- works with dynamic height elements
- respects content flow
- is supported by legacy browsers

Section 6.15: Horizontal and Vertical centering using table layout

One could easily center a child element using `table` display property.

HTML

```
<div class="wrapper">
    <div class="parent">
        <div class="child"></div>
    </div>
</div>
```

CSS

```
.wrapper {
    display: table;
    vertical-align: center;
    width: 200px;
    height: 200px;
    background-color: #9e9e9e;
}
.parent {
    display: table-cell;
    vertical-align: middle;
    text-align: center;
}
.child {
    display: inline-block;
    vertical-align: middle;
    text-align: center;
    width: 100px;
    height: 100px;
    background-color: teal;
}
```

Chapter 7: The Box Model

Parameter	Detail
`content-box`	Width and height of the element only includes content area.
`padding-box`	Width and height of the element includes content and padding.
`border-box`	Width and height of the element includes content, padding and border.
`initial`	Sets the box model to its default state.
`inherit`	Inherits the box model of the parent element.

Section 7.1: What is the Box Model?

The Edges

The browser creates a rectangle for each element in the HTML document. The Box Model describes how the padding, border, and margin are added to the content to create this rectangle.

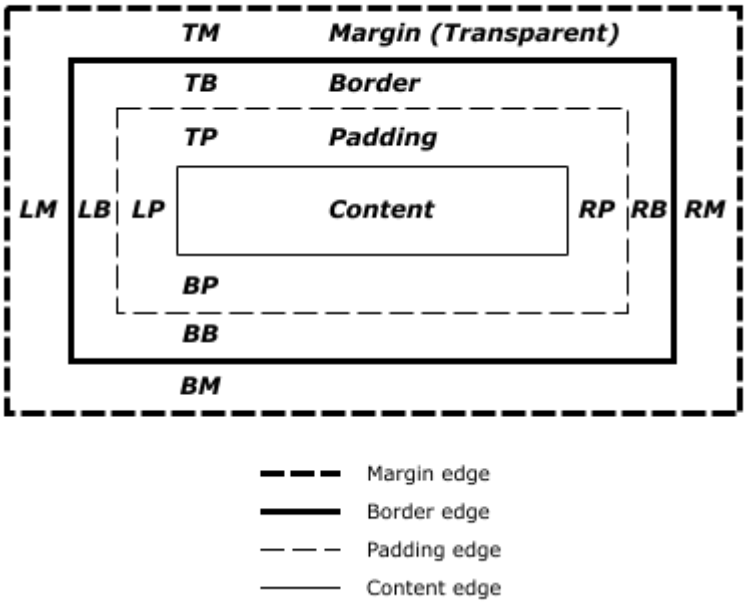


Diagram from [CSS2.2 Working Draft]

The perimeter of each of the four areas is called an *edge*. Each edge defines a *box.*

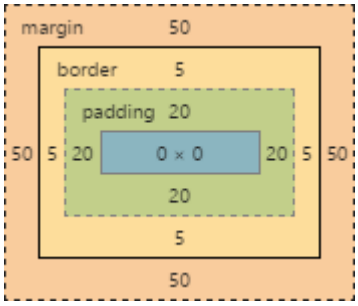
- The innermost rectangle is the **content box**. The width and height of this depends on the element's rendered content (text, images and any child elements it may have).
- Next is the **padding box**, as defined by the `padding` property. If there is no `padding` width defined, the padding edge is equal to the content edge.
- Then we have the **border box**, as defined by the `border` property. If there is no `border` width defined, the border edge is equal to the padding edge.
- The outermost rectangle is the **margin box**, as defined by the `margin` property. If there is no `margin` width defined, the margin edge is equal to the border edge.

Example

```
div {
    border: 5px solid red;
    margin: 50px;
    padding: 20px;
```

```
}
```

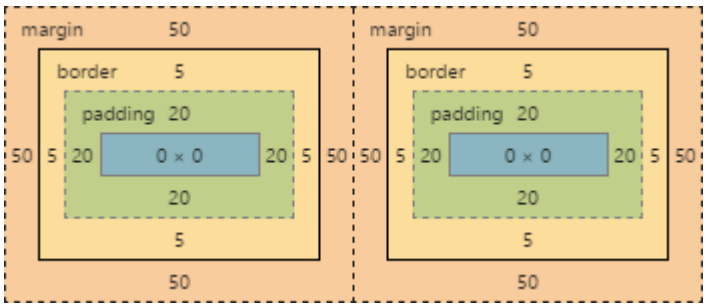
This CSS styles all `div` elements to have a top, right, bottom and left border of `5px` in width; a top, right, bottom and left margin of `50px`; and a top, right, bottom, and left padding of `20px`. Ignoring content, our generated box will look like this:



Screenshot of Google Chrome's Element Styles panel

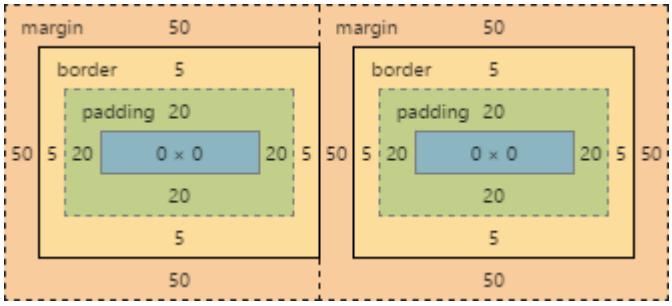
- As there is no content, the content region (the blue box in the middle) has no height or width (0px by 0px).
- The padding box by default is the same size as the content box, plus the 20px width on all four edges we're defining above with the `padding` property (40px by 40px).
- The border box is the same size as the padding box, plus the 5px width we're defining above with the `border` property (50px by 50px).
- Finally the margin box is the same size as the border box, plus the 50px width we're defining above with the `margin` property (giving our element a total size of 150px by 150px).

Now lets give our element a sibling with the same style. The browser looks at the Box Model of both elements to work out where in relation to the previous element's content the new element should be positioned:



The content of each of element is separated by a 150px gap, but the two elements' boxes touch each other.

If we then modify our first element to have no right margin, the right margin edge would be in the same position as the right border edge, and our two elements would now look like this:



Section 7.2: box-sizing

The default box model (`content-box`) can be counter-intuitive, since the `width` / `height` for an element will not represent its actual width or height on screen as soon as you start adding `padding` and `border` styles to the

element.

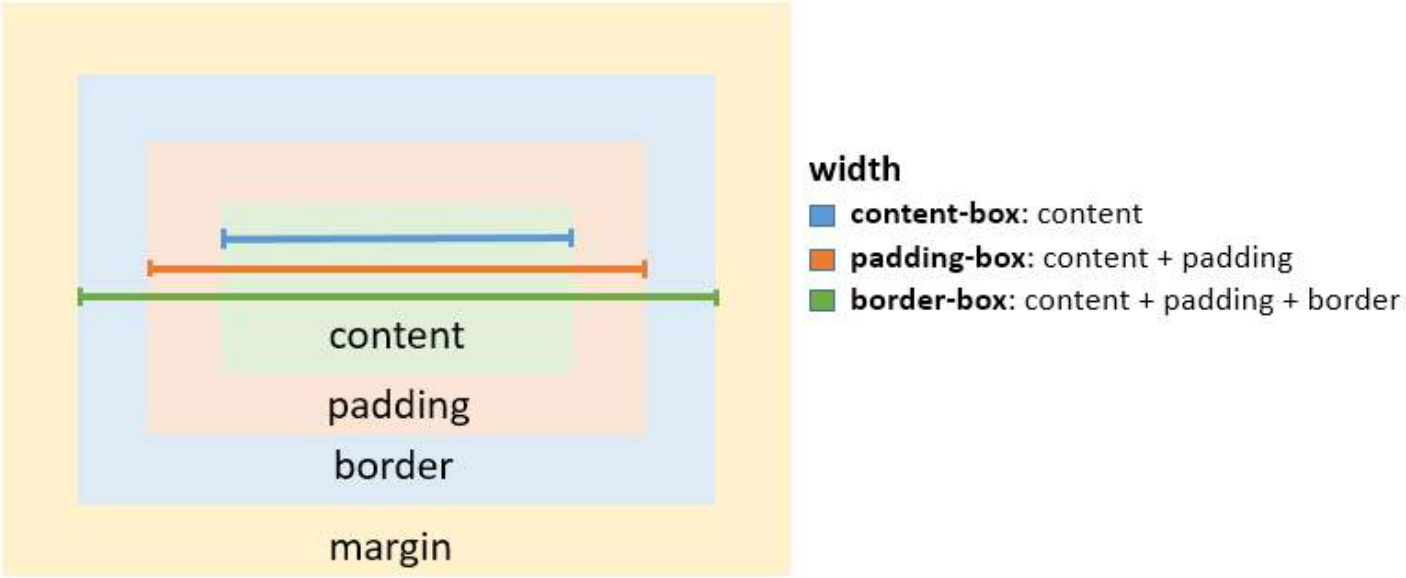
The following example demonstrates this potential issue with `content-box`:

```
textarea {
    width: 100%;
    padding: 3px;
    box-sizing: content-box; /* default value */
}
```

Since the padding will be added to the width of the textarea, the resulting element is a textarea that is wider than 100%.

Fortunately, CSS allows us to change the box model with the `box-sizing` property for an element. There are three different values for the property available:

- `content-box`: The common box model - width and height only includes the content, not the padding or border
- `padding-box`: Width and height includes the content and the padding, but not the border
- `border-box`: Width and height includes the content, the padding as well as the border



To solve the `textarea` problem above, you could just change the `box-sizing` property to `padding-box` or `border-box`. `border-box` is most commonly used.

```
textarea {
    width: 100%;
    padding: 3px;
    box-sizing: border-box;
}
```

To apply a specific box model to every element on the page, use the following snippet:

```
html {
    box-sizing: border-box;
}

*, *:before, *:after {
```

```
    box-sizing: inherit;
}
```

In this coding `box-sizing:border-box;` is not directly applied to `*`, so you can easily overwrite this property on individual elements.

Chapter 8: Margins

Parameter	Details
0	set margin to none
auto	used for centering, by evenly setting values on each side
units (e.g. px)	see parameter section in Units for a list of valid units
inherit	inherit margin value from parent element
initial	restore to initial value

Section 8.1: Margin Collapsing

When two margins are touching each other vertically, they are collapsed. When two margins touch horizontally, they do not collapse.

Example of adjacent vertical margins:

Consider the following styles and markup:

```
div{
    margin: 10px;
}
```

```
<div>
    some content
</div>
<div>
    some more content
</div>
```

They will be 10px apart since vertical margins collapse over one and other. (The spacing will not be the sum of two margins.)

Example of adjacent horizontal margins:

Consider the following styles and markup:

```
span{
    margin: 10px;
}
```

```
<span>some</span><span>content</span>
```

They will be 20px apart since horizontal margins don't collapse over one and other. (The spacing will be the sum of two margins.)

Overlapping with different sizes

```
.top{
    margin: 10px;
}
.bottom{
    margin: 15px;
}
```

```
<div class="top">
    some content
```

```
</div>
<div class="bottom">
    some more content
</div>
```

These elements will be spaced 15px apart vertically. The margins overlap as much as they can, but the larger margin will determine the spacing between the elements.

Overlapping margin gotcha

```
.outer-top{
    margin: 10px;
}
.inner-top{
    margin: 15px;
}
.outer-bottom{
    margin: 20px;
}
.inner-bottom{
    margin: 25px;
}
```

```
<div class="outer-top">
    <div class="inner-top">
        some content
    </div>
</div>
<div class="outer-bottom">
    <div class="inner-bottom">
        some more content
    </div>
</div>
```

What will be the spacing between the two texts? (hover to see answer)

> The spacing will be 25px. Since all four margins are touching each other, they will collapse, thus using the largest margin of the four.

Now, what about if we add some borders to the markup above.

```
div{
    border: 1px solid red;
}
```

What will be the spacing between the two texts? (hover to see answer)

> The spacing will be 59px! Now only the margins of .outer-top and .outer-bottom touch each other, and are the only collapsed margins. The remaining margins are separated by the borders. So we have 1px + 10px + 1px + 15px + 20px + 1px + 25px + 1px. (The 1px's are the borders...)

Collapsing Margins Between Parent and Child Elements:

HTML:

```
<h1>Title</h1>
<div>
  <p>Paragraph</p>
</div>
```

CSS

```
h1 {
  margin: 0;
  background: #cff;
}
div {
  margin: 50px 0 0 0;
  background: #cfc;
}
p {
  margin: 25px 0 0 0;
  background: #cf9;
}
```

In the example above, only the largest margin applies. You may have expected that the paragraph would be located 60px from the h1 (since the div element has a margin-top of 40px and the p has a 20px margin-top). This does not happen because the margins collapse together to form one margin.

Section 8.2: Apply Margin on a Given Side

Direction-Specific Properties

CSS allows you to specify a given side to apply margin to. The four properties provided for this purpose are:

- `margin-left`
- `margin-right`
- `margin-top`
- `margin-bottom`

The following code would apply a margin of 30 pixels to the left side of the selected div. *View Result*

HTML

```
<div id="myDiv"></div>
```

CSS

```
#myDiv {
    margin-left: 30px;
    height: 40px;
    width: 40px;
    background-color: red;
}
```

Parameter	Details
margin-left	The direction in which the margin should be applied.
30px	The width of the margin.

Specifying Direction Using Shorthand Property

The standard `margin` property can be expanded to specify differing widths to each side of the selected elements. The syntax for doing this is as follows:

```
margin: <top> <right> <bottom> <left>;
```

The following example applies a zero-width margin to the top of the div, a 10px margin to the right side, a 50px margin to the left side, and a 100px margin to the left side. *View Result*

HTML

```
<div id="myDiv"></div>
```

CSS

```
#myDiv {
    margin: 0 10px 50px 100px;
    height: 40px;
    width: 40px;
    background-color: red;
}
```

Section 8.3: Margin property simplification

```
p {
    margin:1px;                /* 1px margin in all directions */

    /*equals to:*/

    margin:1px 1px;

    /*equals to:*/

    margin:1px 1px 1px;

    /*equals to:*/

    margin:1px 1px 1px 1px;
}
```

Another exapmle:

```
p{
    margin:10px 15px;        /* 10px margin-top & bottom And 15px margin-right & left*/

    /*equals to:*/

    margin:10px 15px 10px 15px;

    /*equals to:*/

    margin:10px 15px 10px;
    /* margin left will be calculated from the margin right value (=15px) */
}
```

Section 8.4: Horizontally center elements on a page using margin

As long as the element is a **block**, and it has an **explicitly set width value**, margins can be used to center block elements on a page horizontally.

We add a width value that is lower than the width of the window and the auto property of margin then distributes the remaining space to the left and the right:

```
#myDiv {
 width:80%;
 margin:0 auto;
}
```

In the example above we use the shorthand `margin` declaration to first set `0` to the top and bottom margin values (although this could be any value) and then we use `auto` to let the browser allocate the space automatically to the left and right margin values.

In the example above, the #myDiv element is set to 80% width which leaves use 20% leftover. The browser distributes this value to the remaining sides so:

(100% - 80%) / 2 = 10%

Section 8.5: Example 1:

It is obvious to assume that the percentage value of margin to `margin-left` and `margin-right` would be relative to its parent element.

```
.parent {
    width : 500px;
    height: 300px;
}

.child {
    width : 100px;
    height: 100px;
    margin-left: 10%;  /* (parentWidth * 10/100) => 50px */
}
```

But that is not the case, when comes to `margin-top` and `margin-bottom`. Both these properties, in percentages, aren't relative to the height of the parent container but to the **width** of the parent container.

So,

```
.parent {
    width : 500px;
    height: 300px;
}

.child {
    width : 100px;
    height: 100px;
    margin-left: 10%;  /* (parentWidth * 10/100) => 50px  */
    margin-top: 20%;   /* (parentWidth * 20/100) => 100px */
}
```

Section 8.6: Negative margins

Margin is one of a few CSS properties that can be set to negative values. This property can be used to **overlap elements without absolute positioning**.

```
div{
```

```
  display: inline;
}

#over{
  margin-left: -20px;
}

<div>Base div</div>
<div id="over">Overlapping div</div>
```

Chapter 9: Padding

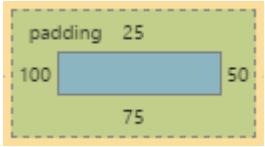
Section 9.1: Padding Shorthand

The padding property sets the padding space on all sides of an element. The padding area is the space between the content of the element and its border. Negative values are not allowed.

To save adding padding to each side individually (using `padding-top`, `padding-left` etc) can you write it as a shorthand, as below:

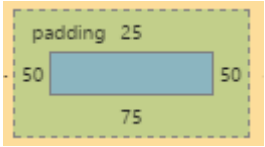
Four values:

```
<style>
    .myDiv {
        padding: 25px 50px 75px 100px; /* top right bottom left; */
    }
</style>
<div class="myDiv"></div>
```



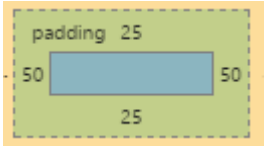
Three values:

```
<style>
    .myDiv {
        padding: 25px 50px 75px; /* top left/right bottom */
    }
</style>
<div class="myDiv"></div>
```



Two values:

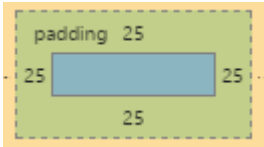
```
<style>
    .myDiv {
        padding: 25px 50px; /* top/bottom left/right */
    }
</style>
<div class="myDiv"></div>
```



One value:

```
<style>
    .myDiv {
```

```
        padding: 25px; /* top/right/bottom/left */
    }
</style>
<div class="myDiv"></div>
```



Section 9.2: Padding on a given side

The padding property sets the padding space on all sides of an element. The padding area is the space between the content of the element and its border. Negative values are not allowed.

You can specify a side individually:

- `padding-top`
- `padding-right`
- `padding-bottom`
- `padding-left`

The following code would add a padding of `5px` to the top of the div:

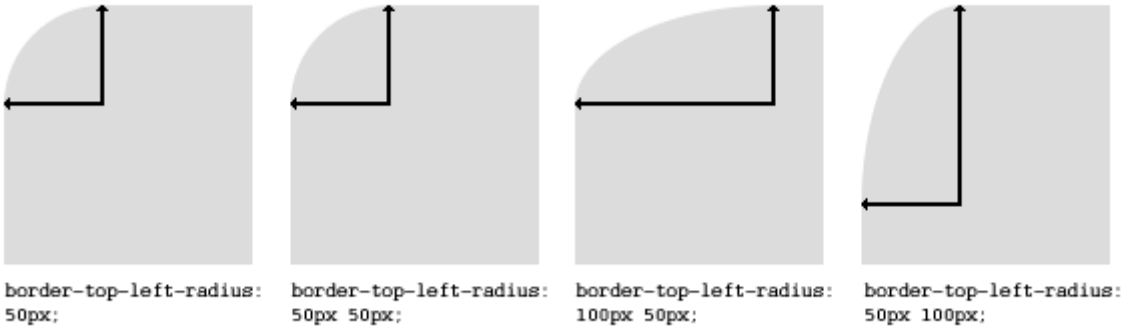
```
<style>
.myClass {
    padding-top: 5px;
}
</style>

<div class="myClass"></div>
```

Chapter 10: Border

Section 10.1: border-radius

The border-radius property allows you to change the shape of the basic box model.

Every corner of an element can have up to two values, for the vertical and horizontal radius of that corner (for a maximum of 8 values).



The first set of values defines the horizontal radius. The optional second set of values, preceded by a '/' , defines the vertical radius. If only one set of values is supplied, it is used for both the vertical and horizontal radius.

```
border-radius: 10px 5% / 20px 25em 30px 35em;
```

The `10px` is the horizontal radius of the top-left-and-bottom-right. And the `5%` is the horizontal radius of the top-right-and-bottom-left. The other four values after '/' are the vertical radii for top-left, top-right, bottom-right and bottom-left.

As with many CSS properties, shorthands can be used for any or all possible values. You can therefore specify anything from one to eight values. The following shorthand allows you to set the horizontal and vertical radius of every corner to the same value:

HTML:

```
<div class='box'></div>
```

CSS:

```
.box {
    width: 250px;
    height: 250px;
    background-color: black;
    border-radius: 10px;
}
```

Border-radius is most commonly used to convert box elements into circles. By setting the border-radius to half of the length of a square element, a circular element is created:

```
.circle {
    width: 200px;
    height: 200px;
    border-radius: 100px;
}
```

Because border-radius accepts percentages, it is common to use 50% to avoid manually calculating the border-radius value:

```
.circle {
    width: 150px;
    height: 150px;
    border-radius: 50%;
}
```

If the width and height properties are not equal, the resulting shape will be an oval rather than a circle.

Browser specific border-radius example:

```
  -webkit-border-top-right-radius: 4px;
  -webkit-border-bottom-right-radius: 4px;
  -webkit-border-bottom-left-radius: 0;
  -webkit-border-top-left-radius: 0;
  -moz-border-radius-topright: 4px;
  -moz-border-radius-bottomright: 4px;
  -moz-border-radius-bottomleft: 0;
  -moz-border-radius-topleft: 0;
  border-top-right-radius: 4px;
  border-bottom-right-radius: 4px;
  border-bottom-left-radius: 0;
  border-top-left-radius: 0;
```

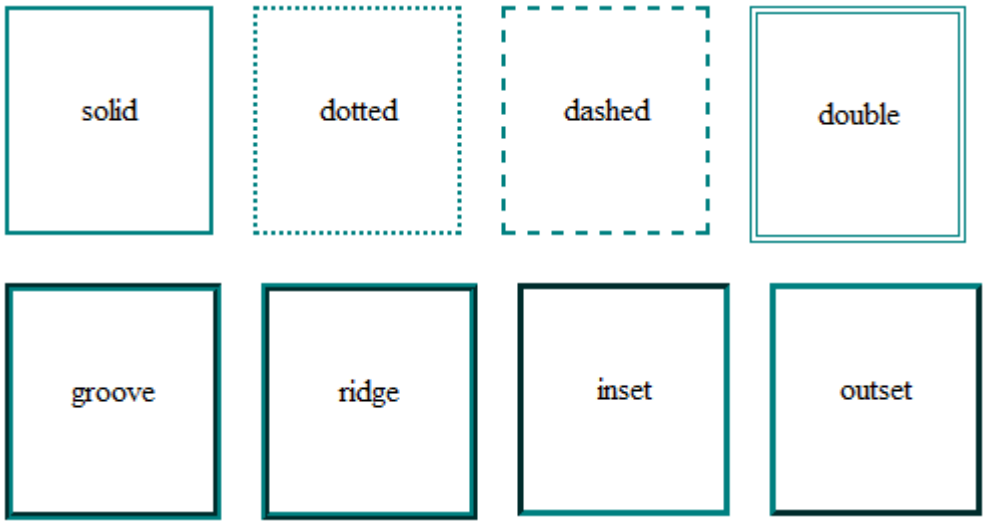
Section 10.2: border-style

The `border-style` property sets the style of an element's border. This property can have from one to four values (for every side of the element one value.)

Examples:

```
border-style: dotted;
```

```
border-style: dotted solid double dashed;
```



`border-style` can also have the values `none` and `hidden`. They have the same effect, except `hidden` works for border conflict resolution for **`<table>`** elements. In a **`<table>`** with multiple borders, `none` has the lowest priority (meaning in a conflict, the border would show), and `hidden` has the highest priority (meaning in a conflict, the border would not show).

Section 10.3: Multiple Borders

Using outline:

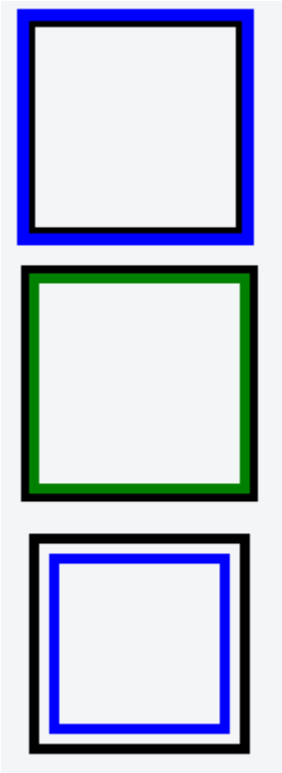
```
.div1{
  border: 3px solid black;
  outline: 6px solid blue;
  width: 100px;
  height: 100px;
  margin: 20px;
}
```

Using box-shadow:

```
.div2{
  border: 5px solid green;
  box-shadow: 0px 0px 0px 4px #000;
  width: 100px;
  height: 100px;
  margin: 20px;
}
```

Using a pseudo element:

```
.div3 {
  position: relative;
  border: 5px solid #000;
  width: 100px;
  height: 100px;
  margin: 20px;
}
.div3:before {
  content: " ";
  position: absolute;
  border: 5px solid blue;
  z-index: -1;
  top: 5px;
  left: 5px;
  right: 5px;
  bottom: 5px;
}
```



http://jsfiddle.net/MadalinaTn/bvqpcohm/2/

Section 10.4: border (shorthands)

In most cases you want to define several border properties (`border-width`, `border-style` and `border-color`) for all sides of an element.

Instead of writing:

```
border-width: 1px;
border-style: solid;
border-color: #000;
```

You can simply write:

```
border: 1px solid #000;
```

These shorthands are also available for every side of an element: `border-top`, `border-left`, `border-right` and `border-bottom`. So you can do:

```
border-top: 2px double #aaaaaa;
```

Section 10.5: border-collapse

The `border-collapse` property applies only to `table`s (and elements displayed as **`display`**: `table` or `inline-table`) and sets whether the table borders are collapsed into a single border or detached as in standard HTML.

```
table {
  border-collapse: separate; /* default */
  border-spacing: 2px; /* Only works if border-collapse is separate */
}
```

Also see Tables - border-collapse documentation entry

Section 10.6: border-image

With the `border-image` property you have the possibility to set an image to be used instead of normal border styles.

A `border-image` essentially consist of a

- `border-image-source`: The path to the image to be used
- `border-image-slice`: Specifies the offset that is used to divide the image into **nine regions** (four **corners**, four **edges** and a **middle**)
- `border-image-repeat`: Specifies how the images for the sides and the middle of the border image are scaled

Consider the following example wheras border.png is a image of 90x90 pixels:

```
border-image: url("border.png") 30 stretch;
```

The image will be split into nine regions with 30x30 pixels. The edges will be used as the corners of the border while the side will be used in between. If the element is higher / wider than 30px this part of the image will be **stretched**. The middle part of the image defaults to be transparent.

Section 10.7: Creating a multi-colored border using border-image

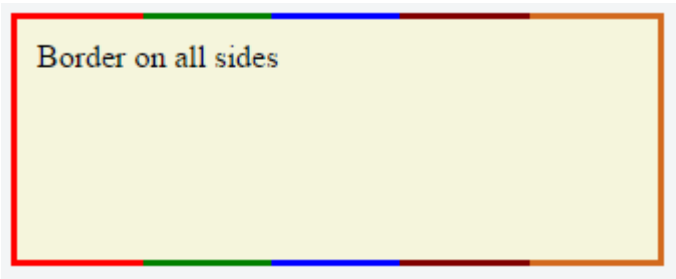
CSS

```
.bordered {
  border-image: linear-gradient(to right, red 20%, green 20%, green 40%, blue 40%, blue 60%, maroon 60%, maroon 80%, chocolate 80%); /* gradient with required colors */
  border-image-slice: 1;
}
```

HTML

```
<div class='bordered'>Border on all sides</div>
```

The above example would produce a border that comprises of 5 different colors. The colors are defined through a `linear-gradient` (you can find more information about gradients in the docs). You can find more information about `border-image-slice` property in the `border-image` example in same page.

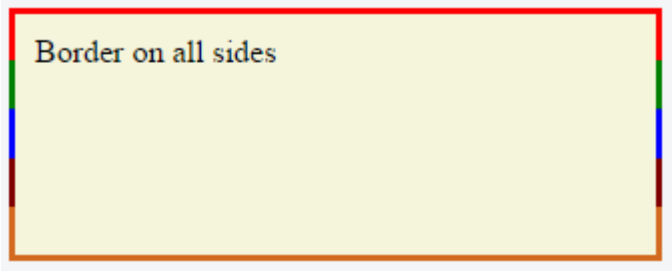


(*Note: Additional properties were added to the element for presentational purpose.*)

You'd have noticed that the left border has only a single color (the start color of the gradient) while the right border also has only a single color (the gradient's end color). This is because of the way that border image property works. It is as though the gradient is applied to the entire box and then the colors are masked from the padding and content areas, thus making it look as though only the border has the gradient.

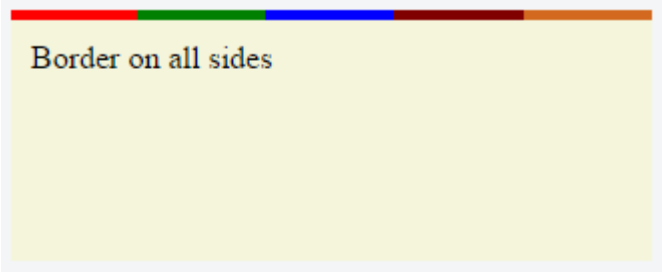
Which border(s) have a single color is dependant on the gradient definition. If the gradient is a `to right` gradient, the left border would be the start color of the gradient and right border would be the end color. If it was a `to bottom` gradient the top border would be the gradient's start color and bottom border would be end color. Below is

the output of a `to bottom` 5 colored gradient.



If the border is required only on specific sides of the element then the `border-width` property can be used just like with any other normal border. For example, adding the below code would produce a border only on the top of the element.

```
border-width: 5px 0px 0px 0px;
```



Note that, any element that has `border-image` property **won't respect the** `border-radius` (that is the border won't curve). This is based on the below statement in the spec:

> A box's backgrounds, but not its border-image, are clipped to the appropriate curve (as determined by 'background-clip').

Section 10.8: border-[left|right|top|bottom]

The `border-[left|right|top|bottom]` property is used to add a border to a specific side of an element.

For example if you wanted to add a border to the left side of an element, you could do:

```
#element {
    border-left: 1px solid black;
}
```

Chapter 11: Outlines

Parameter	Details
dotted	dotted outline
dashed	dashed outline
solid	solid outline
double	double outline
groove	3D grooved outline, depends on the outline-color value
ridge	3D ridged outline, depends on the outline-color value
inset	3D inset outline, depends on the outline-color value
outset	3D outset outline, depends on the outline-color value
none	no outline
hidden	hidden outline

Section 11.1: Overview

Outline is a line that goes around the element, outside of the border. In contrast to `border`, outlines do not take any space in the box model. So adding an outline to an element does not affect the position of the element or other elements.

In addition, outlines can be non-rectangular in some browsers. This can happen if `outline` is applied on a `span` element that has text with different `font-size` properties inside it. Unlike borders, outlines *cannot* have rounded corners.

The essential parts of `outline` are `outline-color`, `outline-style` and `outline-width`.

The definition of an outline is equivalent to the definition of a border:

> An outline is a line around an element. It is displayed around the margin of the element. However, it is different from the border property.

```
outline: 1px solid black;
```

Section 11.2: outline-style

The `outline-style` property is used to set the style of the outline of an element.

```
p {
  border: 1px solid black;
  outline-color:blue;
  line-height:30px;
}
.p1{
  outline-style: dotted;
}
.p2{
  outline-style: dashed;
}
.p3{
  outline-style: solid;
}
```

```
.p4{
  outline-style: double;
}
.p5{
  outline-style: groove;
}
.p6{
  outline-style: ridge;
}
.p7{
  outline-style: inset;
}
.p8{
  outline-style: outset;
}
```

HTML

```
<p class="p1">A dotted outline</p>
<p class="p2">A dashed outline</p>
<p class="p3">A solid outline</p>
<p class="p4">A double outline</p>
<p class="p5">A groove outline</p>
<p class="p6">A ridge outline</p>
<p class="p7">An inset outline</p>
<p class="p8">An outset outline</p>
```

A dotted outline

A dashed outline

A solid outline

A double outline

A groove outline

A ridge outline

An inset outline

An outset outline

Chapter 12: Overflow

`Overflow` Value	Details
`visible`	Shows all overflowing content outside the element
`scroll`	Hides the overflowing content and adds a scroll bar
`hidden`	Hides the overflowing content, both scroll bars disappear and the page becomes fixed
`auto`	Same as `scroll` if content overflows, but doesn't add scroll bar if content fits
`inherit`	Inherit's the parent element's value for this property

Section 12.1: overflow-wrap

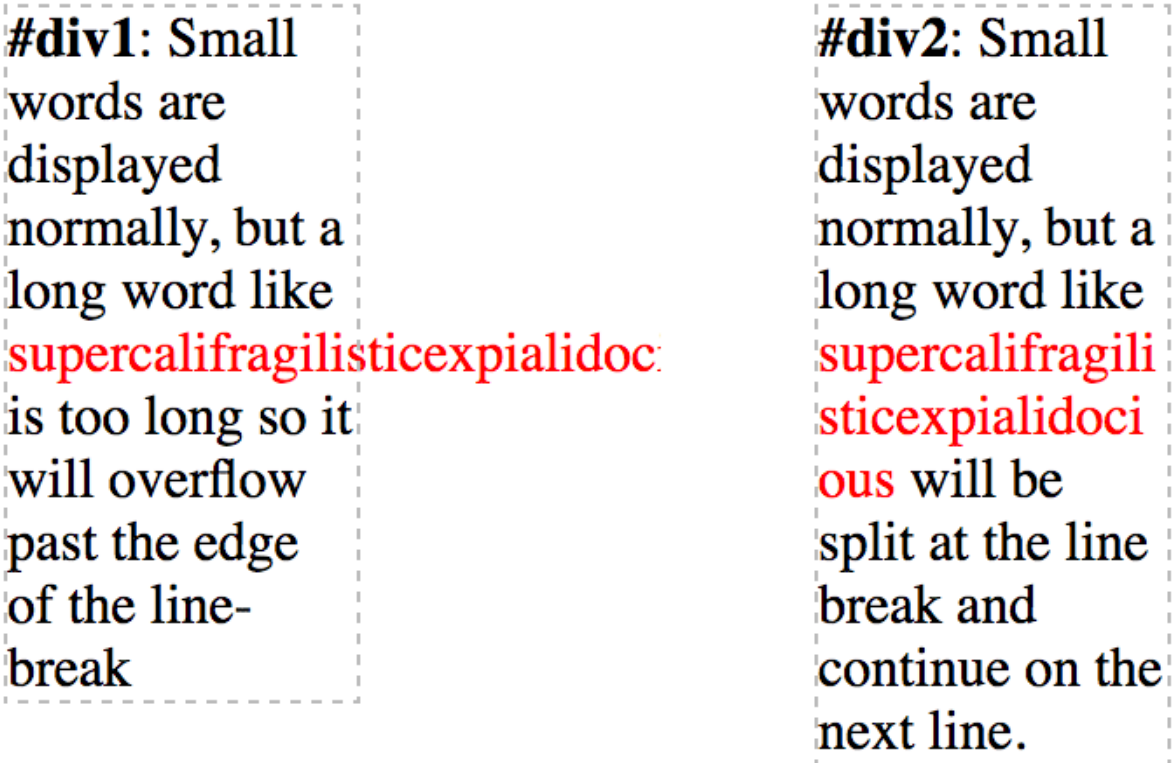
`overflow-wrap` tells a browser that it can break a line of text inside a targeted element onto multiple lines in an otherwise unbreakable place. Helpful in preventing an long string of text causing layout problems due to overflowing it's container.

CSS

```
div {
    width:100px;
    outline: 1px dashed #bbb;
}

#div1 {
    overflow-wrap:normal;
}

#div2 {
    overflow-wrap:break-word;
}
```

HTML

```
<div id="div1">
      <strong>#div1</strong>: Small words are displayed normally, but a long word like <span
style="red;">supercalifragilisticexpialidocious</span> is too long so it will overflow past the
edge of the line-break
</div>

<div id="div2">
      <strong>#div2</strong>: Small words are displayed normally, but a long word like <span
style="red;">supercalifragilisticexpialidocious</span> will be split at the line break and continue
on the next line.
</div>
```



`overflow-wrap` – Value	Details
`normal`	Lets a word overflow if it is longer than the line
`break-word`	Will split a word into multiple lines, if necessary
`inherit`	Inherits the parent element's value for this property

Section 12.2: overflow-x and overflow-y

These two properties work in a similar fashion as the `overflow` property and accept the same values. The `overflow-x` parameter works only on the x or left-to-right axis. The `overflow-y` works on the y or top-to-bottom axis.

HTML

```
<div id="div-x">
    If this div is too small to display its contents,
    the content to the left and right will be clipped.
</div>

<div id="div-y">
    If this div is too small to display its contents,
    the content to the top and bottom will be clipped.
</div>
```

CSS

```
div {
    width: 200px;
    height: 200px;
}

#div-x {
    overflow-x: hidden;
}
```

```
#div-y {
    overflow-y: hidden;
}
```

Section 12.3: overflow: scroll

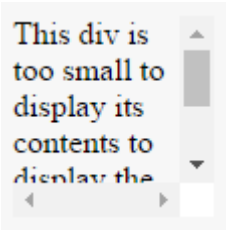
HTML

```
<div>
    This div is too small to display its contents to display the effects of the overflow property.
</div>
```

CSS

```
div {
    width:100px;
    height:100px;
    overflow:scroll;
}
```

Result



The content above is clipped in a 100px by 100px box, with scrolling available to view overflowing content.

Most desktop browsers will display both horizontal and vertical scrollbars, whether or not any content is clipped. This can avoid problems with scrollbars appearing and disappearing in a dynamic environment. Printers may print overflowing content.

Section 12.4: overflow: visible

HTML

```
<div>
     Even if this div is too small to display its contents, the content is not clipped.
</div>
```

CSS

```
div {
     width:50px;
     height:50px;
     overflow:visible;
}
```

Result

Content is not clipped and will be rendered outside the content box if it exceeds its container size.

Section 12.5: Block Formatting Context Created with Overflow

Using the `overflow` property with a value different to `visible` will create a new **block formatting context**. This is useful for aligning a block element next to a floated element.

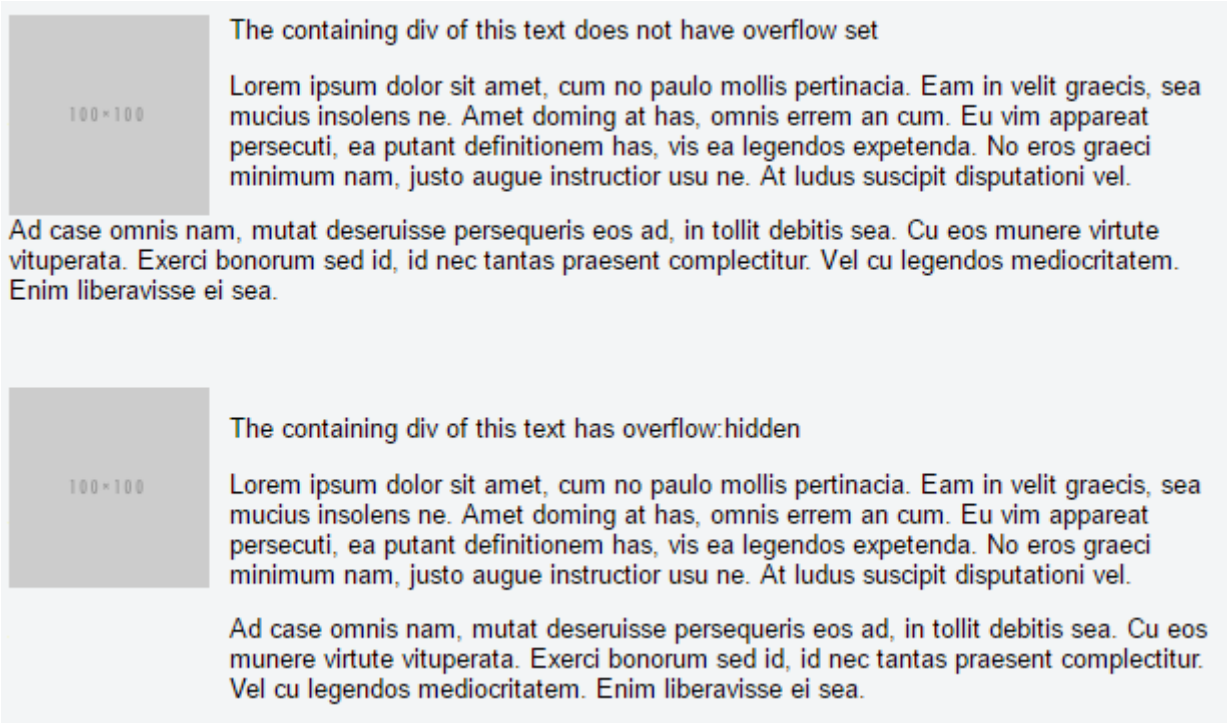
CSS

```
img {
    float:left;
    margin-right: 10px;
}
 div {
     overflow:hidden; /* creates block formatting context */
}
```

HTML

```
<img src="http://placehold.it/100x100">
<div>
  <p>Lorem ipsum dolor sit amet, cum no paulo mollis pertinacia.</p>
  <p>Ad case omnis nam, mutat deseruisse persequeris eos ad, in tollit debitis sea.</p>
</div>
```

Result



This example shows how paragraphs within a div with the `overflow` property set will interact with a floated image.

Chapter 13: Media Queries

Parameter	Details
`mediatype`	(Optional) This is the type of media. Could be anything in the range of `all` to `screen`.
`not`	(Optional) Doesn't apply the CSS for this particular media type and applies for everything else.
`media feature`	Logic to identify use case for CSS. Options outlined below.
Media Feature	**Details**
`aspect-ratio`	Describes the aspect ratio of the targeted display area of the output device.
`color`	Indicates the number of bits per color component of the output device. If the device is not a color device, this value is zero.
`color-index`	Indicates the number of entries in the color look-up table for the output device.
`grid`	Determines whether the output device is a grid device or a bitmap device.
`height`	The height media feature describes the height of the output device's rendering surface.
`max-width`	CSS will not apply on a screen width wider than specified.
`min-width`	CSS will not apply on a screen width narrower than specified.
`max-height`	CSS will not apply on a screen height taller than specified.
`min-height`	CSS will not apply on a screen height shorter than specified.
`monochrome`	Indicates the number of bits per pixel on a monochrome (greyscale) device.
`orientation`	CSS will only display if device is using specified orientation. See remarks for more details.
`resolution`	Indicates the resolution (pixel density) of the output device.
`scan`	Describes the scanning process of television output devices.
`width`	The width media feature describes the width of the rendering surface of the output device (such as the width of the document window, or the width of the page box on a printer).
Deprecated Features	**Details**
`device-aspect-ratio`	**`Deprecated`** CSS will only display on devices whose height/width ratio matches the specified ratio. This is a`deprecated`feature and is not guaranteed to work.
`max-device-width`	**`Deprecated`** Same as `max-width` but measures the physical screen width, rather than the display width of the browser.
`min-device-width`	**`Deprecated`** Same as `min-width` but measures the physical screen width, rather than the display width of the browser.
`max-device-height`	**`Deprecated`** Same as `max-height` but measures the physical screen width, rather than the display width of the browser.
`min-device-height`	**`Deprecated`** Same as `min-height` but measures the physical screen width, rather than the display width of the browser.

Section 13.1: Terminology and Structure

Media queries allow one to apply CSS rules based on the type of device / media (e.g. screen, print or handheld) called **media type**, additional aspects of the device are described with **media features** such as the availability of color or viewport dimensions.

General Structure of a Media Query

```
@media [...] {
    /* One or more CSS rules to apply when the query is satisfied */
}
```

A Media Query containing a Media Type

```
@media print {
    /* One or more CSS rules to apply when the query is satisfied */
```

```
}
```

A Media Query containing a Media Type and a Media Feature

```
@media screen and (max-width: 600px) {
    /* One or more CSS rules to apply when the query is satisfied */
}
```

A Media Query containing a Media Feature (and an implicit Media Type of "all")

```
@media (orientation: portrait) {
    /* One or more CSS rules to apply when the query is satisfied */
}
```

Section 13.2: Basic Example

```
@media screen and (min-width: 720px) {
    body {
        background-color: skyblue;
    }
}
```

The above media query specifies two conditions:

1. The page must be viewed on a normal screen (not a printed page, projector, etc).
2. The width of the user's view port must be at least 720 pixels.

If these conditions are met, the styles inside the media query will be active, and the background color of the page will be sky blue.

Media queries are applied dynamically. If on page load the conditions specified in the media query are met, the CSS will be applied, but will be immediately disabled should the conditions cease to be met. Conversely, if the conditions are initially not met, the CSS will not be applied until the specified conditions are met.

In our example, if the user's view port width is initially greater than 720 pixels, but the user shrinks the browser's width, the background color will cease to be sky blue as soon as the user has resized the view port to less than 720 pixels in width.

Section 13.3: mediatype

Media queries have an optional `mediatype` parameter. This parameter is placed directly after the `@media` declaration (`@media mediatype`), for example:

```
@media print {
    html {
        background-color: white;
    }
}
```

The above CSS code will give the DOM `HTML` element a white background color when being printed.

The `mediatype` parameter has an optional `not` or `only` prefix that will apply the styles to everything except the specified mediatype *or* only the specified media type, respectively. For example, the following code example will apply the style to every media type except `print`.

```
@media not print {
    html {
        background-color: green;
    }
```

```
}
```

And the same way, for just showing it only on the screen, this can be used:

```
@media only screen {
    .fadeInEffects {
        display: block;
    }
}
```

The list of `mediatype` can be understood better with the following table:

Media Type	Description
`all`	Apply to all devices
`screen`	Default computers
`print`	Printers in general. Used to style print-versions of websites
`handheld`	PDA's, cellphones and hand-held devices with a small screen
`projection`	For projected presentation, for example projectors
`aural`	Speech Systems
`braille`	Braille tactile devices
`embossed`	Paged braille printers
`tv`	Television-type devices
`tty`	Devices with a fixed-pitch character grid. Terminals, portables.

Section 13.4: Media Queries for Retina and Non Retina Screens

Although this works only for WebKit based browsers, this is helpful:

```
/* ----------- Non-Retina Screens ----------- */
@media screen
  and (min-width: 1200px)
  and (max-width: 1600px)
  and (-webkit-min-device-pixel-ratio: 1) {
}

/* ----------- Retina Screens ----------- */
@media screen
  and (min-width: 1200px)
  and (max-width: 1600px)
  and (-webkit-min-device-pixel-ratio: 2)
  and (min-resolution: 192dpi) {
}
```

Background Information

There are two types of pixels in the display. One is the logical pixels and the other is the physical pixels. Mostly, the physical pixels always stay the same, because it is the same for all the display devices. The logical pixels change based on the resolution of the devices to display higher quality pixels. The device pixel ratio is the ratio between physical pixels and logical pixels. For instance, the MacBook Pro Retina, iPhone 4 and above report a device pixel ratio of 2, because the physical linear resolution is double the logical resolution.

The reason why this works only with WebKit based browsers is because of:

- The vendor prefix `-webkit-` before the rule.
- This hasn't been implemented in engines other than WebKit and Blink.

Section 13.5: Width vs Viewport

When we are using "width" with media queries it is important to set the meta tag correctly. Basic meta tag looks like this and it needs to be put inside the **`<head>`** tag.

```
<meta name="viewport" content="width=device-width,initial-scale=1">
```

Why this is important?

Based on MDN's definition "width" is

> The width media feature describes the width of the rendering surface of the output device (such as the width of the document window, or the width of the page box on a printer).

What does that mean?

View-port is the width of the device itself. If your screen resolution says the resolution is 1280 x 720, your view-port width is "1280px".

More often many devices allocate different pixel amount to display one pixel. For an example iPhone 6 Plus has 1242 x 2208 resolution. But the actual viewport-width and viewport-height is 414 x 736. That means 3 pixels are used to create 1 pixel.

But if you did not set the `meta` tag correctly it will try to show your webpage with its native resolution which results in a zoomed out view (smaller texts and images).

Section 13.6: Using Media Queries to Target Different Screen Sizes

Often times, responsive web design involves media queries, which are CSS blocks that are only executed if a condition is satisfied. This is useful for responsive web design because you can use media queries to specify different CSS styles for the mobile version of your website versus the desktop version.

```
@media only screen and (min-width: 300px) and (max-width: 767px) {
    .site-title {
        font-size: 80%;
    }

    /* Styles in this block are only applied if the screen size is atleast 300px wide, but no more
than 767px */
}

@media only screen and (min-width: 768px) and (max-width: 1023px) {
    .site-title {
        font-size: 90%;
    }

    /* Styles in this block are only applied if the screen size is atleast 768px wide, but no more
than 1023px */
}

@media only screen and (min-width: 1024px) {
```

```
    .site-title {
        font-size: 120%;
    }

    /* Styles in this block are only applied if the screen size is over 1024px wide. */
}
```

Section 13.7: Use on link tag

```
<link rel="stylesheet" media="min-width: 600px" href="example.css" />
```

This stylesheet is still downloaded but is applied only on devices with screen width larger than 600px.

Section 13.8: Media queries and IE8

Media queries are not supported at all in IE8 and below.

A Javascript based workaround

To add support for IE8, you could use one of several JS solutions. For example, **Respond** can be added to add media query support for IE8 only with the following code :

```
<!--[if lt IE 9]>
<script
   src="respond.min.js">
</script>
<![endif]-->
```

CSS Mediaqueries is another library that does the same thing. The code for adding that library to your HTML would be identical :

```
<!--[if lt IE 9]>
<script
   src="css3-mediaqueries.js">
</script>
<![endif]-->
```

The alternative

If you don't like a JS based solution, you should also consider adding an IE<9 only stylesheet where you adjust your styling specific to IE<9. For that, you should add the following HTML to your code:

```
<!--[if lt IE 9]>
<link rel="stylesheet" type="text/css" media="all" href="style-ielt9.css"/>
<![endif]-->
```

Note :

Technically it's one more alternative: using **CSS hacks** to target IE<9. It has the same impact as an IE<9 only stylesheet, but you don't need a separate stylesheet for that. I do not recommend this option, though, as they produce invalid CSS code (which is but one of several reasons why the use of CSS hacks is generally frowned upon today).

Chapter 14: Floats

Section 14.1: Float an Image Within Text

The most basic use of a float is having text wrap around an image. The below code will produce two paragraphs and an image, with the second paragraph flowing around the image. Notice that it is always content *after* the floated element that flows around the floated element.

HTML:

```
<p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer nec odio. Praesent libero. Sed cursus ante dapibus diam. Sed nisi. Nulla quis sem at nibh elementum imperdiet. Duis sagittis ipsum. Praesent mauris. Fusce nec tellus sed augue semper porta. Mauris massa. Vestibulum lacinia arcu eget nulla. </p>

<img src="http://lorempixel.com/200/100/" />

<p>Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Curabitur sodales ligula in libero. Sed dignissim lacinia nunc. Curabitur tortor. Pellentesque nibh. Aenean quam. In scelerisque sem at dolor. Maecenas mattis. Sed convallis tristique sem. Proin ut ligula vel nunc egestas porttitor. Morbi lectus risus, iaculis vel, suscipit quis, luctus non, massa. Fusce ac turpis quis ligula lacinia aliquet. </p>
```

CSS:

```
img {
  float:left;
  margin-right:1rem;
}
```

This will be the output

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer nec odio. Praesent libero. Sed cursus ante dapibus diam. Sed nisi. Nulla quis sem at nibh elementum imperdiet. Duis sagittis ipsum. Praesent mauris. Fusce nec tellus sed augue semper porta. Mauris massa. Vestibulum lacinia arcu eget nulla.

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Curabitur sodales ligula in libero. Sed dignissim lacinia nunc. Curabitur tortor. Pellentesque nibh. Aenean quam. In scelerisque sem at dolor. Maecenas mattis. Sed convallis tristique sem. Proin ut ligula vel nunc egestas porttitor. Morbi lectus risus, iaculis vel, suscipit quis, luctus non, massa. Fusce ac turpis quis ligula lacinia aliquet.

Codepen Link

Section 14.2: clear property

The clear property is directly related to floats. Property Values:

- none - Default. Allows floating elements on both sides
- left - No floating elements allowed on the left side
- right - No floating elements allowed on the right side
- both - No floating elements allowed on either the left or the right side
- initial - Sets this property to its default value. Read about initial
- inherit - Inherits this property from its parent element. Read about inherit

```
<html>
<head>
<style>
img {
    float: left;
```

```
}

p.clear {
    clear: both;
}
</style>
</head>
<body>

<img src="https://static.pexels.com/photos/69372/pexels-photo-69372-medium.jpeg" width="100">
<p>Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem
ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum </p>
<p class="clear">Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum
Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum Lorem ipsoum </p>

</body>
</html>
```

Section 14.3: Clearfix

> The clearfix hack is a popular way to contain floats (N. Gallagher aka @necolas)

Not to be confused with the `clear` property, clearfix is a *concept* (that is also related to floats, thus the possible confusion). To *contain floats*, you've to add `.cf` or `.clearfix` class on the container (**the parent**) and style this class with a few rules described below.

3 versions with slightly different effects (sources :A new micro clearfix hack by N. Gallagher and clearfix reloaded by T. J. Koblentz):

Clearfix (with top margin collapsing of contained floats still occurring)

```
.cf:after {
    content: "";
    display: table;
}

.cf:after {
    clear: both;
}
```

Clearfix also preventing top margin collapsing of contained floats

```
/**
 * For modern browsers
 * 1. The space content is one way to avoid an Opera bug when the
 *    contenteditable attribute is included anywhere else in the document.
 *    Otherwise it causes space to appear at the top and bottom of elements
 *    that are clearfixed.
 * 2. The use of `table` rather than `block` is only necessary if using
 *    `:before` to contain the top-margins of child elements.
 */
.cf:before,
.cf:after {
    content: " "; /* 1 */
    display: table; /* 2 */
}

.cf:after {
    clear: both;
```

```
}
```

Clearfix with support of outdated browsers IE6 and IE7

```
.cf:before,
.cf:after {
    content: " ";
    display: table;
}

.cf:after {
    clear: both;
}

/**
 * For IE 6/7 only
 * Include this rule to trigger hasLayout and contain floats.
 */
.cf {
    *zoom: 1;
}
```

Codepen showing clearfix effect

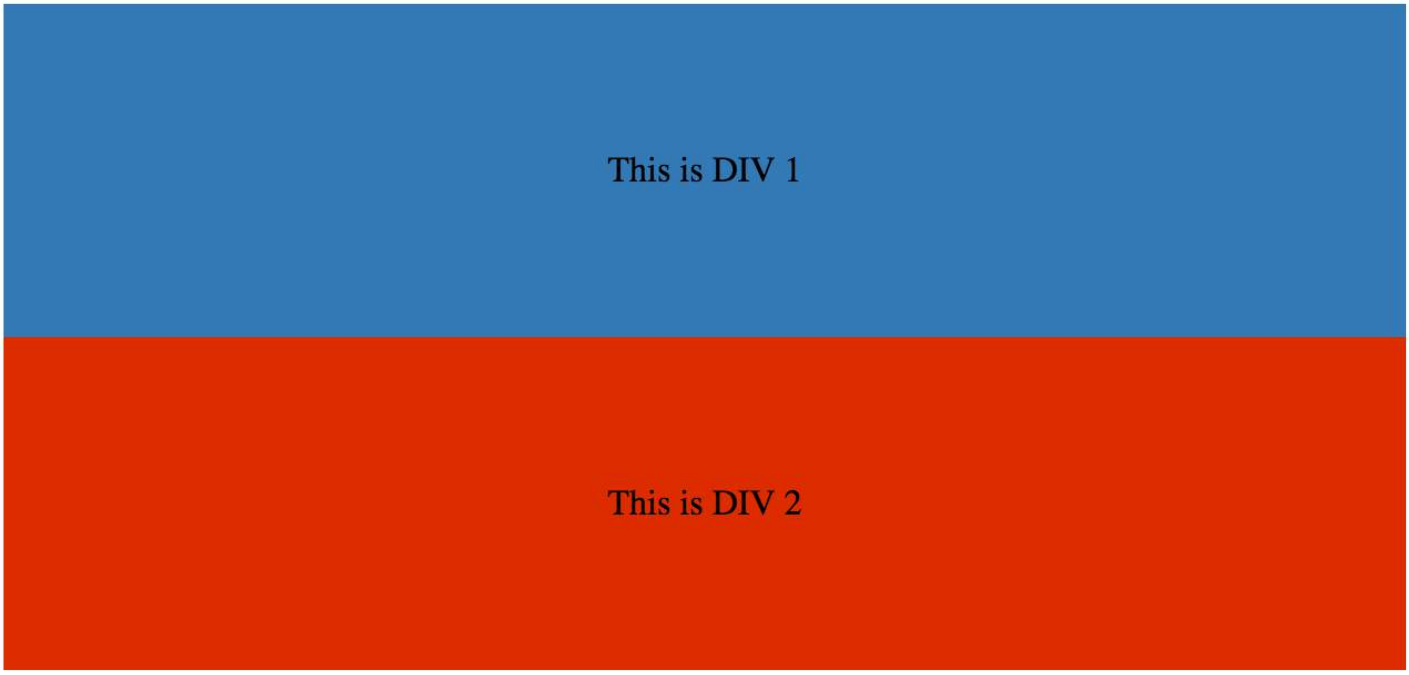
Other resource: Everything you know about clearfix is wrong (clearfix and BFC - Block Formatting Context while hasLayout relates to outdated browsers IE6 maybe 7)

Section 14.4: In-line DIV using float

The `div` is a block-level element, i.e it occupies the whole of the page width and the siblings are place one below the other irrespective of their width.

```
<div>
    <p>This is DIV 1</p>
</div>
<div>
    <p>This is DIV 2</p>
</div>
```

The output of the following code will be



We can make them in-line by adding a `float` css property to the `div`.

HTML:

```
<div class="outer-div">
    <div class="inner-div1">
        <p>This is DIV 1</p>
    </div>
    <div class="inner-div2">
        <p>This is DIV 2</p>
    </div>
</div>
```

CSS

```
.inner-div1 {
    width: 50%;
    margin-right:0px;
    float:left;
    background : #337ab7;
    padding:50px 0px;
}

.inner-div2 {
    width: 50%;
    margin-right:0px;
    float:left;
    background : #dd2c00;
    padding:50px 0px;
}

p {
    text-align:center;
}
```

Codepen Link

Section 14.5: Use of overflow property to clear floats

Setting `overflow` value to `hidden`,`auto` or `scroll` to an element, will clear all the floats within that element.

Note: using `overflow:scroll` will always show the scrollbox

Section 14.6: Simple Two Fixed-Width Column Layout

A simple two-column layout consists of two fixed-width, floated elements. Note that the sidebar and content area are not the same height in this example. This is one of the tricky parts with multi-column layouts using floats, and requires workarounds to make multiple columns appear to be the same height.

HTML:

```
<div class="wrapper">

<div class="sidebar">
  <h2>Sidebar</h2>

  <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer nec odio.</p>
</div>

<div class="content">
  <h1>Content</h1>

  <p>Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos.
Curabitur sodales ligula in libero. Sed dignissim lacinia nunc. Curabitur tortor. Pellentesque
nibh. Aenean quam. In scelerisque sem at dolor. Maecenas mattis. Sed convallis tristique sem. Proin
ut ligula vel nunc egestas porttitor. Morbi lectus risus, iaculis vel, suscipit quis, luctus non,
massa. Fusce ac turpis quis ligula lacinia aliquet. </p>
</div>

</div>
```

CSS:

```
.wrapper {
  width:600px;
  padding:20px;
  background-color:pink;

  /* Floated elements don't use any height. Adding "overflow:hidden;" forces the
     parent element to expand to contain its floated children. */
  overflow:hidden;
}

.sidebar {
  width:150px;
  float:left;
  background-color:blue;
```

```
}

.content {
  width:450px;
  float:right;
  background-color:yellow;
}
```

Section 14.7: Simple Three Fixed-Width Column Layout

HTML:

```
<div class="wrapper">
  <div class="left-sidebar">
    <h1>Left Sidebar</h1>
    <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. </p>
  </div>
  <div class="content">
    <h1>Content</h1>
    <p>Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos.
Curabitur sodales ligula in libero. Sed dignissim lacinia nunc. Curabitur tortor. Pellentesque
nibh. Aenean quam. In scelerisque sem at dolor. Maecenas mattis. Sed convallis tristique sem. Proin
ut ligula vel nunc egestas porttitor. Morbi lectus risus, iaculis vel, suscipit quis, luctus non,
massa. </p>
  </div>
  <div class="right-sidebar">
    <h1>Right Sidebar</h1>
    <p>Fusce ac turpis quis ligula lacinia aliquet.</p>
  </div>
</div>
```

CSS:

```
.wrapper {
  width:600px;
  background-color:pink;
  padding:20px;

  /* Floated elements don't use any height. Adding "overflow:hidden;" forces the
     parent element to expand to contain its floated children. */
  overflow:hidden;
}

.left-sidebar {
  width:150px;
  background-color:blue;
  float:left;
}

.content {
  width:300px;
  background-color:yellow;
  float:left;
}

.right-sidebar {
  width:150px;
  background-color:green;
  float:right;
}
```

Section 14.8: Two-Column Lazy/Greedy Layout

This layout uses one floated column to create a two-column layout with no defined widths. In this example the left sidebar is "lazy," in that it only takes up as much space as it needs. Another way to say this is that the left sidebar is "shrink-wrapped." The right content column is "greedy," in that it takes up all the remaining space.

HTML:

```
<div class="sidebar">
<h1>Sidebar</h1>
<img src="http://lorempixel.com/150/200/" />
</div>

<div class="content">
<h1>Content</h1>
<p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer nec odio. Praesent libero. Sed cursus ante dapibus diam. Sed nisi. Nulla quis sem at nibh elementum imperdiet. Duis sagittis ipsum. Praesent mauris. Fusce nec tellus sed augue semper porta. Mauris massa. Vestibulum lacinia arcu eget nulla. </p>
<p>Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Curabitur sodales ligula in libero. Sed dignissim lacinia nunc. Curabitur tortor. Pellentesque nibh. Aenean quam. In scelerisque sem at dolor. Maecenas mattis. Sed convallis tristique sem. Proin ut ligula vel nunc egestas porttitor. Morbi lectus risus, iaculis vel, suscipit quis, luctus non, massa. Fusce ac turpis quis ligula lacinia aliquet. Mauris ipsum. Nulla metus metus, ullamcorper vel, tincidunt sed, euismod in, nibh. </p>
</div>
```

CSS:

```
.sidebar {
  /* `display:table;` shrink-wraps the column */
  display:table;
  float:left;
  background-color:blue;
}

.content {
  /* `overflow:hidden;` prevents `.content` from flowing under `.sidebar` */
  overflow:hidden;
  background-color:yellow;
}
```

Fiddle

Chapter 15: Typography

Parameter	Details
font-style	`italics` or `oblique`
font-variant	`normal` or `small-caps`
font-weight	`normal`, `bold` or numeric from 100 to 900.
font-size	The font size given in `%`, `px`, `em`, or any other valid CSS measurement
line-height	The line height given in `%`, `px`, `em`, or any other valid CSS measurement
font-family	This is for defining the family's name.
color	Any valid CSS color representation, like `red`, `#00FF00`, `hsl(240, 100%, 50%)` etc.
font-stretch	Whether or not to use a confenced or expanded face from font. Valid values are `normal`, `ultra-condensed`, `extra-condensed`, `condensed`, `semi-condensed`, `semi-expanded`, `expanded`, `extra-expanded` or `ultra-expanded`
text-align	`start`, `end`, `left`, `right`, `center`, `justify`, `match-parent`
text-decoration	`none`, `underline`, `overline`, `line-through`, `initial`, `inherit`;

Section 15.1: The Font Shorthand

With the syntax:

```
element {
    font: [font-style] [font-variant] [font-weight] [font-size/line-height] [font-family];
}
```

You can have all your font-related styles in one declaration with the `font` shorthand. Simply use the `font` property, and put your values in the correct order.

For example, to make all `p` elements bold with a font size of 20px and using Arial as the font family typically you would code it as follows:

```
p {
    font-weight: bold;
    font-size: 20px;
    font-family: Arial, sans-serif;
}
```

However with the font shorthand it can be condensed as follows:

```
p {
    font: bold 20px Arial, sans-serif;
}
```

Note: that since `font-style`, `font-variant`, `font-weight` and `line-height` are optional, the three of them are skipped in this example. It is important to note that using the shortcut **resets** the other attributes not given. Another important point is that the two necessary attributes for the font shortcut to work are `font-size` and `font-family`. If they are not both included the shortcut is ignored.

Initial value for each of the properties:

- `font-style: normal;`
- `font-variant: normal;`
- `font-weight: normal;`

- `font-stretch: normal;`
- `font-size: medium;`
- `line-height: normal;`
- `font-family` – depends on user agent

Section 15.2: Quotes

The `quotes` property is used to customize the opening and closing quotation marks of the `<q>` tag.

```
q {
    quotes: "«" "»";
}
```

Section 15.3: Font Size

HTML:

```
<div id="element-one">Hello I am some text.</div>
<div id="element-two">Hello I am some smaller text.</div>
```

CSS:

```
#element-one {
    font-size: 30px;
}

#element-two {
    font-size: 10px;
}
```

The text inside `#element-one` will be `30px` in size, while the text in `#element-two` will be `10px` in size.

Section 15.4: Text Direction

```
div {
    direction: ltr; /* Default, text read read from left-to-right */
}
.ex {
    direction: rtl; /* text read from right-to-left */
}
.horizontal-tb {
    writing-mode: horizontal-tb; /* Default, text read from left-to-right and top-to-bottom. */
}
.vertical-rtl {
    writing-mode: vertical-rl; /* text read from right-to-left and top-to-bottom */
}
.vertical-ltr {
    writing-mode: vertical-rl; /* text read from left-to-right and top to bottom */
}
```

The `direction` property is used to change the horizontal text direction of an element.

Syntax: `direction: ltr | rtl | initial | inherit;`

The `writing-mode` property changes the alignment of text so it can be read from top-to-bottom or from left-to-right, depending on the language.

Syntax: `direction: horizontal-tb | vertical-rl | vertical-lr;`

Section 15.5: Font Stacks

```
font-family: 'Segoe UI', Tahoma, sans-serif;
```

The browser will attempt to apply the font face "Segoe UI" to the characters within the elements targeted by the above property. If this font is not available, or the font does not contain a glyph for the required character, the browser will fall back to Tahoma, and, if necessary, any sans-serif font on the user's computer. Note that any font names with more than one word such as "Segoe UI" need to have single or double quotes around them.

```
font-family: Consolas, 'Courier New', monospace;
```

The browser will attempt to apply the font face "Consolas" to the characters within the elements targeted by the above property. If this font is not available, or the font does not contain a glyph for the required character, the browser will fall back to "Courier New," and, if necessary, any monospace font on the user's computer.

Section 15.6: Text Overflow

The `text-overflow` property deals with how overflowed content should be signaled to users. In this example, the `ellipsis` represents clipped text.

```
.text {
  overflow: hidden;
  text-overflow: ellipsis;
}
```

Unfortunately, `text-overflow: ellipsis` only works on a single line of text. There is no way to support ellipsis on the last line in standard CSS, but it can be achieved with non-standard webkit-only implementation of flexboxes.

```
.giveMeEllipsis {
   overflow: hidden;
   text-overflow: ellipsis;
   display: -webkit-box;
   -webkit-box-orient: vertical;
   -webkit-line-clamp: N; /* number of lines to show */
   line-height: X;        /* fallback */
   max-height: X*N;       /* fallback */
}
```

Example (open in Chrome or Safari):

http://jsfiddle.net/csYjC/1131/

Resources:

https://www.w3.org/TR/2012/WD-css3-ui-20120117/#text-overflow0

Section 15.7: Text Shadow

To add shadows to text, use the `text-shadow` property. The syntax is as follows:

```
text-shadow: horizontal-offset vertical-offset blur color;
```

Shadow without blur radius

```
h1 {
  text-shadow: 2px 2px #0000FF;
}
```

This creates a blue shadow effect around a heading

Shadow with blur radius

To add a blur effect, add an option `blur radius` argument

```
h1 {
  text-shadow: 2px 2px 10px #0000FF;
}
```

Multiple Shadows

To give an element multiple shadows, separate them with commas

```
h1 {
    text-shadow: 0 0 3px #FF0000, 0 0 5px #0000FF;
}
```

Section 15.8: Text Transform

The `text-transform` property allows you to change the capitalization of text. Valid values are: `uppercase`, `capitalize`, `lowercase`, `initial`, `inherit`, and `none`

CSS

```
.example1 {
    text-transform: uppercase;
}
.example2 {
    text-transform: capitalize;
}
.example3 {
    text-transform: lowercase;
}
```

HTML

```
<p class="example1">
    all letters in uppercase  <!-- "ALL LETTERS IN UPPERCASE" -->
</p>
<p class="example2">
    all letters in capitalize <!-- "All Letters In Capitalize (Sentence Case)" -->
</p>
<p class="example3">
    all letters in lowercase  <!-- "all letters in lowercase" -->
</p>
```

Section 15.9: Letter Spacing

```
h2 {
  /* adds a 1px space horizontally between each letter;
     also known as tracking */
  letter-spacing: 1px;
```

```
}
```

The letter-spacing property is used to specify the space between the characters in a text.

! letter-spacing also supports negative values:

```
p {
  letter-spacing: -1px;
}
```

Resources: https://developer.mozilla.org/en-US/docs/Web/CSS/letter-spacing

Section 15.10: Text Indent

```
p {
    text-indent: 50px;
}
```

The `text-indent` property specifies how much horizontal space text should be moved before the beginning of the first line of the text content of an element.

Resources:

- Indenting only the first line of text in a paragraph?
- https://www.w3.org/TR/CSS21/text.html#propdef-text-indent
- https://developer.mozilla.org/en-US/docs/Web/CSS/text-indent

Section 15.11: Text Decoration

The `text-decoration` property is used to set or remove decorations from text.

```
h1 { text-decoration: none; }
h2 { text-decoration: overline; }
h3 { text-decoration: line-through; }
h4 { text-decoration: underline; }
```

text-decoration can be used in combination with text-decoration-style and text-decoration-color as a shorthand property:

```
.title { text-decoration: underline dotted blue; }
```

This is a shorthand version of

```
.title {
  text-decoration-style: dotted;
  text-decoration-line: underline;
  text-decoration-color: blue;
}
```

It should be noted that the following properties are only supported in Firefox

- text-decoration-color
- text-decoration-line
- text-decoration-style
- text-decoration-skip

Section 15.12: Word Spacing

The word-spacing property specifies the spacing behavior between tags and words.

Possible values

- a positive or negative *length* (using `em` `px` `vh` `cm` etc.) or *percentage* (using `%`)
- the keyword `normal` uses the font's default word spacing
- the keyword `inherit` takes the value from the parent element

CSS

```
.normal    { word-spacing: normal; }
.narrow    { word-spacing: -3px; }
.extensive { word-spacing: 10px; }
```

HTML

```
<p>
  <span class="normal">This is an example, showing the effect of "word-spacing".</span><br>
  <span class="narrow">This is an example, showing the effect of "word-spacing".</span><br>
  <span class="extensive">This is an example, showing the effect of "word-spacing".</span><br>
</p>
```

Online-Demo

Try it yourself

Further reading:

- word-spacing – MDN
- word-spacing – w3.org

Section 15.13: Font Variant

Attributes:

normal

Default attribute of fonts.

small-caps

Sets every letter to uppercase, **but** makes the lowercase letters(from original text) smaller in size than the letters that originally uppercase.

CSS:

```
.smallcaps{
   font-variant: small-caps;
}
```

HTML:

```
<p class="smallcaps">
    Documentation about CSS Fonts
```

```
    <br>
    aNd ExAmpLe
</p>
```

Output:

DOCUMENTATION ABOUT CSS FONTS
ANDEXAMPLE

Note: The font-variant property is a shorthand for the properties: font-variant-caps, font-variant-numeric, font-variant-alternates, font-variant-ligatures, and font-variant-east-asian.

Chapter 16: Flexible Box Layout (Flexbox)

The Flexible Box module, or just 'flexbox' for short, is a box model designed for user interfaces, and it allows users to align and distribute space among items in a container such that elements behave predictably when the page layout must accommodate different, unknown screen sizes. A flex container expands items to fill available space and shrinks them to prevent overflow.

Section 16.1: Dynamic Vertical and Horizontal Centering (align-items, justify-content)

Simple Example (centering a single element)

HTML

```
<div class="aligner">
  <div class="aligner-item">…</div>
</div>
```

CSS

```
.aligner {
  display: flex;
  align-items: center;
  justify-content: center;
}

.aligner-item {
  max-width: 50%; /*for demo. Use actual width instead.*/
}
```

Here is a demo.

Reasoning

Property	Value	Description
`align-items`	`center`	This centers the elements along the axis other than the one specified by `flex-direction`, i.e., vertical centering for a horizontal flexbox and horizontal centering for a vertical flexbox.
`justify-content`	`center`	This centers the elements along the axis specified by `flex-direction`. I.e., for a horizontal (**`flex-direction`**`: row`) flexbox, this centers horizontally, and for a vertical flexbox (**`flex-direction`**`: column`) flexbox, this centers vertically)

Individual Property Examples

All of the below styles are applied onto this simple layout:

```
<div id="container">
  <div></div>
  <div></div>
  <div></div>
</div>
```

where `#container` is the `flex-box`.

Example: `justify-content: center` on a horizontal flexbox

CSS:

```
div#container {
  display: flex;
  flex-direction: row;
  justify-content: center;
}
```

Outcome:

Here is a demo.

Example: `justify-content: center` on a vertical flexbox

CSS:

```
div#container {
  display: flex;
  flex-direction: column;
  justify-content: center;
}
```

Outcome:

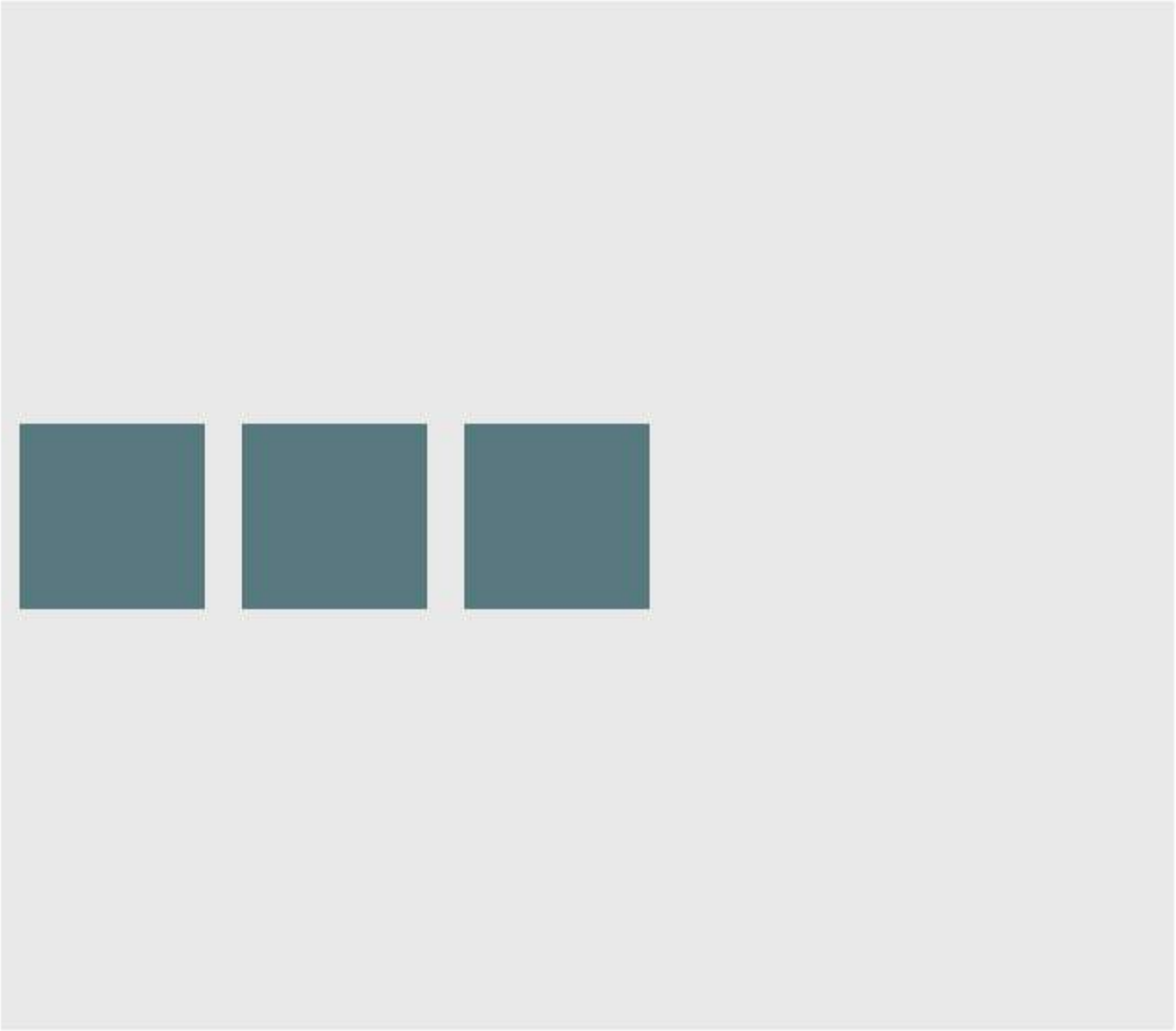
Here is a demo.

Example: `align-content: center` on a horizontal flexbox

CSS:

```
div#container {
  display: flex;
  flex-direction: row;
  align-items: center;
}
```

Outcome:



Here is a demo.

Example: `align-content: center` on a vertical flexbox

CSS:

```
div#container {
  display: flex;
  flex-direction: column;
  align-items: center;
}
```

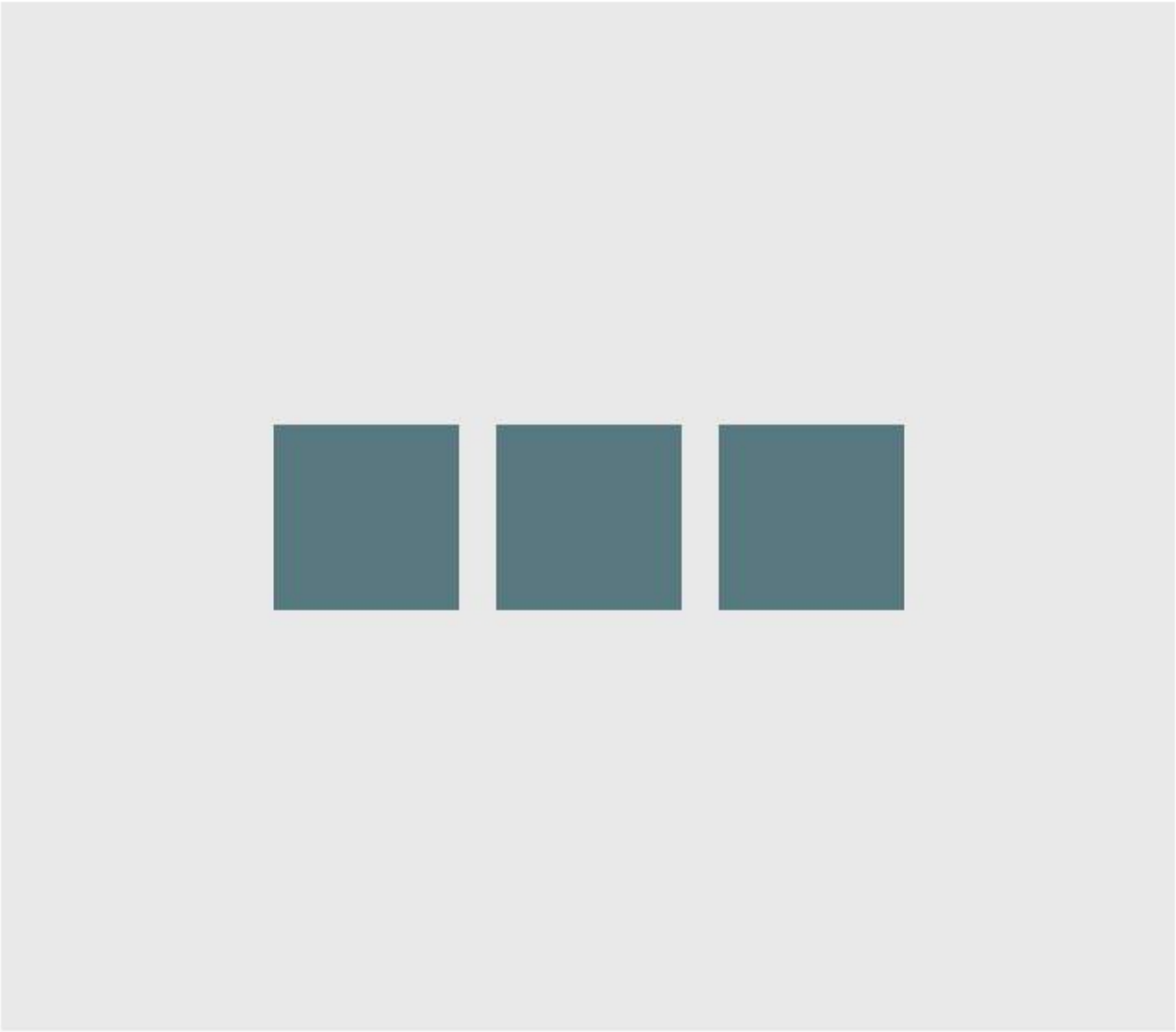
Outcome:

Here is a demo.

Example: Combination for centering both on horizontal flexbox

```
div#container {
  display: flex;
  flex-direction: row;
  justify-content: center;
  align-items: center;
}
```

Outcome:



Here is a demo.

Example: Combination for centering both on vertical flexbox

```
div#container {
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
}
```

Outcome:

Here is a demo.

Section 16.2: Sticky Variable-Height Footer

This code creates a sticky footer. When the content doesn't reach the end of the viewport, the footer sticks to the bottom of the viewport. When the content extends past the bottom of the viewport, the footer is also pushed out of the viewport. *View Result*

HTML:

```
<div class="header">
  <h2>Header</h2>
</div>

<div class="content">
  <h1>Content</h1>
  <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer nec odio. Praesent libero.
Sed cursus ante dapibus diam. Sed nisi. Nulla quis sem at nibh elementum imperdiet. Duis sagittis
ipsum. Praesent mauris. Fusce nec tellus sed augue semper porta. Mauris massa. Vestibulum lacinia
arcu eget nulla. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos
himenaeos. Curabitur sodales ligula in libero. </p>
</div>

<div class="footer">
```

```
  <h4>Footer</h4>
</div>
```

CSS:

```
html, body {
  height: 100%;
}

body {
  display: flex;
  flex-direction: column;
}

.content {
  /* Include `0 auto` for best browser compatibility. */
  flex: 1 0 auto;
}

.header, .footer {
  background-color: grey;
  color: white;
  flex: none;
}
```

Section 16.3: Optimally fit elements to their container

One of the nicest features of flexbox is to allow optimally fitting containers to their parent element.

Live demo.

HTML:

```
<div class="flex-container">
  <div class="flex-item">1</div>
  <div class="flex-item">2</div>
  <div class="flex-item">3</div>
  <div class="flex-item">4</div>
  <div class="flex-item">5</div>
</div>
```

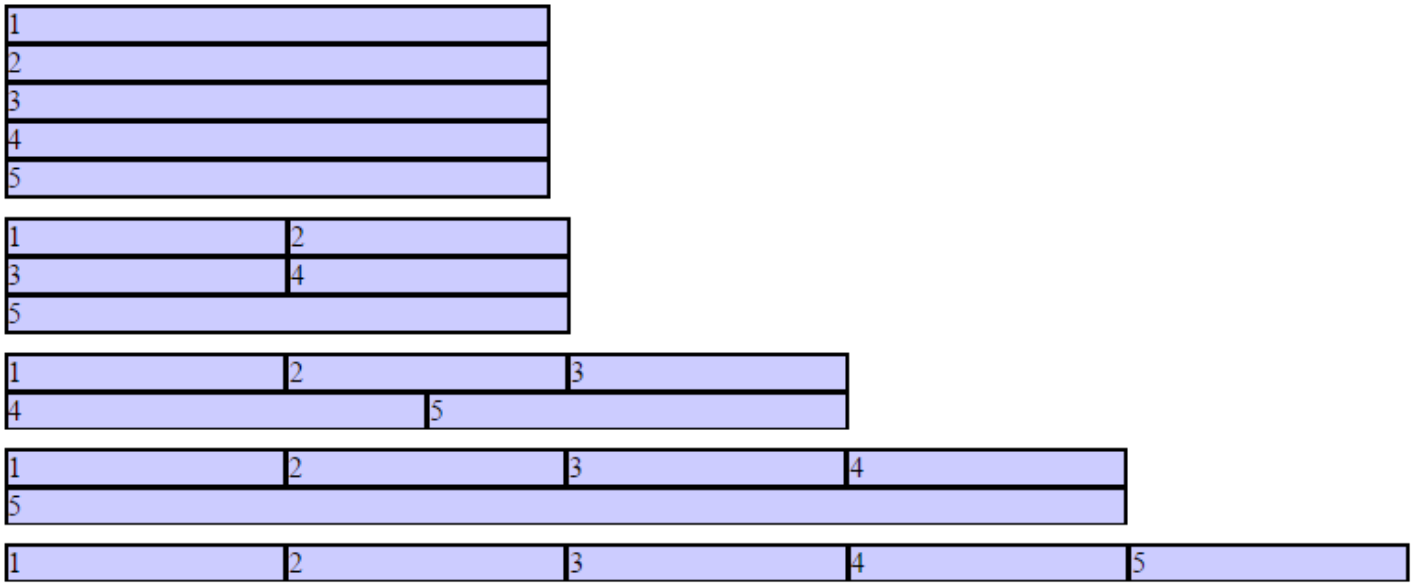
CSS:

```
.flex-container {
    background-color: #000;
    height: 100%;
    display:flex;
    flex-direction: row;
    flex-wrap: wrap;
    justify-content: flex-start;
    align-content: stretch;
    align-items: stretch;
}

.flex-item {
    background-color: #ccf;
    margin: 0.1em;
    flex-grow: 1;
    flex-shrink: 0;
```

```
    flex-basis: 200px; /* or % could be used to ensure a specific layout */
}
```

Outcome:

Columns adapt as screen is resized.



Section 16.4: Holy Grail Layout using Flexbox

Holy Grail layout is a layout with a fixed height header and footer, and a center with 3 columns. The 3 columns include a fixed width sidenav, a fluid center, and a column for other content like ads (the fluid center appears first in the markup). CSS Flexbox can be used to achieve this with a very simple markup:

HTML Markup:

```
<div class="container">
  <header class="header">Header</header>
  <div class="content-body">
    <main class="content">Content</main>
    <nav class="sidenav">Nav</nav>
    <aside class="ads">Ads</aside>
  </div>
  <footer class="footer">Footer</footer>
</div>
```

CSS:

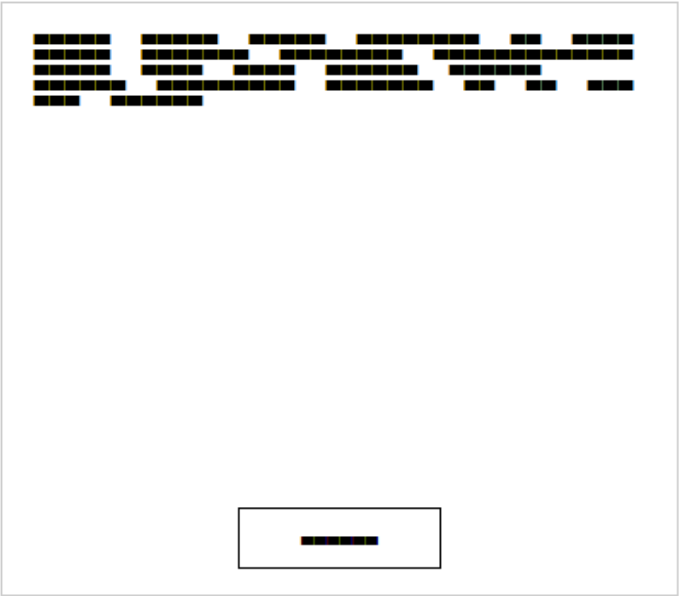
```
body {
  margin: 0;
  padding: 0;
}

.container {
  display: flex;
  flex-direction: column;
  height: 100vh;
}

.header {
  flex: 0 0 50px;
```

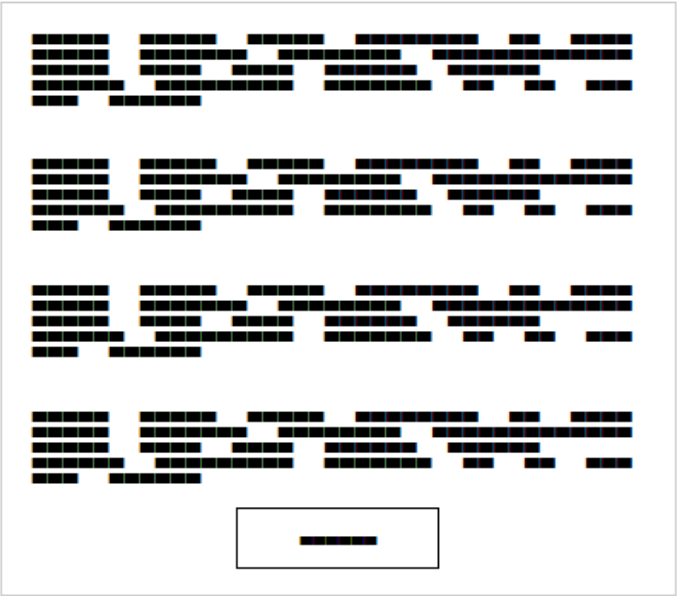
```
}

.content-body {
  flex: 1 1 auto;

  display: flex;
  flex-direction: row;
}

.content-body .content {
  flex: 1 1 auto;
  overflow: auto;
}

.content-body .sidenav {
  order: -1;
  flex: 0 0 100px;
  overflow: auto;
}

.content-body .ads {
  flex: 0 0 100px;
  overflow: auto;
}

.footer {
  flex: 0 0 50px;
}
```

Demo

Section 16.5: Perfectly aligned buttons inside cards with flexbox

It's a regular pattern in design these days to vertically align **call to actions** inside its containing cards like this:





This can be achieved using a special trick with `flexbox`

HTML

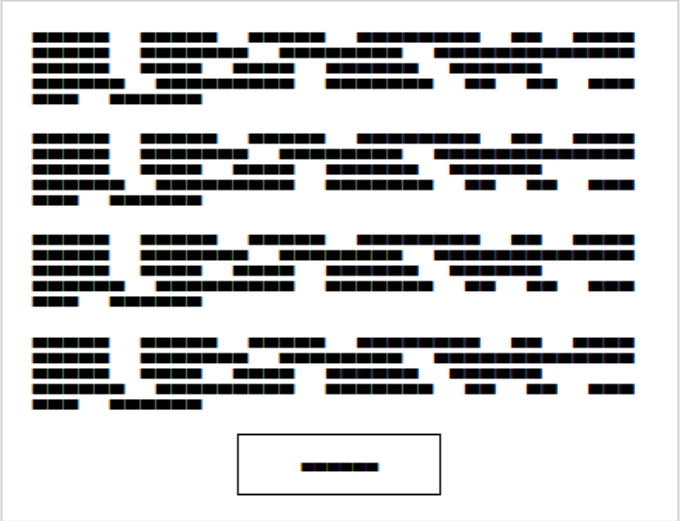
```
<div class="cards">
  <div class="card">
    <p>Lorem ipsum Magna proident ex anim dolor ullamco pariatur reprehenderit culpa esse enim
mollit labore dolore voluptate ullamco et ut sed qui minim.</p>
    <p><button>Action</button></p>
  </div>
  <div class="card">
    <p>Lorem ipsum Magna proident ex anim dolor ullamco pariatur reprehenderit culpa esse enim
mollit labore dolore voluptate ullamco et ut sed qui minim.</p>
    <p>Lorem ipsum Magna proident ex anim dolor ullamco pariatur reprehenderit culpa esse enim
mollit labore dolore voluptate ullamco et ut sed qui minim.</p>
    <p>Lorem ipsum Magna proident ex anim dolor ullamco pariatur reprehenderit culpa esse enim
mollit labore dolore voluptate ullamco et ut sed qui minim.</p>
    <p>Lorem ipsum Magna proident ex anim dolor ullamco pariatur reprehenderit culpa esse enim
mollit labore dolore voluptate ullamco et ut sed qui minim.</p>
    <p><button>Action</button></p>
  </div>
</div>
```

First of all, we use CSS to apply `display: flex;` to the container. This will create 2 columns equal in height with the content flowing naturally inside it

CSS

```
.cards {
    display: flex;
}
.card {
    border: 1px solid #ccc;
    margin: 10px 10px;
    padding: 0 20px;
}
button {
    height: 40px;
    background: #fff;
    padding: 0 40px;
    border: 1px solid #000;
}
p:last-child {
    text-align: center;
}
```

The layout will change and become like this:



In order to move the buttons to the bottom of the block, we need to apply `display: flex;` to the card itself with the direction set to `column`. After that, we should select the last element inside the card and set the `margin-top` to `auto`. This will push the last paragraph to the bottom of the card and achieve the required result.

Final CSS:

```
.cards {
    display: flex;
}
.card {
    border: 1px solid #ccc;
    margin: 10px 10px;
    padding: 0 20px;
    display: flex;
    flex-direction: column;
}
button {
    height: 40px;
    background: #fff;
    padding: 0 40px;
    border: 1px solid #000;
}
p:last-child {
    text-align: center;
    margin-top: auto;
}
```

Section 16.6: Same height on nested containers

This code makes sure that all nested containers are always the same height. This is done by assuring that all nested elements are the same height as the containing parent div. See working example: https://jsfiddle.net/3wwh7ewp/

This effect is achieved due to the property `align-items` being set to `stretch` by default.

HTML

```
<div class="container">
  <div style="background-color: red">
    Some <br />
    data <br />
    to make<br />
```

```
    a height <br />
 </div>
  <div style="background-color: blue">
    Fewer <br />
    lines <br />
 </div>
</div>
```

CSS

```
.container {
    display: flex;
    align-items: stretch; // Default value
}
```

Note: Does not work on IE versions under 10

Chapter 17: Cascading and Specificity

Section 17.1: Calculating Selector Specificity

Each individual CSS Selector has its own specificity value. Every selector in a sequence increases the sequence's overall specificity. Selectors fall into one of three different specificity groups: *A*, *B* and *c*. When multiple selector sequences select a given element, the browser uses the styles applied by the sequence with the highest overall specificity.

Group	Comprised of	Examples
A	id selectors	`#foo`
B	class selectors	`.bar`
	attribute selectors	`[title]`, `[colspan="2"]`
	pseudo-classes	`:hover`, `:nth-child(2)`
c	type selectors	`div`, `li`
	pseudo-elements	`::before`, `::first-letter`

Group *A* is the most specific, followed by Group *B*, then finally Group *c*.

The universal selector (`*`) and combinators (like `>` and `~`) have no specificity.

Example 1: Specificity of various selector sequences

```
#foo #baz {}       /* a=2, b=0, c=0 */

#foo.bar {}        /* a=1, b=1, c=0 */

#foo {}            /* a=1, b=0, c=0 */

.bar:hover {}      /* a=0, b=2, c=0 */

div.bar {}         /* a=0, b=1, c=1 */

:hover {}          /* a=0, b=1, c=0 */

[title] {}         /* a=0, b=1, c=0 */

.bar {}            /* a=0, b=1, c=0 */

div ul + li {}     /* a=0, b=0, c=3 */

p::after {}        /* a=0, b=0, c=2 */

*::before {}       /* a=0, b=0, c=1 */

::before {}        /* a=0, b=0, c=1 */

div {}             /* a=0, b=0, c=1 */

* {}               /* a=0, b=0, c=0 */
```

Example 2: How specificity is used by the browser

Imagine the following CSS implementation:

```
#foo {
  color: blue;
}
```

```
.bar {
  color: red;
  background: black;
}
```

Here we have an ID selector which declares `color` as *blue*, and a class selector which declares `color` as *red* and `background` as *black*.

An element with an ID of `#foo` and a class of `.bar` will be selected by both declarations. ID selectors have a Group *A* specificity and class selectors have a Group *B* specificity. An ID selector outweighs any number of class selectors. Because of this, `color:blue;` from the `#foo` selector and the `background:black;` from the `.bar` selector will be applied to the element. The higher specificity of the ID selector will cause the browser to ignore the `.bar` selector's `color` declaration.

Now imagine a different CSS implementation:

```
.bar {
  color: red;
  background: black;
}

.baz {
  background: white;
}
```

Here we have two class selectors; one of which declares `color` as *red* and `background` as *black*, and the other declares `background` as *white*.

An element with both the `.bar` and `.baz` classes will be affected by both of these declarations, however the problem we have now is that both `.bar` and `.baz` have an identical Group *B* specificity. The cascading nature of CSS resolves this for us: as `.baz` is defined *after* `.bar`, our element ends up with the *red* `color` from `.bar` but the *white* `background` from `.baz`.

Example 3: How to manipulate specificity

The last snippet from Example 2 above can be manipulated to ensure our `.bar` class selector's `color` declaration is used instead of that of the `.baz` class selector.

```
.bar {}         /* a=0, b=1, c=0 */
.baz {}         /* a=0, b=1, c=0 */
```

The most common way to achieve this would be to find out what other selectors can be applied to the `.bar` selector sequence. For example, if the `.bar` class was only ever applied to `span` elements, we could modify the `.bar` selector to `span.bar`. This would give it a new Group *C* specificity, which would override the `.baz` selector's lack thereof:

```
span.bar {}     /* a=0, b=1, c=1 */
.baz {}         /* a=0, b=1, c=0 */
```

However it may not always possible to find another common selector which is shared between any element which uses the `.bar` class. Because of this, CSS allows us to duplicate selectors to increase specificity. Instead of just `.bar`, we can use `.bar.bar` instead (See [The grammar of Selectors, W3C Recommendation]). This still selects any element with a class of `.bar`, but now has double the Group *B* specificity:

```
.bar.bar {}     /* a=0, b=2, c=0 */
.baz {}         /* a=0, b=1, c=0 */
```

`!important` and inline style declarations

The `!important` flag on a style declaration and styles declared by the HTML `style` attribute are considered to have a greater specificity than any selector. If these exist, the style declaration they affect will overrule other declarations regardless of their specificity. That is, unless you have more than one declaration that contains an `!important` flag for the same property that apply to the same element. Then, normal specificity rules will apply to those properties in reference to each other.

Because they completely override specificity, the use of `!important` is frowned upon in most use cases. One should use it as little as possible. To keep CSS code efficient and maintainable in the long run, it's almost always better to increase the specificity of the surrounding selector than to use `!important`.

One of those rare exceptions where `!important` is not frowned upon, is when implementing generic helper classes like a `.hidden` or `.background-yellow` class that are supposed to always override one or more properties wherever they are encountered. And even then, you need to know what you're doing. The last thing you want, when writing maintainable CSS, is to have `!important` flags throughout your CSS.

A final note

A common misconception about CSS specificity is that the Group *A*, *B* and *c* values should be combined with each other (`a=1, b=5, c=1` => 151). This is **not** the case. If this were the case, having 20 of a Group *B* or *c* selector would be enough to override a single Group *A* or *B* selector respectively. The three groups should be regarded as individual levels of specificity. Specificity cannot be represented by a single value.

When creating your CSS style sheet, you should maintain the lowest specificity as possible. If you need to make the specificity a little higher to overwrite another method, make it higher but as low as possible to make it higher. You shouldn't need to have a selector like this:

```
body.page header.container nav div#main-nav li a {}
```

This makes future changes harder and pollutes that css page.

You can calculate the specificity of your selector here

Section 17.2: The !important declaration

The `!important` declaration is used to override the usual specificity in a style sheet by giving a higher priority to a rule. Its usage is: `property : value !important;`

```
#mydiv {
    font-weight: bold !important;    /* This property won't be overridden
                                        by the rule below */
}

#outerdiv #mydiv {
    font-weight: normal;             /* #mydiv font-weight won't be set to normal
                                     even if it has a higher specificity because
                                     of the !important declaration above */
}
```

Avoiding the usage of `!important` is strongly recommended (unless absolutely necessary), because it will disturb the natural flow of css rules which can bring uncertainty in your style sheet. Also it is important to note that when multiple `!important` declarations are applied to the same rule on a certain element, the one with the higher specificity will be the ona applied.

Here are some examples where using `!important` declaration can be justified:

- If your rules shouldn't be overridden by any inline style of the element which is written inside `style` attribute of the html element.
- To give the user more control over the web accessibility, like increasing or decreasing size of the font-size, by overriding the author style using `!important`.
- For testing and debugging using inspect element.

See also:

- **W3C - 6 Assigning property values, Cascading, and Inheritance -- 6.4.2 !important rules**

Section 17.3: Cascading

Cascading and specificity are used together to determine the final value of a CSS styling property. They also define the mechanisms for resolving conflicts in CSS rule sets.

CSS Loading order

Styles are read from the following sources, in this order:

1. User Agent stylesheet (The styles supplied by the browser vendor)
2. User stylesheet (The additional styling a user has set on his/her browser)
3. Author stylesheet (Author here means the creator of the webpage/website)
    - Maybe one or more `.css` files
    - In the **`<style>`** element of the HTML document
4. Inline styles (In the `style` attribute on an HTML element)

The browser will lookup the corresponding style(s) when rendering an element.

How are conflicts resolved?

When only one CSS rule set is trying to set a style for an element, then there is no conflict, and that rule set is used.

When multiple rule sets are found with conflicting settings, first the Specificty rules, and then the Cascading rules are used to determine what style to use.

Example 1 - Specificity rules

```
.mystyle { color: blue; }  /* specificity: 0, 0, 1, 0 */
div { color: red; }        /* specificity: 0, 0, 0, 1 */
```

```
<div class="mystyle">Hello World</div>
```

What color will the text be? (hover to see the answer)

> blue

First the specificity rules are applied, and the one with the highest specificity "wins".

Example 2 - Cascade rules with identical selectors

External css file

```
.class {
```

```
  background: #FFF;
}
```

Internal css (in HTML file)

```
<style>
.class {
  background: #000;
}
<style>
```

In this case, where you have identical selectors, the cascade kicks in, and determines that the last one loaded "wins".

Example 3 - Cascade rules after Specificity rules

```
body > .mystyle { background-color: blue; }   /* specificity: 0, 0, 1, 1 */
.otherstyle > div { background-color: red; }  /* specificity: 0, 0, 1, 1 */
```

```
<body class="otherstyle">
  <div class="mystyle">Hello World</div>
</body>
```

What color will the background be?

> red

After applying the specificity rules, there's still a conflict between blue and red, so the cascading rules are applied on top of the specificity rules. Cascading looks at the load order of the rules, whether inside the same `.css` file or in the collection of style sources. The last one loaded overrides any earlier ones. In this case, the `.otherstyle > div` rule "wins".

A final note

- Selector specificity always take precedence.
- Stylesheet order break ties.
- Inline styles trump everything.

Section 17.4: More complex specificity example

```
div {
    font-size: 7px;
    border: 3px dotted pink;
    background-color: yellow;
    color: purple;
}

body.mystyle > div.myotherstyle {
    font-size: 11px;
    background-color: green;
}

#elmnt1 {
    font-size: 24px;
    border-color: red;
}
```

```
.mystyle .myotherstyle {
    font-size: 16px;
    background-color: black;
    color: red;
}
```

```
<body class="mystyle">
    <div id="elmnt1" class="myotherstyle">
        Hello, world!
    </div>
</body>
```

What borders, colors, and font-sizes will the text be?

font-size:

> `font-size: 24;`, since `#elmnt1` rule set has the highest specificity for the `<div>` in question, every property here is set.

border:

> `border: 3px dotted red;`. The border-color `red` is taken from `#elmnt1` rule set, since it has the highest specificity. The other properties of the border, border-thickness, and border-style are from the `div` rule set.

background-color:

> `background-color: green;`. The `background-color` is set in the `div`, `body.mystyle > div.myotherstyle`, and `.mystyle .myotherstyle` rule sets. The specificities are (0, 0, 1) vs. (0, 2, 2) vs. (0, 2, 0), so the middle one "wins".

color:

> `color: red;`. The color is set in both the `div` and `.mystyle .myotherstyle` rule sets. The latter has the higher specificity of (0, 2, 0) and "wins".

Chapter 18: Colors

Section 18.1: currentColor

`currentColor` returns the computed color value of the current element.

Use in same element

Here currentColor evaluates to red since the `color` property is set to `red`:

```
div {
   color: red;
   border: 5px solid currentColor;
   box-shadow: 0 0 5px currentColor;
}
```

In this case, specifying currentColor for the border is most likely redundant because omitting it should produce identical results. Only use currentColor inside the border property within the same element if it would be overwritten otherwise due to a more specific selector.

Since it's the computed color, the border will be green in the following example due to the second rule overriding the first:

```
div {
   color: blue;
   border: 3px solid currentColor;
   color: green;
}
```

Inherited from parent element

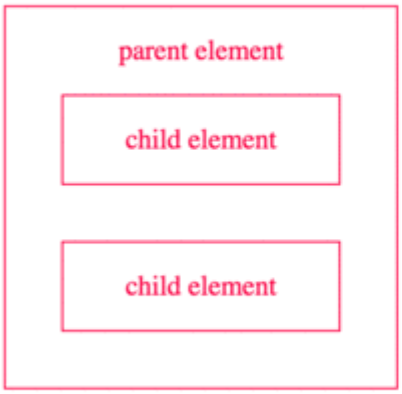
The parent's color is inherited, here currentColor evaluates to 'blue', making the child element's border-color blue.

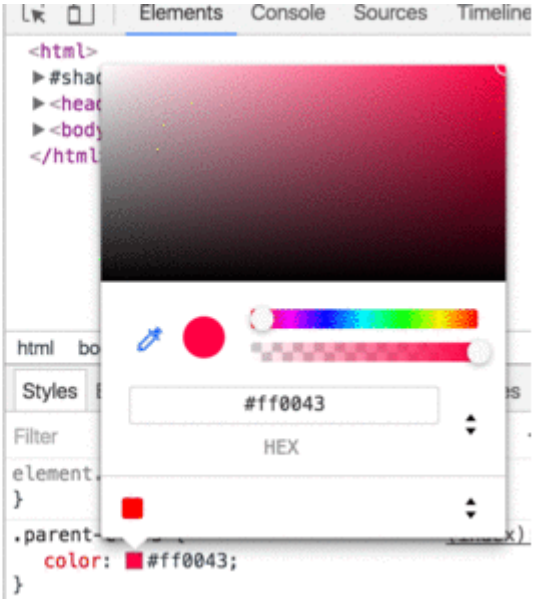
```
.parent-class {
    color: blue;
}

.parent-class .child-class {
    border-color: currentColor;
}
```

currentColor can also be used by other rules which normally would not inherit from the color property, such as background-color. The example below shows the children using the color set in the parent as its background:

```
.parent-class {
    color: blue;
}

.parent-class .child-class {
    background-color: currentColor;
}
```

Possible Result:





Section 18.2: Color Keywords

Most browsers support using color keywords to specify a color. For example, to set the `color` of an element to blue, use the `blue` keyword:

```
.some-class {
    color: blue;
}
```

CSS keywords are not case sensitive—`blue`, `Blue` and `BLUE` will all result in `#0000FF`.

Color Keywords

Color name	Hex value	RGB values	Color
AliceBlue	#F0F8FF	rgb(240,248,255)	
AntiqueWhite	#FAEBD7	rgb(250,235,215)	
Aqua	#00FFFF	rgb(0,255,255)	
Aquamarine	#7FFFD4	rgb(127,255,212)	
Azure	#F0FFFF	rgb(240,255,255)	
Beige	#F5F5DC	rgb(245,245,220)	
Bisque	#FFE4C4	rgb(255,228,196)	
Black	#000000	rgb(0,0,0)	
BlanchedAlmond	#FFEBCD	rgb(255,235,205)	
Blue	#0000FF	rgb(0,0,255)	
BlueViolet	#8A2BE2	rgb(138,43,226)	
Brown	#A52A2A	rgb(165,42,42)	

BurlyWood	#DEB887	rgb(222,184,135)
CadetBlue	#5F9EA0	rgb(95,158,160)
Chartreuse	#7FFF00	rgb(127,255,0)
Chocolate	#D2691E	rgb(210,105,30)
Coral	#FF7F50	rgb(255,127,80)
CornflowerBlue	#6495ED	rgb(100,149,237)
Cornsilk	#FFF8DC	rgb(255,248,220)
Crimson	#DC143C	rgb(220,20,60)
Cyan	#00FFFF	rgb(0,255,255)
DarkBlue	#00008B	rgb(0,0,139)
DarkCyan	#008B8B	rgb(0,139,139)
DarkGoldenRod	#B8860B	rgb(184,134,11)
DarkGray	#A9A9A9	rgb(169,169,169)
DarkGrey	#A9A9A9	rgb(169,169,169)
DarkGreen	#006400	rgb(0,100,0)
DarkKhaki	#BDB76B	rgb(189,183,107)
DarkMagenta	#8B008B	rgb(139,0,139)
DarkOliveGreen	#556B2F	rgb(85,107,47)
DarkOrange	#FF8C00	rgb(255,140,0)
DarkOrchid	#9932CC	rgb(153,50,204)
DarkRed	#8B0000	rgb(139,0,0)
DarkSalmon	#E9967A	rgb(233,150,122)
DarkSeaGreen	#8FBC8F	rgb(143,188,143)
DarkSlateBlue	#483D8B	rgb(72,61,139)
DarkSlateGray	#2F4F4F	rgb(47,79,79)
DarkSlateGrey	#2F4F4F	rgb(47,79,79)
DarkTurquoise	#00CED1	rgb(0,206,209)
DarkViolet	#9400D3	rgb(148,0,211)

DeepPink	#FF1493	rgb(255,20,147)
DeepSkyBlue	#00BFFF	rgb(0,191,255)
DimGray	#696969	rgb(105,105,105)
DimGrey	#696969	rgb(105,105,105)
DodgerBlue	#1E90FF	rgb(30,144,255)
FireBrick	#B22222	rgb(178,34,34)
FloralWhite	#FFFAF0	rgb(255,250,240)
ForestGreen	#228B22	rgb(34,139,34)
Fuchsia	#FF00FF	rgb(255,0,255)
Gainsboro	#DCDCDC	rgb(220,220,220)
GhostWhite	#F8F8FF	rgb(248,248,255)
Gold	#FFD700	rgb(255,215,0)
GoldenRod	#DAA520	rgb(218,165,32)
Gray	#808080	rgb(128,128,128)
Grey	#808080	rgb(128,128,128)
Green	#008000	rgb(0,128,0)
GreenYellow	#ADFF2F	rgb(173,255,47)
HoneyDew	#F0FFF0	rgb(240,255,240)
HotPink	#FF69B4	rgb(255,105,180)
IndianRed	#CD5C5C	rgb(205,92,92)
Indigo	#4B0082	rgb(75,0,130)
Ivory	#FFFFF0	rgb(255,255,240)
Khaki	#F0E68C	rgb(240,230,140)
Lavender	#E6E6FA	rgb(230,230,250)
LavenderBlush	#FFF0F5	rgb(255,240,245)
LawnGreen	#7CFC00	rgb(124,252,0)
LemonChiffon	#FFFACD	rgb(255,250,205)
LightBlue	#ADD8E6	rgb(173,216,230)

LightCoral	#F08080	rgb(240,128,128)
LightCyan	#E0FFFF	rgb(224,255,255)
LightGoldenRodYellow	#FAFAD2	rgb(250,250,210)
LightGray	#D3D3D3	rgb(211,211,211)
LightGrey	#D3D3D3	rgb(211,211,211)
LightGreen	#90EE90	rgb(144,238,144)
LightPink	#FFB6C1	rgb(255,182,193)
LightSalmon	#FFA07A	rgb(255,160,122)
LightSeaGreen	#20B2AA	rgb(32,178,170)
LightSkyBlue	#87CEFA	rgb(135,206,250)
LightSlateGray	#778899	rgb(119,136,153)
LightSlateGrey	#778899	rgb(119,136,153)
LightSteelBlue	#B0C4DE	rgb(176,196,222)
LightYellow	#FFFFE0	rgb(255,255,224)
Lime	#00FF00	rgb(0,255,0)
LimeGreen	#32CD32	rgb(50,205,50)
Linen	#FAF0E6	rgb(250,240,230)
Magenta	#FF00FF	rgb(255,0,255)
Maroon	#800000	rgb(128,0,0)
MediumAquaMarine	#66CDAA	rgb(102,205,170)
MediumBlue	#0000CD	rgb(0,0,205)
MediumOrchid	#BA55D3	rgb(186,85,211)
MediumPurple	#9370DB	rgb(147,112,219)
MediumSeaGreen	#3CB371	rgb(60,179,113)
MediumSlateBlue	#7B68EE	rgb(123,104,238)
MediumSpringGreen	#00FA9A	rgb(0,250,154)
MediumTurquoise	#48D1CC	rgb(72,209,204)
MediumVioletRed	#C71585	rgb(199,21,133)

MidnightBlue	#191970	rgb(25,25,112)
MintCream	#F5FFFA	rgb(245,255,250)
MistyRose	#FFE4E1	rgb(255,228,225)
Moccasin	#FFE4B5	rgb(255,228,181)
NavajoWhite	#FFDEAD	rgb(255,222,173)
Navy	#000080	rgb(0,0,128)
OldLace	#FDF5E6	rgb(253,245,230)
Olive	#808000	rgb(128,128,0)
OliveDrab	#6B8E23	rgb(107,142,35)
Orange	#FFA500	rgb(255,165,0)
OrangeRed	#FF4500	rgb(255,69,0)
Orchid	#DA70D6	rgb(218,112,214)
PaleGoldenRod	#EEE8AA	rgb(238,232,170)
PaleGreen	#98FB98	rgb(152,251,152)
PaleTurquoise	#AFEEEE	rgb(175,238,238)
PaleVioletRed	#DB7093	rgb(219,112,147)
PapayaWhip	#FFEFD5	rgb(255,239,213)
PeachPuff	#FFDAB9	rgb(255,218,185)
Peru	#CD853F	rgb(205,133,63)
Pink	#FFC0CB	rgb(255,192,203)
Plum	#DDA0DD	rgb(221,160,221)
PowderBlue	#B0E0E6	rgb(176,224,230)
Purple	#800080	rgb(128,0,128)
RebeccaPurple	#663399	rgb(102,51,153)
Red	#FF0000	rgb(255,0,0)
RosyBrown	#BC8F8F	rgb(188,143,143)
RoyalBlue	#4169E1	rgb(65,105,225)
SaddleBrown	#8B4513	rgb(139,69,19)

Salmon	#FA8072	rgb(250,128,114)
SandyBrown	#F4A460	rgb(244,164,96)
SeaGreen	#2E8B57	rgb(46,139,87)
SeaShell	#FFF5EE	rgb(255,245,238)
Sienna	#A0522D	rgb(160,82,45)
Silver	#C0C0C0	rgb(192,192,192)
SkyBlue	#87CEEB	rgb(135,206,235)
SlateBlue	#6A5ACD	rgb(106,90,205)
SlateGray	#708090	rgb(112,128,144)
SlateGrey	#708090	rgb(112,128,144)
Snow	#FFFAFA	rgb(255,250,250)
SpringGreen	#00FF7F	rgb(0,255,127)
SteelBlue	#4682B4	rgb(70,130,180)
Tan	#D2B48C	rgb(210,180,140)
Teal	#008080	rgb(0,128,128)
Thistle	#D8BFD8	rgb(216,191,216)
Tomato	#FF6347	rgb(255,99,71)
Turquoise	#40E0D0	rgb(64,224,208)
Violet	#EE82EE	rgb(238,130,238)
Wheat	#F5DEB3	rgb(245,222,179)
White	#FFFFFF	rgb(255,255,255)
WhiteSmoke	#F5F5F5	rgb(245,245,245)
Yellow	#FFFF00	rgb(255,255,0)
YellowGreen	#9ACD32	rgb(154,205,50)

In addition to the named colors, there is also the keyword `transparent`, which represents a fully-transparent black: `rgba(0,0,0,0)`

Section 18.3: Hexadecimal Value

Background

CSS colors may also be represented as a hex triplet, where the members represent the red, green and blue components of a color. Each of these values represents a number in the range of `00` to `FF`, or `0` to `255` in decimal notation. Uppercase and/or lowercase Hexadecimal values may be used (i.e. `#3fc` = `#3FC` = `#33ffCC`). The browser interprets `#369` as `#336699`. If that is not what you intended but rather wanted `#306090`, you need to specify that explicitly.

The total number of colors that can be represented with hex notation is 256 ^ 3 or 16,777,216.

Syntax

```
color: #rrggbb;
color: #rgb
```

Value	Description
`rr`	`00` - `FF` for the amount of red
`gg`	`00` - `FF` for the amount of green
`bb`	`00` - `FF` for the amount of blue

```
.some-class {
    /* This is equivalent to using the color keyword 'blue' */
    color: #0000FF;
}

.also-blue {
    /* If you want to specify each range value with a single number, you can!
       This is equivalent to '#0000FF' (and 'blue') */
    color: #00F;
}
```

Hexadecimal notation is used to specify color values in the RGB color format, per the W3C's 'Numerical color values'.

There are a lot of tools available on the Internet for looking up hexadecimal (or simply hex) color values.

Search for "**hex color palette**" or "**hex color picker**" with your favorite web browser to find a bunch of options!

Hex values always start with a pound sign (#), are up to six "digits" long, and are case-insensitive: that is, they don't care about capitalization. `#FFC125` and `#ffc125` are the same color.

Section 18.4: rgb() Notation

RGB is an additive color model which represents colors as mixtures of red, green, and blue light. In essence, the RGB representation is the decimal equivalent of the Hexadecimal Notation. In Hexadecimal each number ranges from 00-FF which is equivalent to 0-255 in decimal and 0%-100% in percentages.

```
.some-class {
    /* Scalar RGB, equivalent to 'blue'*/
    color: rgb(0, 0, 255);
}

.also-blue {
    /* Percentile RGB values*/
    color: rgb(0%, 0%, 100%);
}
```

Syntax

```
rgb(<red>, <green>, <blue>)
```

Value	Description
`<red>`	an integer from 0 - 255 or percentage from 0 - 100%
`<green>`	an integer from 0 - 255 or percentage from 0 - 100%
`<blue>`	an integer from 0 - 255 or percentage from 0 - 100%

Section 18.5: rgba() Notation

Similar to rgb() notation, but with an additional alpha (opacity) value.

```
.red {
    /* Opaque red */
    color: rgba(255, 0, 0, 1);
}

.red-50p {
    /* Half-translucent red. */
    color: rgba(255, 0, 0, .5);
}
```

Syntax

```
rgba(<red>, <green>, <blue>, <alpha>);
```

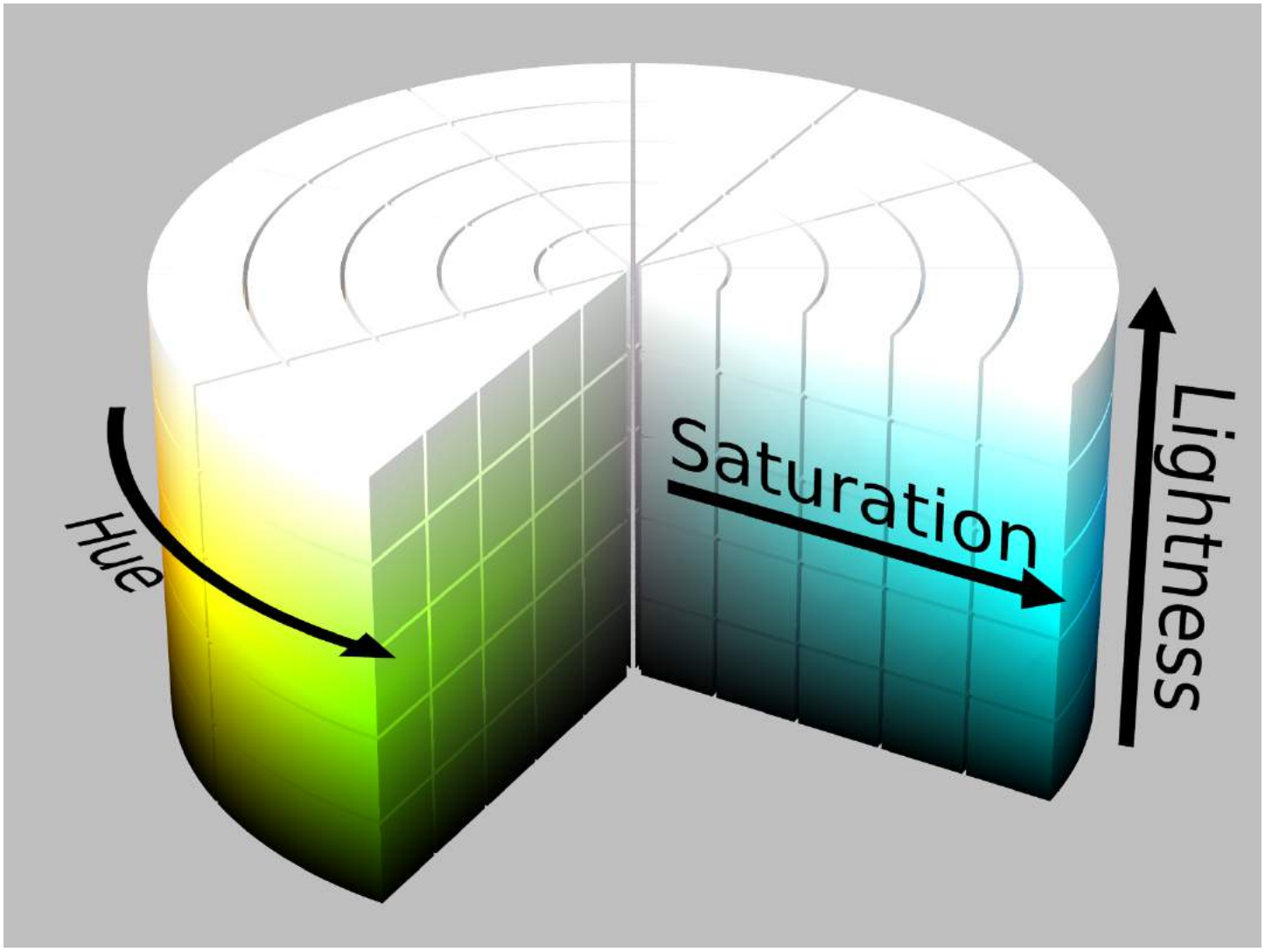
Value	Description
`<red>`	an integer from 0 - 255 or percentage from 0 - 100%
`<green>`	an integer from 0 - 255 or percentage from 0 - 100%
`<blue>`	an integer from 0 - 255 or percentage from 0 - 100%
`<alpha>`	a number from 0 - 1, where 0.0 is fully transparent and 1.0 is fully opaque

Section 18.6: hsl() Notation

HSL stands for **hue** ("which color"), **saturation** ("how much color") and **lightness** ("how much white").

Hue is represented as an angle from 0° to 360° (without units), while saturation and lightness are represented as percentages.

```
p {
    color: hsl(240, 100%, 50%); /* Blue */
}
```



Syntax

```
color: hsl(<hue>, <saturation>%, <lightness>%);
```

Value	Description
`<hue>`	specified in degrees around the color wheel (without units), where 0° is red, 60° is yellow, 120° is green, 180° is cyan, 240° is blue, 300° is magenta, and 360° is red
`<saturation>`	specified in percentage where 0% is fully desaturated (grayscale) and 100% is fully saturated (vividly colored)
`<lightness>`	specified in percentage where 0% is fully black and 100% is fully white

Notes

- A saturation of 0% always produces a grayscale color; changing the hue has no effect.
- A lightness of 0% always produces black, and 100% always produces white; changing the hue or saturation has no effect.

Section 18.7: hsla() Notation

Similar to hsl() notation, but with an added alpha (opacity) value.

```
hsla(240, 100%, 50%, 0)     /* transparent */
hsla(240, 100%, 50%, 0.5)   /* half-translucent blue */
hsla(240, 100%, 50%, 1)     /* fully opaque blue */
```

Syntax

```
hsla(<hue>, <saturation>%, <lightness>%, <alpha>);
```

Value	Description
`<hue>`	specified in degrees around the color wheel (without units), where 0° is red, 60° is yellow, 120° is green, 180° is cyan, 240° is blue, 300° is magenta, and 360° is red
`<saturation>`	percentage where 0% is fully desaturated (grayscale) and 100% is fully saturated (vividly colored)
`<lightness>`	percentage where 0% is fully black and 100% is fully white
`<alpha>`	a number from 0 - 1 where 0 is fully transparent and 1 is fully opaque

Chapter 19: Opacity

Section 19.1: Opacity Property

An element's opacity can be set using the `opacity` property. Values can be anywhere from `0.0` (transparent) to `1.0` (opaque).

Example Usage

```
<div style="opacity:0.8;">
    This is a partially transparent element
</div>
```

Property Value	Transparency
`opacity: 1.0;`	Opaque
`opacity: 0.75;`	25% transparent (75% Opaque)
`opacity: 0.5;`	50% transparent (50% Opaque)
`opacity: 0.25;`	75% transparent (25% Opaque)
`opacity: 0.0;`	Transparent

Section 19.2: IE Compatibility for `opacity`

To use `opacity` in all versions of IE, the order is:

```
.transparent-element {
  /* for IE 8 & 9 */
  -ms-filter:"progid:DXImageTransform.Microsoft.Alpha(Opacity=60)"; // IE8
  /* works in IE 8 & 9 too, but also 5, 6, 7 */
  filter: alpha(opacity=60); // IE 5-7
  /* Modern Browsers */
  opacity: 0.6;
}
```

Chapter 20: Length Units

Unit	Description
%	Define sizes in terms of parent objects or current object dependent on property
em	Relative to the font-size of the element (2em means 2 times the size of the current font)
rem	Relative to font-size of the root element
vw	Relative to 1% of the width of the viewport*
vh	Relative to 1% of the height of the viewport*
vmin	Relative to 1% of viewport's* smaller dimension
vmax	Relative to 1% of viewport's* larger dimension
cm	centimeters
mm	millimeters
in	inches (1in = 96px = 2.54cm)
px	pixels (1px = 1/96th of 1in)
pt	points (1pt = 1/72 of 1in)
pc	picas (1pc = 12 pt)
s	seconds (used for animations and transitions)
ms	milliseconds (used for animations and transitions)
ex	Relative to the x-height of the current font
ch	Based on the width of the zero (0) character
fr	fractional unit (used for CSS Grid Layout)

A CSS distance measurement is a number immediately followed by a length unit (px, em, pc, in, …)

CSS supports a number of length measurements units. They are absolute or relative.

Section 20.1: Creating scalable elements using rems and ems

Version ≥ 3

You can use `rem` defined by the `font-size` of your `html` tag to style elements by setting their `font-size` to a value of `rem` and use `em` inside the element to create elements that scale with your global `font-size`.

HTML:

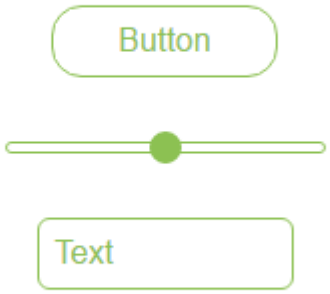
```
<input type="button" value="Button">
<input type="range">
<input type="text" value="Text">
```

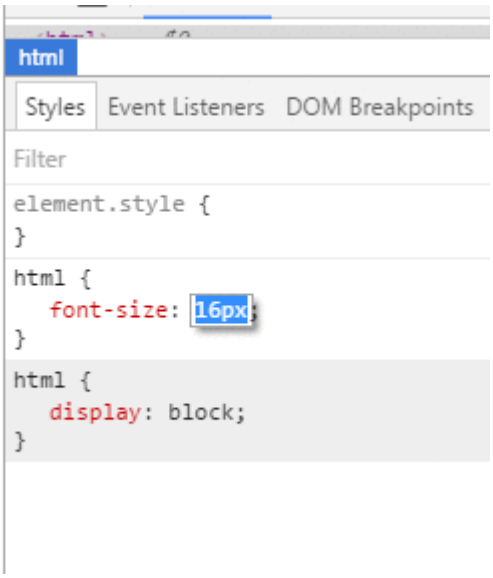
Relevant CSS:

```
html {
  font-size: 16px;
}

input[type="button"] {
  font-size: 1rem;
  padding: 0.5em 2em;
}

input[type="range"] {
  font-size: 1rem;
```

```
  width: 10em;
}

input[type=text] {
  font-size: 1rem;
  padding: 0.5em;
}
```

Possible Result:





Section 20.2: Font size with rem

CSS3 introduces a few new units, including the `rem` unit, which stands for "root em". Let's look at how `rem` works.

First, let's look at the differences between `em` and `rem`.

- **em**: Relative to the font size of the parent. This causes the compounding issue
- **rem**: Relative to the font size of the root or **`<html>`** element. This means it's possible to declare a single font size for the html element and define all `rem` units to be a percentage of that.

The main issue with using `rem` for font sizing is that the values are somewhat difficult to use. Here is an example of some common font sizes expressed in `rem` units, assuming that the base size is 16px :

- 10px = 0.625rem
- 12px = 0.75rem
- 14px = 0.875rem
- 16px = 1rem (base)
- 18px = 1.125rem
- 20px = 1.25rem
- 24px = 1.5rem
- 30px = 1.875rem
- 32px = 2rem

CODE:

Version ≥ 3

```
html {
  font-size: 16px;
}
```

```
h1 {
  font-size: 2rem;          /* 32px */
}

p {
  font-size: 1rem;          /* 16px */
}

li {
  font-size: 1.5em;         /* 24px */
}
```

Section 20.3: vmin and vmax

- **vmin**: Relative to 1 percent of the viewport's smaller dimension
- **vmax**: Relative to 1 percent of the viewport's larger dimension

In other words, 1 `vmin` is equal to the smaller of *1 vh* and *1 vw*

1 `vmax` is equal to the larger of *1 vh* and *1 vw*

Note: `vmax` is not supported in:

- any version of Internet Explorer
- Safari before version 6.1

Section 20.4: vh and vw

CSS3 introduced two units for representing size.

- `vh`, which stands for `viewport height` is relative to 1% of the viewport height
- `vw`, which stands for `viewport width` is relative to 1% of the viewport width

Version ≥ 3

```
div {
  width: 20vw;
  height: 20vh;
}
```

Above, the size for the div takes up 20% of the width and height of the viewport

Section 20.5: using percent %

One of the useful unit when creating a responsive application.

Its size depends on its parent container.

Equation:

> (Parent Container`s width) * (Percentage(%)) = Output

For Example:

Parent has **100px** width while the *Child* has **50%**.

On the output, the *Child*'s width will be half(50%) of the *Parent*'s, which is **50px**.

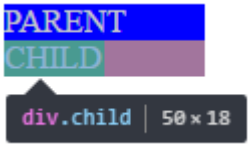
HTML

```
<div class="parent">
   PARENT
   <div class="child">
     CHILD
   </div>
</div>
```

CSS

```
<style>

*{
  color: #CCC;
}

.parent{
  background-color: blue;
  width: 100px;
}

.child{
  background-color: green;
  width: 50%;
}

</style>
```

OUTPUT



Chapter 21: Pseudo-Elements

pseudo-element	Description
`::after`	Insert content after the content of an element
`::before`	Insert content before the content of an element
`::first-letter`	Selects the first letter of each element
`::first-line`	Selects the first line of each element
`::selection`	Matches the portion of an element that is selected by a user
`::backdrop`	Used to create a backdrop that hides the underlying document for an element in the top layer's stack
`::placeholder`	Allows you to style the placeholder text of a form element (Experimental)
`::marker`	For applying list-style attributes on a given element (Experimental)
`::spelling-error`	Represents a text segment which the browser has flagged as incorrectly spelled (Experimental)
`::grammar-error`	Represents a text segment which the browser has flagged as grammatically incorrect (Experimental)

Pseudo-elements, just like pseudo-classes, are added to a CSS selectors but instead of describing a special state, they allow you to scope and style certain parts of an html element.

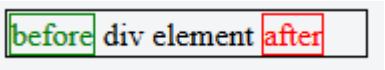
For example, the ::first-letter pseudo-element targets only the first letter of a block element specified by the selector.

Section 21.1: Pseudo-Elements

Pseudo-elements are added to selectors but instead of describing a special state, they allow you to style certain parts of a document.

The `content` attribute is required for pseudo-elements to render; however, the attribute can have an empty value (e.g. `content: ""`).

```
div::after {
  content: 'after';
  color: red;
  border: 1px solid red;
}

div {
  color: black;
  border: 1px solid black;
  padding: 1px;
}

div::before {
  content: 'before';
  color: green;
  border: 1px solid green;
}
```



Section 21.2: Pseudo-Elements in Lists

Pseudo-elements are often used to change the look of lists (mostly for unordered lists, `ul`).

The first step is to remove the default list bullets:

```
ul {
  list-style-type: none;
}
```

Then you add the custom styling. In this example, we will create gradient boxes for bullets.

```
li:before {
  content: "";
  display: inline-block;
  margin-right: 10px;
  height: 10px;
  width: 10px;
  background: linear-gradient(red, blue);
}
```

HTML

```
<ul>
   <li>Test I</li>
   <li>Test II</li>
</ul>
```

Result

Chapter 22: Positioning

Parameter	Details
static	Default value. Elements render in order, as they appear in the document flow. The top, right, bottom, left and z-index properties do not apply.
relative	The element is positioned relative to its normal position, so `left:20px` adds 20 pixels to the element's LEFT position
fixed	The element is positioned relative to the browser window
absolute	The element is positioned relative to its first positioned (not static) ancestor element
initial	Sets this property to its default value.
inherit	Inherits this property from its parent element.
sticky	Experimental feature. It behaves like `position: static` within its parent until a given offset threshold is reached, then it acts as `position: fixed`.
unset	Combination of initial and inherit. More info here.

Section 22.1: Overlapping Elements with z-index

To change the default stack order positioned elements (`position` property set to `relative`, `absolute` or `fixed`), use the `z-index` property.

The higher the z-index, the higher up in the stacking context (on the z-axis) it is placed.

Example

In the example below, a z-index value of 3 puts green on top, a z-index of 2 puts red just under it, and a z-index of 1 puts blue under that.

HTML

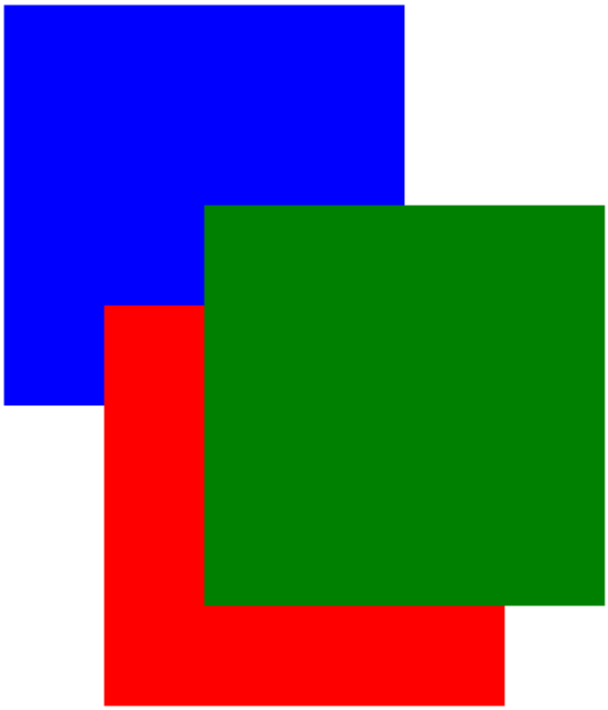
```
<div id="div1"></div>
<div id="div2"></div>
<div id="div3"></div>
```

CSS

```
div {
    position: absolute;
    height: 200px;
    width: 200px;
}
div#div1 {
    z-index: 1;
    left: 0px;
    top: 0px;
    background-color: blue;
}
div#div2 {
    z-index: 3;
    left: 100px;
    top: 100px;
    background-color: green;
}
div#div3 {
    z-index: 2;
    left: 50px;
```

```
    top: 150px;
    background-color: red;
}
```

This creates the following effect:



See a working example at JSFiddle.

Syntax

```
z-index: [ number ] | auto;
```

Parameter	Details
`number`	An integer value. A higher number is higher on the `z-index` stack. `0` is the default value. Negative values are allowed.
`auto`	Gives the element the same stacking context as its parent. (**Default**)

Remarks

All elements are laid out in a 3D axis in CSS, including a depth axis, measured by the `z-index` property. `z-index` only works on positioned elements: (see: Why does z-index need a defined position to work?). The only value where it is ignored is the default value, `static`.

Read about the z-index property and Stacking Contexts in the CSS Specification on layered presentation and at the Mozilla Developer Network.

Section 22.2: Absolute Position

When absolute positioning is used the box of the desired element is taken out of the *Normal Flow* and it no longer affects the position of the other elements on the page. Offset properties:

1. top
2. left
3. right

4. bottom

specify the element should appear in relation to its next non-static containing element.

```
.abspos{
    position:absolute;
    top:0px;
    left:500px;
}
```

This code will move the box containing element with attribute `class="abspos"` down 0px and right 500px relative to its containing element.

Section 22.3: Fixed position

Defining position as fixed we can remove an element from the document flow and set its position relatively to the browser window. One obvious use is when we want something to be visible when we scroll to the bottom of a long page.

```
#stickyDiv {
    position:fixed;
    top:10px;
    left:10px;
}
```

Section 22.4: Relative Position

Relative positioning moves the element in relation to where it would have been in *normal flow* .Offset properties:

1. top
2. left
3. right
4. bottom

are used to indicate how far to move the element from where it would have been in normal flow.

```
.relpos{
    position:relative;
    top:20px;
    left:30px;
}
```

This code will move the box containing element with attribute class="relpos" 20px down and 30px to the right from where it would have been in normal flow.

Section 22.5: Static positioning

The default position of an element is `static`. To quote MDN:

> This keyword lets the element use the normal behavior, that is it is laid out in its current position in the flow. The top, right, bottom, left and z-index properties do not apply.

```
.element{
  position:static;
```

```
}
```

Chapter 23: Layout Control

Value	Effect
`none`	Hide the element and prevent it from occupying space.
`block`	Block element, occupy 100% of the available width, break after element.
`inline`	Inline element, occupy no width, no break after element.
`inline-block`	Taking special properties from both inline and block elements, no break, but can have width.
`inline-flex`	Displays an element as an inline-level flex container.
`inline-table`	The element is displayed as an inline-level table.
`grid`	Behaves like a block element and lays out its content according to the grid model.
`flex`	Behaves like a block element and lays out its content according to the flexbox model.
`inherit`	Inherit the value from the parent element.
`initial`	Reset the value to the default value taken from behaviors described in the HTML specifications or from the browser/user default stylesheet.
`table`	Behaves like the HTML `table` element.
`table-cell`	Let the element behave like a **`<td>`** element
`table-column`	Let the element behave like a **`<col>`** element
`table-row`	Let the element behave like a **`<tr>`** element
`list-item`	Let the element behave like a **`<li>`** element.

Section 23.1: The display property

The `display` CSS property is fundamental for controlling the layout and flow of an HTML document. Most elements have a default `display` value of either `block` or `inline` (though some elements have other default values).

Inline

An `inline` element occupies only as much width as necessary. It stacks horizontally with other elements of the same type and may not contain other non-inline elements.

```
<span>This is some <b>bolded</b> text!</span>
```

This is some **bolded** text!

As demonstrated above, two `inline` elements, **`<span>`** and **`<b>`**, are in-line (hence the name) and do not break the flow of the text.

Block

A `block` element occupies the maximum available width of its' parent element. It starts with a new line and, in contrast to `inline` elements, it does not restrict the type of elements it may contain.

```
<div>Hello world!</div><div>This is an example!</div>
```

Hello world!
This is an example!

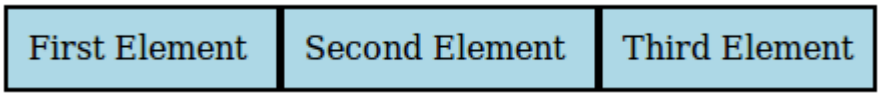
The `div` element is block-level by default, and as shown above, the two `block` elements are vertically stacked and, unlike the `inline` elements, the flow of the text breaks.

Inline Block

The `inline-block` value gives us the best of both worlds: it blends the element in with the flow of the text while allowing us to use `padding`, `margin`, `height` and similar properties which have no visible effect on `inline` elements.

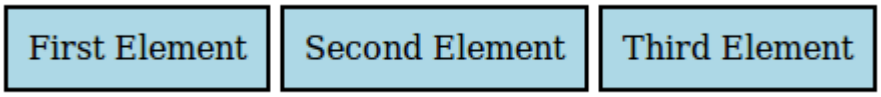
Elements with this display value act as if they were regular text and as a result are affected by rules controlling the flow of text such as `text-align`. By default they are also shrunk to the the smallest size possible to accommodate their content.

```
<!--Inline: unordered list-->
<style>
li {
    display : inline;
    background : lightblue;
    padding:10px;

    border-width:2px;
    border-color:black;
    border-style:solid;
    }
</style>

<ul>
<li>First Element </li>
<li>Second Element </li>
<li>Third Element </li>
</ul>
```




```
<!--block: unordered list-->
<style>
li {
    display : block;
    background : lightblue;
    padding:10px;

    border-width:2px;
    border-color:black;
    border-style:solid;
    }
</style>

<ul>
<li>First Element </li>
<li>Second Element </li>
<li>Third Element </li>
</ul>
```

```
<!--Inline-block: unordered list-->
<style>
li {
    display : inline-block;
    background : lightblue;
    padding:10px;

    border-width:2px;
    border-color:black;
    border-style:solid;
    }

</style>

<ul>
<li>First Element </li>
<li>Second Element </li>
<li>Third Element </li>
</ul>
```



none

An element that is given the none value to its display property will not be displayed at all.

For example let's create a div-element that has an id of `myDiv`:

```
<div id="myDiv"></div>
```

This can now be marked as not being displayed by the following CSS rule:

```
#myDiv {
    display: none;
}
```

When an element has been set to be `display:none;` the browser ignores every other layout property for that specific element (both `position` and `float`). No box will be rendered for that element and its existence in html does not affect the position of following elements.

Note that this is different from setting the `visibility` property to `hidden`. Setting `visibility: hidden;` for an element would not display the element on the page but the element would still take up the space in the rendering process as if it would be visible. This will therefore affect how following elements are displayed on the page.

The `none` value for the display property is commonly used along with JavaScript to show or hide elements at will, eliminating the need to actually delete and re-create them.

Section 23.2: To get old table structure using div

This is the normal HTML table structure

```
<style>
    table {
        width: 100%;
    }
```

```
</style>

<table>
  <tr>
    <td>
        I'm a table
    </td>
  </tr>
</table>
```

You can do same implementation like this

```
<style>
    .table-div {
        display: table;
    }
    .table-row-div {
        display: table-row;
    }
    .table-cell-div {
        display: table-cell;
    }
</style>

<div class="table-div">
  <div class="table-row-div">
    <div class="table-cell-div">
      I behave like a table now
    </div>
  </div>
</div>
```

Chapter 24: Grid

Grid layout is a new and powerful CSS layout system that allows to divide a web page content into rows and columns in an easy way.

Section 24.1: Basic Example

Property	Possible Values
display	grid / inline-grid

The CSS Grid is defined as a display property. It applies to a parent element and its immediate children only.

Consider the following markup:

```
<section class="container">
  <div class="item1">item1</div>
  <div class="item2">item2</div>
  <div class="item3">item3</div>
  <div class="item4">item4</div>
</section>
```

The easiest way to define the markup structure above as a grid is to simply set its `display` property to `grid`:

```
.container {
  display: grid;
}
```

However, doing this will invariably cause all the child elements to collapse on top of one another. This is because the children do not currently know how to position themselves within the grid. But we can explicitly tell them.

First we need to tell the grid element `.container` how many rows and columns will make up its structure and we can do this using the `grid-columns` and `grid-rows` properties (note the pluralisation):

```
.container {
  display: grid;
  grid-columns: 50px 50px 50px;
  grid-rows: 50px 50px;
}
```

However, that still doesn't help us much because we need to give an order to each child element. We can do this by specifying the `grid-row` and `grid-column` values which will tell it where it sits in the grid:

```
.container .item1 {
  grid-column: 1;
  grid-row: 1;
}
.container .item2 {
  grid-column: 2;
  grid-row: 1;
}
.container .item3 {
  grid-column: 1;
  grid-row: 2;
}
.container .item4 {
  grid-column: 2;
```

```
  grid-row: 2;
}
```

By giving each item a column and row value it identifies the items order within the container.

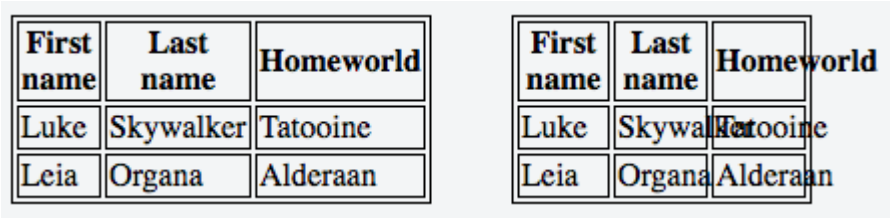
View a working example on JSFiddle. You'll need to view this in IE10, IE11 or Edge for it to work as these are currently the only browsers supporting Grid Layout (with vendor prefix `-ms-`) or enable a flag in Chrome, Opera and Firefox according to caniuse in order to test with them.

Chapter 25: Tables

Section 25.1: table-layout

The `table-layout` property changes the algorithm that is used for the layout of a table.

Below an example of two tables both set to `width: 150px`:



First name	Last name	Homeworld
Luke	Skywalker	Tatooine
Leia	Organa	Alderaan

First name	Last name	Homeworld
Luke	Skywalker	Tatooine
Leia	Organa	Alderaan

The table on the left has `table-layout: auto` while the one on the right has `table-layout: fixed`. The former is wider than the specified width (210px instead of 150px) but the contents fit. The latter takes the defined width of 150px, regardless if the contents overflow or not.

Value	Description
auto	This is the default value. It defines the layout of the table to be determined by the contents of its' cells.
fixed	This value sets the table layout to be determined by the width property provided to the table. If the content of a cell exceeds this width, the cell will not resize but instead, let the content overflow.

Section 25.2: empty-cells

The `empty-cells` property determines if cells with no content should be displayed or not. This has no effect unless `border-collapse` is set to `separate`.

Below an example with two tables with different values set to the `empty-cells` property:

First name	Last name	Homeworld
Luke		Tatooine
Leia	Organa	

First name	Last name	Homeworld
Luke		Tatooine
Leia	Organa	

The table on the left has `empty-cells: show` while the one on the right has `empty-cells: hide`. The former does display the empty cells whereas the latter does not.

Value	Description
show	This is the default value. It shows cells even if they are empty.
hide	This value hides a cell altogether if there are no contents in the cell.

More Information:

- https://www.w3.org/TR/CSS21/tables.html#empty-cells
- https://developer.mozilla.org/en-US/docs/Web/CSS/empty-cells
- http://codepen.io/SitePoint/pen/yfhtq
- https://css-tricks.com/almanac/properties/e/empty-cells/

Section 25.3: border-collapse

The `border-collapse` property determines if a tables' borders should be separated or merged.

Below an example of two tables with different values to the `border-collapse` property:

First name	Last name	Homeworld
Luke	Skywalker	Tatooine
Leia	Organa	Alderaan

First name	Last name	Homeworld
Luke	Skywalker	Tatooine
Leia	Organa	Alderaan

The table on the left has **`border-collapse`**: `separate` while the one on the right has **`border-collapse`**: `collapse`.

Value	Description
separate	This is the default value. It makes the borders of the table separate from each other.
collapse	This value sets the borders of the table to merge together, rather than being distinct.

Section 25.4: border-spacing

The `border-spacing` property determines the spacing between cells. This has no effect unless `border-collapse` is set to `separate`.

Below an example of two tables with different values to the `border-spacing` property:

First name	Last name	Homeworld
Luke	Skywalker	Tatooine
Leia	Organa	Alderaan

First name	Last name	Homeworld
Luke	Skywalker	Tatooine
Leia	Organa	Alderaan

The table on the left has **`border-spacing`**: `2px` (default) while the one on the right has **`border-spacing`**: `8px`.

Value	Description
<length>	This is the default behavior, though the exact value can vary between browsers.
<length> <length>	This syntax allows specifying separate horizontal and vertical values respectively.

Section 25.5: caption-side

The `caption-side` property determines the vertical positioning of the **`<caption>`** element within a table. This has no effect if such element does not exist.

Below an example with two tables with different values set to the `caption-side` property:

Star Wars figures

First name	Last name	Homeworld
Luke	Skywalker	Tatooine
Leia	Organa	Alderaan

First name	Last name	Homeworld
Luke	Skywalker	Tatooine
Leia	Organa	Alderaan

Star Wars figures

The table on the left has **`caption-side`**: `top` while the one on the right has **`caption-side`**: `bottom`.

Value	Description
top	This is the default value. It places the caption above the table.
bottom	This value places the caption below the table.

Chapter 26: Transitions

Parameter	Details
transition-property	The specific CSS property whose value change needs to be transitioned (or) `all`, if all the transitionable properties need to be transitioned.
transition-duration	The duration (or period) in seconds (`s`) or milliseconds (`ms`) over which the transition must take place.
transition-timing-function	A function that describes how the intermediate values during the transition are calculated. Commonly used values are `ease`, `ease-in`, `ease-out`, `ease-in-out`, `linear`, `cubic-bezier()`, `steps()`. More information about the various timing functions can be found in the W3C specs.
transition-delay	The amount of time that must have elapsed before the transition can start. Can be specified in seconds (`s`) or milliseconds (`ms`)

Section 26.1: Transition shorthand

CSS

```
div{
    width: 150px;
    height:150px;
    background-color: red;
    transition: background-color 1s;
}
div:hover{
    background-color: green;
}
```

HTML

```
<div></div>
```

This example will change the background color when the div is hovered the background-color change will last 1 second.

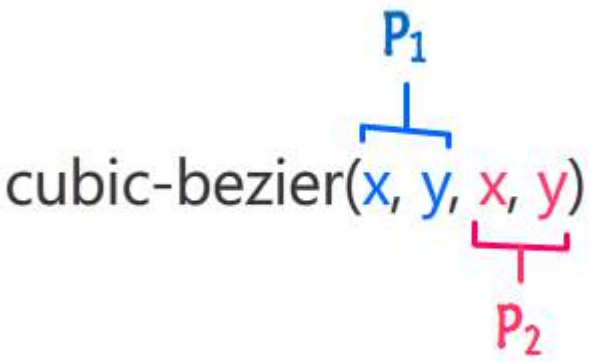
Section 26.2: cubic-bezier

The `cubic-bezier` function is a transition timing function which is often used for custom and smooth transitions.

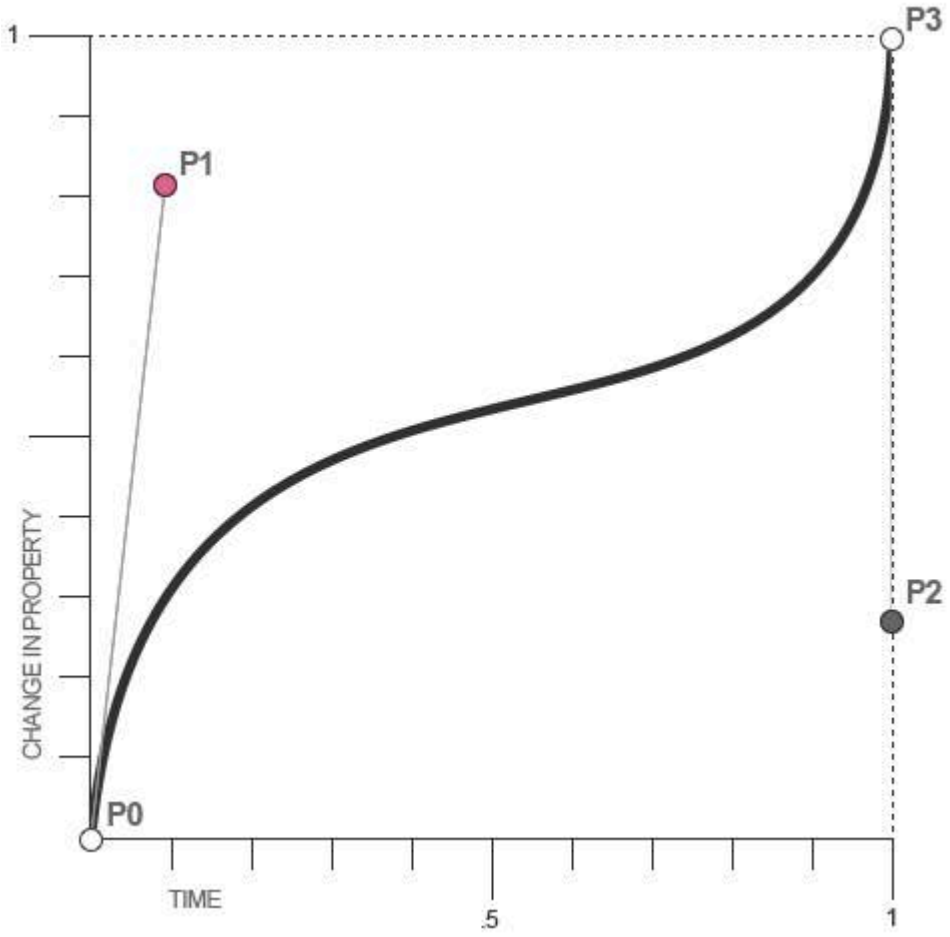
```
transition-timing-function: cubic-bezier(0.1, 0.7, 1.0, 0.1);
```

The function takes four parameters:

`cubic-bezier(P1_x, P1_y, P2_x, P2_y)`



These parameters will be mapped to points which are part of a Bézier curve:



For CSS Bézier Curves, P0 and P3 are always in the same spot. P0 is at (0,0) and P3 is at (1,1), which menas that the parameters passed to the cubic-bezier function can only be between 0 and 1.

If you pass parameters which aren't in this interval the function will default to a `linear` transition.

Since cubic-bezier is the most flexible transition in CSS, you can translate all other transition timing function to cubic-bezier functions:

`linear`: `cubic-bezier(0,0,1,1)`

`ease-in`: `cubic-bezier(0.42, 0.0, 1.0, 1.0)`

`ease-out`: `cubic-bezier(0.0, 0.0, 0.58, 1.0)`

`ease-in-out`: `cubic-bezier(0.42, 0.0, 0.58, 1.0)`

Section 26.3: Transition (longhand)

CSS

```
div {
  height: 100px;
  width: 100px;
  border: 1px solid;
  transition-property: height, width;
  transition-duration: 1s, 500ms;
  transition-timing-function: linear;
  transition-delay: 0s, 1s;
}
div:hover {
  height: 200px;
  width: 200px;
}
```

HTML

```
<div></div>
```

- **transition-property**: Specifies the CSS properties the transition effect is for. In this case, the div will expand both horizontally and vertically when hovered.
- **transition-duration**: Specifies the length of time a transition takes to complete. In the above example, the height and width transitions will take 1 second and 500 milliseconds respectively.
- **transition-timing-function**: Specifies the speed curve of the transition effect. A *linear* value indicates the transition will have the same speed from start to finish.
- **transition-delay**: Specifies the amount of time needed to wait before the transition effect starts. In this case, the height will start transitioning immediately, whereas the width will wait 1 second.

Chapter 27: Animations

Transition

Parameter	Details
property	Either the CSS property to transition on, or `all`, which specifies all transition-able properties.
duration	Transition time, either in seconds or milliseconds.
timing-function	Specifies a function to define how intermediate values for properties are computed. Common values are `ease`, `linear`, and `step-end`. Check out the easing function cheat-sheet for more.
delay	Amount of time, in seconds or milliseconds, to wait before playing the animation.

@keyframes

[from | to | `<percentage>`]	You can either specify a set time with a percentage value, or two percentage values, ie `10%, 20%`, for a period of time where the keyframe's set attributes are set.
`block`	Any amount of CSS attributes for the keyframe.

Section 27.1: Animations with keyframes

For multi-stage CSS animations, you can create CSS `@keyframes`. Keyframes allow you to define multiple animation points, called a keyframe, to define more complex animations.

Basic Example

In this example, we'll make a basic background animation that cycles between all colors.

```
@keyframes rainbow-background {
         0%      { background-color: #ff0000; }
     8.333%      { background-color: #ff8000; }
    16.667%      { background-color: #ffff00; }
    25.000%      { background-color: #80ff00; }
    33.333%      { background-color: #00ff00; }
    41.667%      { background-color: #00ff80; }
    50.000%      { background-color: #00ffff; }
    58.333%      { background-color: #0080ff; }
    66.667%      { background-color: #0000ff; }
    75.000%      { background-color: #8000ff; }
    83.333%      { background-color: #ff00ff; }
    91.667%      { background-color: #ff0080; }
    100.00%      { background-color: #ff0000; }
}

.RainbowBackground {
    animation: rainbow-background 5s infinite;
}
```

View Result

There's a few different things to note here. First, the actual `@keyframes` syntax.

```
@keyframes rainbow-background{
```

This sets the name of the animation to `rainbow-background`.

```
0%      { background-color: #ff0000; }
```

This is the definition for a keyframe within the animation. The first part, the `0%` in the case, defines where the keyframe is during the animation. The `0%` implies it is 0% of the total animation time from the beginning.

The animation will automatically transition between keyframes. So, by setting the next background color at `8.333%`, the animation will smoothly take 8.333% of the time to transition between those keyframes.

```
.RainbowBackground {
    animation: rainbow-background 5s infinite;
}
```

This code attaches our animation to all elements which have the `.RainbowBackground` class.

The actual animation property takes the following arguments.

- **animation-name**: The name of our animation. In this case, `rainbow-background`
- **animation-duration**: How long the animation will take, in this case 5 seconds.
- **animation-iteration-count (Optional)**: The number of times the animation will loop. In this case, the animation will go on indefinitely. By default, the animation will play once.
- **animation-delay (Optional)**: Specifies how long to wait before the animation starts. It defaults to 0 seconds, and can take negative values. For example, `-2s` would start the animation 2 seconds into its loop.
- **animation-timing-function (Optional)**: Specifies the speed curve of the animation. It defaults to `ease`, where the animation starts slow, gets faster and ends slow.

In this particular example, both the `0%` and `100%` keyframes specify `{ background-color: #ff0000; }`. Wherever two or more keyframes share a state, one may specify them in a single statement. In this case, the two `0%` and `100%` lines could be replaced with this single line:

```
0%, 100%      { background-color: #ff0000; }
```

Cross-browser compatibility

For older WebKit-based browsers, you'll need to use the vendor prefix on both the `@keyframes` declaration and the `animation` property, like so:

```
@-webkit-keyframes{}

-webkit-animation: ...
```

Section 27.2: Animations with the transition property

Useful for simple animations, the CSS `transition` property allows number-based CSS properties to animate between states.

Example

```
.Example{
    height: 100px;
    background: #fff;
}

.Example:hover{
    height: 120px;
```

```
    background: #ff0000;
}
```

View Result

By default, hovering over an element with the `.Example` class would immediately cause the element's height to jump to `120px` and its background color to red (`#ff0000`).

By adding the `transition` property, we can cause these changes to occur over time:

```
.Example{
    ...
    transition: all 400ms ease;
}
```

View Result

The `all` value applies the transition to all compatible (numbers-based) properties. Any compatible property name (such as `height` or `top`) can be substituted for this keyword.

`400ms` specifies the amount of time the transition takes. In this case, the element's change in height will take 400 milliseconds to complete.

Finally, the value `ease` is the animation function, which determines how the animation is played. `ease` means start slow, speed up, then end slow again. Other values are `linear`, `ease-out`, and `ease-in`.

Cross-Browser Compatibility

The `transition` property is generally well-supported across all major browsers, excepting IE 9. For earlier versions of Firefox and Webkit-based browsers, use vendor prefixes like so:

```
.Example{
    transition:         all 400ms ease;
    -moz-transition:    all 400ms ease;
    -webkit-transition: all 400ms ease;
}
```

Note: The `transition` property can animate changes between any two numerical values, regardless of unit. It can also transition between units, such as `100px` to `50vh`. However, it cannot transition between a number and a default or automatic value, such as transitioning an element's height from `100px` to `auto`.

Section 27.3: Syntax Examples

Our first syntax example shows the animation shorthand property using all of the available properties/parameters:

```
  animation: 3s         ease-in           1s      2                 reverse     both        paused
      slidein;
  /*         duration | timing-function | delay | iteration-count | direction | fill-mode | play-
state | name   */
```

Our second example is a little more simple, and shows that some properties can be omitted:

```
  animation: 3s         linear            1s      slidein;
  /*         duration | timing-function | delay | name   */
```

Our third example shows the most minimal declaration. Note that the animation-name and animation-duration must be declared:

```
animation: 3s         slidein;
/*         duration | name */
```

It's also worth mentioning that when using the animation shorthand the order of the properties makes a difference. Obviously the browser may confuse your duration with your delay.

If brevity isn't your thing, you can also skip the shorthand property and write out each property individually:

```
animation-duration: 3s;
animation-timing-function: ease-in;
animation-delay: 1s;
animation-iteration-count: 2;
animation-direction: reverse;
animation-fill-mode: both;
animation-play-state: paused;
animation-name: slidein;
```

Section 27.4: Increasing Animation Performance Using the `will-change` Attribute

When creating animations and other GPU-heavy actions, it's important to understand the `will-change` attribute.

Both CSS keyframes and the `transition` property use GPU acceleration. Performance is increased by offloading calculations to the device's GPU. This is done by creating paint layers (parts of the page that are individually rendered) that are offloaded to the GPU to be calculated. The `will-change` property tells the browser what will animate, allowing the browser to create smaller paint areas, thus increasing performance.

The `will-change` property accepts a comma-separated list of properties to be animated. For example, if you plan on transforming an object and changing its opacity, you would specify:

```
.Example{
    ...
    will-change: transform, opacity;
}
```

Note: Use `will-change` sparingly. Setting `will-change` for every element on a page can cause performance problems, as the browser may attempt to create paint layers for every element, significantly increasing the amount of processing done by the GPU.

Chapter 28: 2D Transforms

Function/Parameter	Details
`rotate(x)`	Defines a transformation that moves the element around a fixed point on the Z axis
`translate(x,y)`	Moves the position of the element on the X and Y axis
`translateX(x)`	Moves the position of the element on the X axis
`translateY(y)`	Moves the position of the element on the Y axis
`scale(x,y)`	Modifies the size of the element on the X and Y axis
`scaleX(x)`	Modifies the size of the element on the X axis
`scaleY(y)`	Modifies the size of the element on the Y axis
`skew(x,y)`	Shear mapping, or transvection, distorting each point of an element by a certain angle in each direction
`skewX(x)`	Horizontal shear mapping distorting each point of an element by a certain angle in the horizontal direction
`skewY(y)`	Vertical shear mapping distorting each point of an element by a certain angle in the vertical direction
`matrix()`	Defines a 2D transformation in the form of a transformation matrix.
angle	The angle by which the element should be rotated or skewed (depending on the function with which it is used). Angle can be provided in degrees (`deg`), gradians (`grad`), radians (`rad`) or turns (`turn`). In `skew()` function, the second angle is optional. If not provided, there will be no (0) skew in Y-axis.
length-or-percentage	The distance expressed as a length or a percentage by which the element should be translated. In `translate()` function, the second length-or-percentage is optional. If not provided, then there would be no (0) translation in Y-axis.
scale-factor	A number which defines how many times the element should be scaled in the specified axis. In `scale()` function, the second scale-factor is optional. If not provided, the first scale-factor will be applied for Y-axis also.

Section 28.1: Rotate

HTML

```
<div class="rotate"></div>
```

CSS

```
.rotate {
    width: 100px;
    height: 100px;
    background: teal;
    transform: rotate(45deg);
}
```

This example will rotate the div by 45 degrees clockwise. The center of rotation is in the center of the div, `50%` from left and `50%` from top. You can change the center of rotation by setting the `transform-origin` property.

```
transform-origin: 100% 50%;
```

The above example will set the center of rotation to the middle of the right side end.

Section 28.2: Scale

HTML

```
<div class="scale"></div>
```

CSS

```
.scale {
    width: 100px;
    height: 100px;
    background: teal;
    transform: scale(0.5, 1.3);
}
```

This example will scale the div to `100px * 0.5 = 50px` on the X axis and to `100px * 1.3 = 130px` on the Y axis.

The center of the transform is in the center of the div, `50%` from left and `50%` from top.

Section 28.3: Skew

HTML

```
<div class="skew"></div>
```

CSS

```
.skew {
    width: 100px;
    height: 100px;
    background: teal;
    transform: skew(20deg, -30deg);
}
```

This example will skew the div by 20 degrees on the X axis and by - 30 degrees on the Y axis.
The center of the transform is in the center of the div, `50%` from left and `50%` from top.

See the result here.

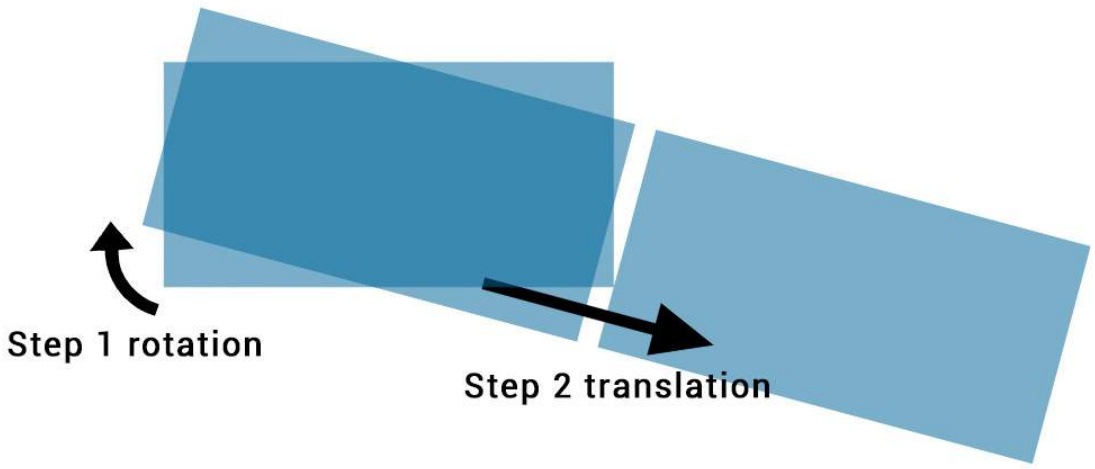
Section 28.4: Multiple transforms

Multiple transforms can be applied to an element in one property like this:

```
transform: rotate(15deg) translateX(200px);
```

This will rotate the element 15 degrees clockwise and then translate it 200px to the right.

In chained transforms, **the coordinate system moves with the element**. This means that the translation won't be horizontal but on an axis rotate 15 degrees clockwise as shown in the following image:



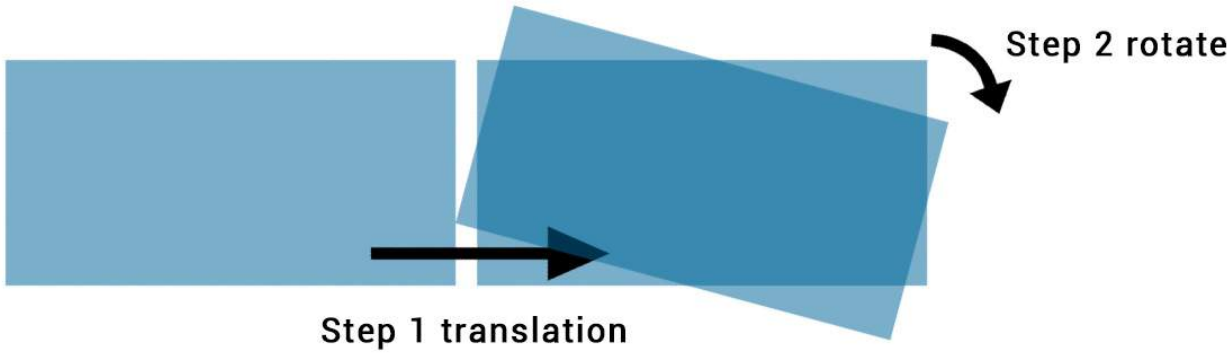
Changing the order of the transforms will change the output. The first example will be different to

```
transform: translateX(200px) rotate(15deg);
```

```
<div class="transform"></div>
```

```
.transform {
  transform: rotate(15deg) translateX(200px);
}
```

As shown in this image:



Section 28.5: Translate

HTML

```
<div class="translate"></div>
```

CSS

```
.translate {
    width: 100px;
    height: 100px;
    background: teal;
    transform: translate(200px, 50%);
```

```
}
```

This example will move the div by 200px on the X axis and by `100px * 50% = 50px` on the Y axis.

You can also specify translations on a single axis.

On the X axis:

```
.translate {
    transform: translateX(200px);
}
```

On the Y axis:

```
.translate {
    transform: translateY(50%);
}
```

Section 28.6: Transform Origin

Transformations are done with respect to a point which is defined by the `transform-origin` property.

The property takes 2 values : `transform-origin: X Y;`

In the following example the first div (`.tl`) is rotate around the top left corner with `transform-origin: 0 0;` and the second (`.tr`)is transformed around it's top right corner with `transform-origin: 100% 0`. The rotation is applied **on hover** :

HTML:

```
<div class="transform originl"></div>
<div class="transform origin2"></div>
```

CSS:

```
.transform {
    display: inline-block;
    width: 200px;
    height: 100px;
    background: teal;
    transition: transform 1s;
}

.origin1 {
    transform-origin: 0 0;
}

.origin2 {
    transform-origin: 100% 0;
}

.transform:hover {
    transform: rotate(30deg);
}
```

The default value for the transform-origin property is `50% 50%` which is the center of the element.

Chapter 29: 3D Transforms

Section 29.1: Compass pointer or needle shape using 3D transforms

CSS

```
div.needle {
  margin: 100px;
  height: 150px;
  width: 150px;
  transform: rotateY(85deg) rotateZ(45deg);
  /* presentational */
  background-image: linear-gradient(to top left, #555 0%, #555 40%, #444 50%, #333 97%);
  box-shadow: inset 6px 6px 22px 8px #272727;
}
```

HTML

```
<div class='needle'></div>
```

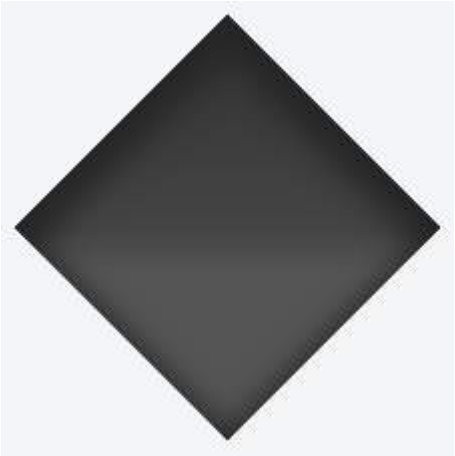
In the above example, a needle or compass pointer shape is created using 3D transforms. Generally when we apply the `rotate` transform on an element, the rotation happens only in the Z-axis and at best we will end up with diamond shapes only. But when a `rotateY` transform is added on top of it, the element gets squeezed in the Y-axis and thus ends up looking like a needle. The more the rotation of the Y-axis the more squeezed the element looks.

The output of the above example would be a needle resting on its tip. For creating a needle that is resting on its base, the rotation should be along the X-axis instead of along Y-axis. So the `transform` property's value would have to be something like `rotateX(85deg) rotateZ(45deg);`.

This pen uses a similar approach to create something that resembles the Safari logo or a compass dial.

Screenshot of element with no transform:

Screenshot of element with only 2D transform:



Screenshot of element with 3D transform:

Section 29.2: 3D text effect with shadow

HTML:

```
<div id="title">
  <h1 data-content="HOVER">HOVER</h1>
</div>
```

CSS:

```
*{margin:0;padding:0;}
html,body{height:100%;width:100%;overflow:hidden;background:#0099CC;}
#title{
  position:absolute;
  top:50%; left:50%;
  transform:translate(-50%,-50%);
  perspective-origin:50% 50%;
  perspective:300px;
}
h1{
  text-align:center;
  font-size:12vmin;
  font-family: 'Open Sans', sans-serif;
  color:rgba(0,0,0,0.8);
  line-height:1em;
  transform:rotateY(50deg);
  perspective:150px;
  perspective-origin:0% 50%;
```

```
}
h1:after{
  content:attr(data-content);
  position:absolute;
  left:0;top:0;
  transform-origin:50% 100%;
  transform:rotateX(-90deg);
  color:#0099CC;
}
#title:before{
  content:'';
  position:absolute;
  top:-150%; left:-25%;
  width:180%; height:328%;
  background:rgba(255,255,255,0.7);
  transform-origin: 0 100%;
  transform: translatez(-200px) rotate(40deg) skewX(35deg);
  border-radius:0 0 100% 0;
}
```

View example with additional hover effect

In this example, the text is transformed to make it look like it is going into the screen away from the user.

The shadow is transformed accordingly so it follows the text. As it is made with a pseudo element and the `data` attribute, it inherits the transforms form it's parent (the H1 tag).

The white "light" is made with a pseudo element on the `#title` element. It is skewed and uses border-radius for the rounded corner.

Section 29.3: backface-visibility

The `backface-visibility` property relates to 3D transforms.

With 3D transforms and the `backface-visibility` property, you're able to rotate an element such that the original front of an element no longer faces the screen.

For example, this would flip an element away from the screen:

JSFIDDLE

```
<div class="flip">Loren ipsum</div>
<div class="flip back">Lorem ipsum</div>
```

```
.flip {
  -webkit-transform: rotateY(180deg);
  -moz-transform: rotateY(180deg);
  -ms-transform: rotateY(180deg);
  -webkit-backface-visibility: visible;
  -moz-backface-visibility:    visible;
  -ms-backface-visibility:     visible;
}

.flip.back {
  -webkit-backface-visibility: hidden;
  -moz-backface-visibility:    hidden;
  -ms-backface-visibility:     hidden;
}
```

Firefox 10+ and IE 10+ support `backface-visibility` without a prefix. Opera, Chrome, Safari, iOS, and Android all need `-webkit-backface-visibility`.

It has 4 values:

1. **visible** (default) - the element will always be visible even when not facing the screen.
2. **hidden** - the element is not visible when not facing the screen.
3. **inherit** - the property will gets its value from the its parent element
4. **initial** - sets the property to its default, which is visible

Section 29.4: 3D cube

3D transforms can be use to create many 3D shapes. Here is a simple 3D CSS cube example:

HTML:

```
<div class="cube">
  <div class="cubeFace"></div>
  <div class="cubeFace face2"></div>
</div>
```

CSS:

```
body {
  perspective-origin: 50% 100%;
  perspective: 1500px;
  overflow: hidden;
}
.cube {
  position: relative;
  padding-bottom: 20%;
  transform-style: preserve-3d;
  transform-origin: 50% 100%;
  transform: rotateY(45deg) rotateX(0);
}
.cubeFace {
```

```
  position: absolute;
  top: 0;
  left: 40%;
  width: 20%;
  height: 100%;
  margin: 0 auto;
  transform-style: inherit;
  background: #C52329;
  box-shadow: inset 0 0 0 5px #333;
  transform-origin: 50% 50%;
  transform: rotateX(90deg);
  backface-visibility: hidden;
}
.face2 {
  transform-origin: 50% 50%;
  transform: rotatez(90deg) translateX(100%) rotateY(90deg);
}
.cubeFace:before, .cubeFace:after {
  content: '';
  position: absolute;
  width: 100%;
  height: 100%;
  transform-origin: 0 0;
  background: inherit;
  box-shadow: inherit;
  backface-visibility: inherit;
}
.cubeFace:before {
  top: 100%;
  left: 0;
  transform: rotateX(-90deg);
}
.cubeFace:after {
  top: 0;
  left: 100%;
  transform: rotateY(90deg);
}
```

View this example

Additional styling is added in the demo and a transform is applied on hover to view the 6 faces of the cube.

Should be noted that:

- 4 faces are made with pseudo elements
- chained transforms are applied

Chapter 30: Filter Property

Value	Description
blur(x)	Blurs the image by x pixels.
brightness(x)	Brightens the image at any value above 1.0 or 100%. Below that, the image will be darkened.
contrast(x)	Provides more contrast to the image at any value above 1.0 or 100%. Below that, the image will get less saturated.
drop-shadow(h, v, x, y, z)	Gives the image a drop-shadow. h and v can have negative values. x, y, and z are optional.
greyscale(x)	Shows the image in greyscale, with a maximum value of 1.0 or 100%.
hue-rotate(x)	Applies a hue-rotation to the image.
invert(x)	Inverts the color of the image with a maximum value of 1.0 or 100%.
opacity(x)	Sets how opaque/transparent the image is with a maximum value of 1.0 or 100%.
saturate(x)	Saturates the image at any value above 1.0 or 100%. Below that, the image will start to de-saturate.
sepia(x)	Converts the image to sepia with a maximum value of 1.0 or 100%.

Section 30.1: Blur

HTML

```
<img src='donald-duck.png' alt='Donald Duck' title='Donald Duck' />
```

CSS

```
img {
    -webkit-filter: blur(1px);
    filter: blur(1px);
}
```

Result

Makes you wanna rub your glasses.

Section 30.2: Drop Shadow (use box-shadow instead if

possible)

HTML

```
<p>My shadow always follows me.</p>
```

CSS

```
p {
    -webkit-filter: drop-shadow(10px 10px 1px green);
    filter: drop-shadow(10px 10px 1px green);
}
```

Result

My shadow always follows me.

Section 30.3: Hue Rotate

HTML

```
<img src='donald-duck.png' alt='Donald Duck' title='Donald Duck' />
```

CSS

```
img {
    -webkit-filter: hue-rotate(120deg);
    filter: hue-rotate(120deg);
}
```

Result

Section 30.4: Multiple Filter Values

To use multiple filters, separate each value with a space.

HTML

```
<img src='donald-duck.png' alt='Donald Duck' title='Donald Duck' />
```

CSS

```
img {
    -webkit-filter: brightness(200%) grayscale(100%) sepia(100%) invert(100%);
    filter: brightness(200%) grayscale(100%) sepia(100%) invert(100%);
}
```

Result

Section 30.5: Invert Color

HTML

```
<div></div>
```

CSS

```
div {
    width: 100px;
    height: 100px;
    background-color: white;
    -webkit-filter: invert(100%);
    filter: invert(100%);
}
```

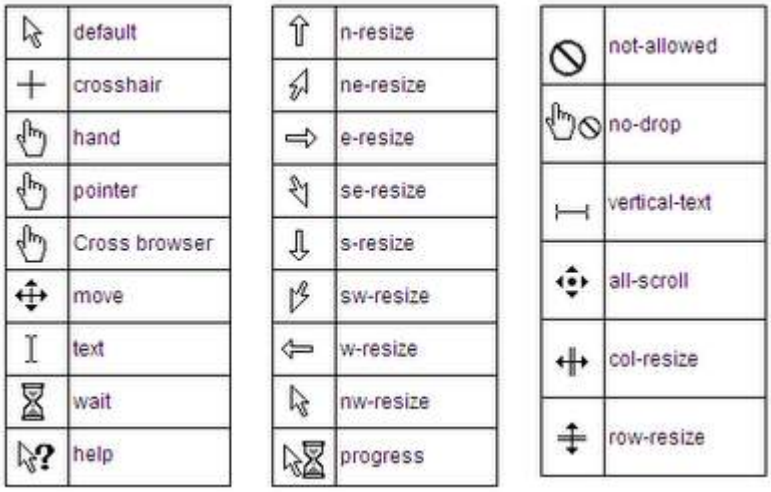
Result

Turns from white to black.

Chapter 31: Cursor Styling

Section 31.1: Changing cursor type

```
cursor: value;
```



Examples:

Value	Description
none	No cursor is rendered for the element
auto	Default. The browser sets a cursor
help	The cursor indicates that help is available
wait	The cursor indicates that the program is busy
move	The cursor indicates something is to be moved
pointer	The cursor is a pointer and indicates a link

Section 31.2: pointer-events

The pointer-events property allows for control over how HTML elements respond to mouse/touch events.

```
.disabled {
  pointer-events: none;
}
```

In this example,

> 'none' prevents all click, state and cursor options on the specified HTML element [[1]]

Other valid values for HTMl elements are:

- auto;
- inherit.

1. https://css-tricks.com/almanac/properties/p/pointer-events/

Other resources:

- https://developer.mozilla.org/en-US/docs/Web/CSS/pointer-events
- https://davidwalsh.name/pointer-events

Section 31.3: caret-color

The caret-color CSS property specifies the color of the caret, the visible indicator of the insertion point in an element where text and other content is inserted by the user's typing or editing.

HTML

```
<input id="example" />
```

CSS

```
#example {
  caret-color: red;
}
```

Resources:

- https://developer.mozilla.org/en-US/docs/Web/CSS/caret-color

Chapter 32: box-shadow

Parameters	Details
inset	by default, the shadow is treated as a drop shadow. the inset keyword draws the shadow inside the frame/border.
offset-x	the horizontal distance
offset-y	the vertical distance
blur-radius	0 by default. value cannot be negative. the bigger the value, the bigger and lighter the shadow becomes.
spread-radius	0 by default. positive values will cause the shadow to expand. negative values will cause the shadow to shrink.
color	can be of various notations: a color keyword, hexadecimal, `rgb()`, `rgba()`, `hsl()`, `hsla()`

Section 32.1: bottom-only drop shadow using a pseudo-element

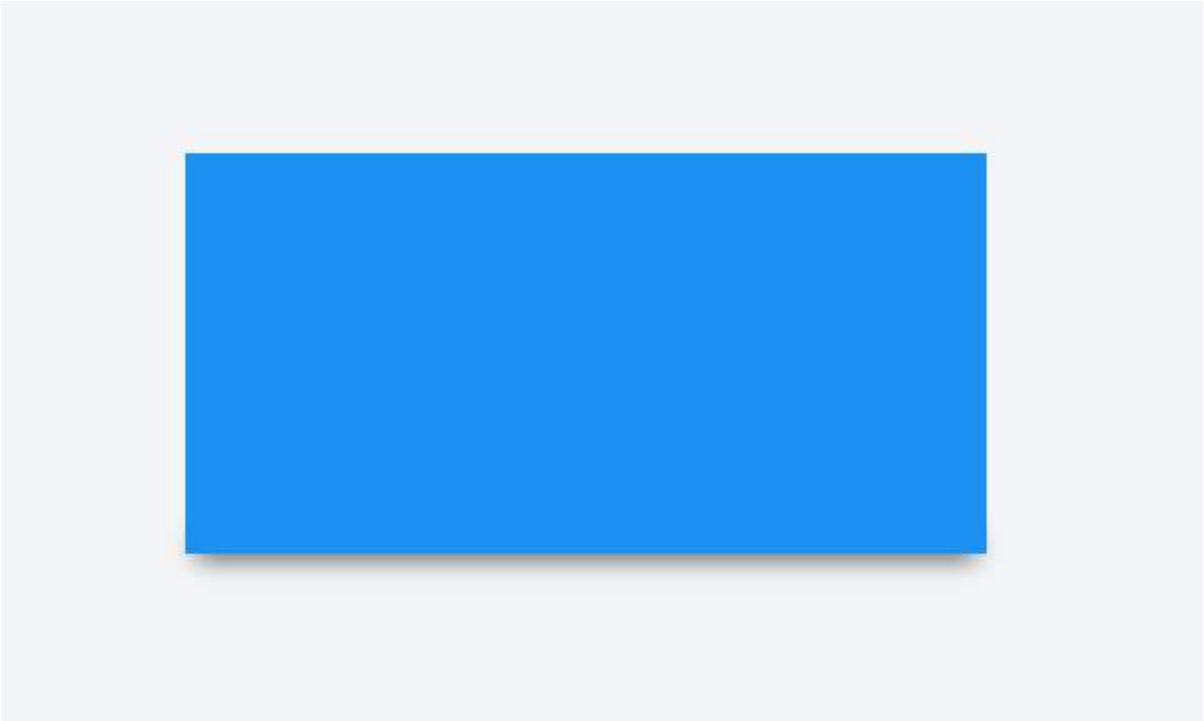
JSFiddle: https://jsfiddle.net/UnsungHero97/80qod7aL/2/

HTML

```
<div class="box_shadow"></div>
```

CSS

```
.box_shadow {
  background-color: #1C90F3;
  width: 200px;
  height: 100px;
  margin: 50px;
}

.box_shadow:after {
  content: "";
  width: 190px;
  height: 1px;
  margin-top: 98px;
  margin-left: 5px;
  display: block;
  position: absolute;
  z-index: -1;
  -webkit-box-shadow: 0px 0px 8px 2px #444444;
     -moz-box-shadow: 0px 0px 8px 2px #444444;
          box-shadow: 0px 0px 8px 2px #444444;
}
```



Section 32.2: drop shadow

JSFiddle: https://jsfiddle.net/UnsungHero97/80qod7aL/

HTML

```
<div class="box_shadow"></div>
```

CSS

```
.box_shadow {
  -webkit-box-shadow: 0px 0px 10px -1px #444444;
     -moz-box-shadow: 0px 0px 10px -1px #444444;
          box-shadow: 0px 0px 10px -1px #444444;
}
```

Section 32.3: inner drop shadow

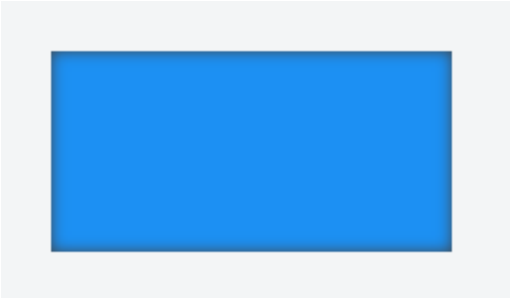
HTML

```
<div class="box_shadow"></div>
```

CSS

```
.box_shadow {
  background-color: #1C90F3;
  width: 200px;
  height: 100px;
  margin: 50px;
  -webkit-box-shadow: inset 0px 0px 10px 0px #444444;
     -moz-box-shadow: inset 0px 0px 10px 0px #444444;
          box-shadow: inset 0px 0px 10px 0px #444444;
}
```

Result:



JSFiddle: https://jsfiddle.net/UnsungHero97/80qod7aL/1/

Section 32.4: multiple shadows

JSFiddle: https://jsfiddle.net/UnsungHero97/80qod7aL/5/

HTML

```
<div class="box_shadow"></div>
```

CSS

```
.box_shadow {
  width: 100px;
  height: 100px;
  margin: 100px;
  box-shadow:
    -52px -52px 0px 0px #f65314,
    52px -52px 0px 0px #7cbb00,
    -52px 52px 0px 0px #00a1f1,
    52px 52px 0px 0px #ffbb00;
}
```

Chapter 33: Shapes for Floats

Parameter	Details
none	A value of `none` means that the float area (the area that is used for wrapping content around a float element) is unaffected. This is the default/initial value.
basic-shape	Refers to one among `inset()`, `circle()`, `ellipse()` or `polygon()`. Using one of these functions and its values the shape is defined.
shape-box	Refers to one among `margin-box`, `border-box`, `padding-box`, `content-box`. When only <shape-box> is provided (without <basic-shape>) this box *is the* shape. When it is used along with <basic-shape>, this acts as the reference box.
image	When an image is provided as value, the shape is computed based on the alpha channel of the image specified.

Section 33.1: Shape Outside with Basic Shape – circle()

With the `shape-outside` CSS property one can define shape values for the float area so that the inline content wraps around the shape instead of the float's box.

CSS

```
img:nth-of-type(1) {
  shape-outside: circle(80px at 50% 50%);
  float: left;
  width: 200px;
}
img:nth-of-type(2) {
  shape-outside: circle(80px at 50% 50%);
  float: right;
  width: 200px;
}
p {
  text-align: center;
  line-height: 30px; /* purely for demo */
}
```

HTML

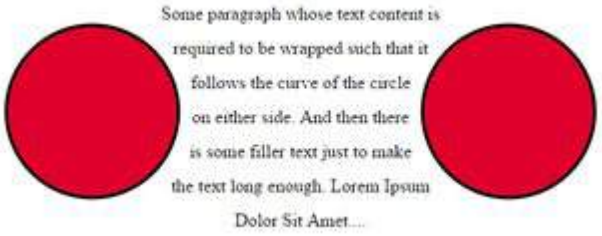
```
<img src="http://images.clipartpanda.com/circle-clip-art-circlergb.jpg">
<img src="http://images.clipartpanda.com/circle-clip-art-circlergb.jpg">
<p>Some paragraph whose text content is required to be wrapped such that it follows the curve of
the circle on either side. And then there is some filler text just to make the text long enough.
Lorem Ipsum Dolor Sit Amet....</p>
```

In the above example, both the images are actually square images and when the text is placed without the `shape-outside` property, it will not flow around the circle on either side. It will flow around the containing box of the image only. With `shape-outside` the float area is re-defined as a *circle* and the content is made to flow around this *imaginary circle* that is created using `shape-outside`.

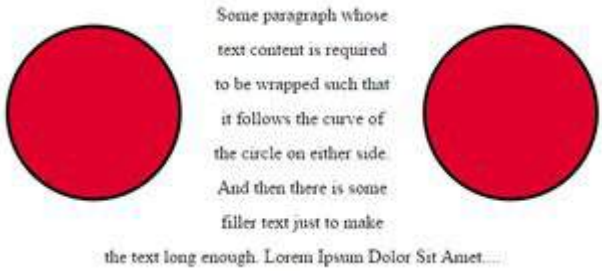
The *imaginary circle* that is used to re-define the float area is a circle with radius of 80px drawn from the center-mid point of the image's reference box.

Below are a couple of screenshots to illustrate how the content would be wrapped around when `shape-outside` is used and when it is not used.

Output with `shape-outside`



Output without `shape-outside`



Section 33.2: Shape margin

The `shape-margin` CSS property adds a *margin* to `shape-outside`.

CSS

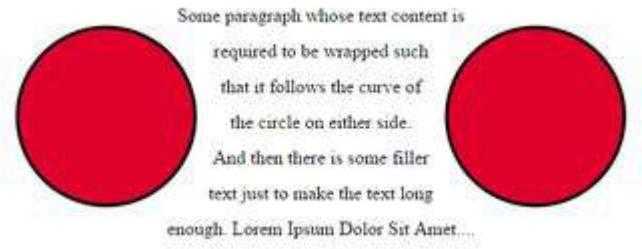
```
img:nth-of-type(1) {
  shape-outside: circle(80px at 50% 50%);
  shape-margin: 10px;
  float: left;
  width: 200px;
}
img:nth-of-type(2) {
  shape-outside: circle(80px at 50% 50%);
  shape-margin: 10px;
  float: right;
  width: 200px;
}
p {
  text-align: center;
  line-height: 30px; /* purely for demo */
}
```

HTML

```
<img src="http://images.clipartpanda.com/circle-clip-art-circlergb.jpg">
<img src="http://images.clipartpanda.com/circle-clip-art-circlergb.jpg">
<p>Some paragraph whose text content is required to be wrapped such that it follows the curve of
the circle on either side. And then there is some filler text just to make the text long enough.
Lorem Ipsum Dolor Sit Amet....</p>
```

In this example, a 10px margin is added around the **shape** using `shape-margin`. This creates a bit more space between the *imaginary circle* that defines the float area and the actual content that is flowing around.

Output:


Some paragraph whose text content is
required to be wrapped such
that it follows the curve of
the circle on either side.
And then there is some filler
text just to make the text long
enough. Lorem Ipsum Dolor Sit Amet....

Chapter 34: List Styles

Value	Description
list-style-type	the type of list-item marker.
list-style-position	specifies where to place the marker
list-style-image	specifies the type of list-item marker
initial	sets this property to its default value
inherit	inherits this property from its parent element

Section 34.1: Bullet Position

A list consists of `<li>` elements inside a containing element (`<ul>` or `<ol>`). Both the list items and the container can have margins and paddings which influence the exact position of the list item content in the document. The default values for the margin and padding may be different for each browser. In order to get the same layout cross-browser, these need to be set specifically.

Each list item gets a 'marker box', which contains the bullet marker. This box can either be placed inside or outside of the list item box.

```
list-style-position: inside;
```

places the bullet within the `<li>` element, pushing the content to the right as needed.

```
list-style-position: outside;
```

places the bullet left of the `<li>` element. If there is not enough space in the padding of the containing element, the marker box will extend to the left even if it would fall off the page.

Showing the result of `inside` and `outside` positioning: jsfiddle

Section 34.2: Removing Bullets / Numbers

Sometimes, a list should just not display any bullet points or numbers. In that case, remember to specify margin and padding.

```
<ul>
    <li>first item</li>
    <li>second item</li>
</ul>
```

CSS

```
ul {
    list-style-type: none;
}
li {
    margin: 0;
    padding: 0;
}
```

Section 34.3: Type of Bullet or Numbering

Specific for `<li>` tags within an unordered list (`<ul>`):

```
list-style: disc;                /* A filled circle (default) */
list-style: circle;              /* A hollow circle */
list-style: square;              /* A filled square */
list-style: '-';                 /* any string */
```

Specific for `<li>` tags within an ordered list (`<ol>`):

```
list-style: decimal;             /* Decimal numbers beginning with 1 (default) */
list-style: decimal-leading-zero;/* Decimal numbers padded by initial zeros (01, 02, 03, … 10) */
list-style: lower-roman;         /* Lowercase roman numerals (i., ii., iii., iv., ...) */
list-style: upper-roman;         /* Uppercase roman numerals (I., II., III., IV., ...) */
list-style-type: lower-greek;    /* Lowercase roman letters (α., β., γ., δ., ...) */
list-style-type: lower-alpha;    /* Lowercase letters (a., b., c., d., ...) */
list-style-type: lower-latin;    /* Lowercase letters (a., b., c., d., ...) */
list-style-type: upper-alpha;    /* Uppercase letters (A., B., C., D., ...) */
list-style-type: upper-latin;    /* Uppercase letters (A., B., C., D., ...) */
```

Non-specific:

```
list-style: none;                /* No visible list marker */
list-style: inherit;             /* Inherits from parent */
```

Chapter 35: Counters

Parameter	Details
counter-name	This is the name of the counter that needs to be created or incremented or printed. It can be any custom name as the developer wishes.
integer	This integer is an optional value that when provided next to the counter name will represent the initial value of the counter (in `counter-set`, `counter-reset` properties) or the value by which the counter should be incremented (in `counter-increment`).
none	This is the initial value for all 3 `counter-*` properties. When this value is used for `counter-increment`, the value of none of the counters are affected. When this is used for the other two, no counter is created.
counter-style	This specifies the style in which the counter value needs to be displayed. It supports all values supported by the `list-style-type` property. If `none` is used then the counter value is not printed at all.
connector-string	This represents the string that must be placed between the values of two different counter levels (like the "." in "2.1.1").

Section 35.1: Applying roman numerals styling to the counter output

CSS

```
body {
  counter-reset: item-counter;
}

.item {
  counter-increment: item-counter;
}

.item:before {
  content: counter(item-counter, upper-roman) ". "; /* by specifying the upper-roman as style the output would be in roman numbers */
}
```

HTML

```
<div class='item'>Item No: 1</div>
<div class='item'>Item No: 2</div>
<div class='item'>Item No: 3</div>
```

In the above example, the counter's output would be displayed as I, II, III (roman numbers) instead of the usual 1, 2, 3 as the developer has explicitly specified the counter's style.

Section 35.2: Number each item using CSS Counter

CSS

```
body {
  counter-reset: item-counter; /* create the counter */
}
.item {
  counter-increment: item-counter; /* increment the counter every time an element with class "item" is encountered */
}
.item-header:before {
  content: counter(item-counter) ". "; /* print the value of the counter before the header and append a "." to it */
}
```

```
/* just for demo */

.item {
  border: 1px solid;
  height: 100px;
  margin-bottom: 10px;
}
.item-header {
  border-bottom: 1px solid;
  height: 40px;
  line-height: 40px;
  padding: 5px;
}
.item-content {
  padding: 8px;
}
```

HTML

```
<div class='item'>
  <div class='item-header'>Item 1 Header</div>
  <div class='item-content'>Lorem Ipsum Dolor Sit Amet....</div>
</div>
<div class='item'>
  <div class='item-header'>Item 2 Header</div>
  <div class='item-content'>Lorem Ipsum Dolor Sit Amet....</div>
</div>
<div class='item'>
  <div class='item-header'>Item 3 Header</div>
  <div class='item-content'>Lorem Ipsum Dolor Sit Amet....</div>
</div>
```

The above example numbers every "item" in the page and adds the item's number before its header (using `content` property of `.item-header` element's `:before` pseudo). A live demo of this code is available here.

Section 35.3: Implementing multi-level numbering using CSS counters

CSS

```
ul {
  list-style: none;
  counter-reset: list-item-number; /* self nesting counter as name is same for all levels */
}
li {
  counter-increment: list-item-number;
}
li:before {
  content: counters(list-item-number, ".") " "; /* usage of counters() function means value of
counters at all higher levels are combined before printing */
}
```

HTML

```
<ul>
  <li>Level 1
    <ul>
      <li>Level 1.1
        <ul>
          <li>Level 1.1.1</li>
        </ul>
      </li>
    </ul>
```

```
  </li>
  <li>Level 2
    <ul>
      <li>Level 2.1
        <ul>
          <li>Level 2.1.1</li>
          <li>Level 2.1.2</li>
        </ul>
      </li>
    </ul>
  </li>
  <li>Level 3</li>
</ul>
```

The above is an example of multi-level numbering using CSS counters. It makes use of the ***self-nesting*** concept of counters. Self nesting is a concept where if an element already has a counter with the given name but is having to create another then it creates it as a child of the existing counter. Here, the second level `ul` already inherits the `list-item-number` counter from its parent but then has to create its own `list-item-number` (for its children `li`) and so creates `list-item-number[1]` (counter for second level) and nests it under `list-item-number[0]` (counter for first level). Thus it achieves the multi-level numbering.

The output is printed using the `counters()` function instead of the `counter()` function because the `counters()` function is designed to prefix the value of all higher level counters (parent) when printing the output.

Chapter 36: Functions

Section 36.1: calc() function

Accepts a mathematical expression and returns a numerical value.

It is especially useful when working with different types of units (e.g. subtracting a px value from a percentage) to calculate the value of an attribute.

`+`, `-`, `/`, and `*` operators can all be used, and parentheses can be added to specify the order of operations if necessary.

Use `calc()` to calculate the width of a div element:

```
#div1 {
    position: absolute;
    left: 50px;
    width: calc(100% - 100px);
    border: 1px solid black;
    background-color: yellow;
    padding: 5px;
    text-align: center;
}
```

Use `calc()` to determine the position of a background-image:

```
background-position: calc(50% + 17px) calc(50% + 10px), 50% 50%;
```

Use `calc()` to determine the height of an element:

```
height: calc(100% - 20px);
```

Section 36.2: attr() function

Returns the value of an attribute of the selected element.

Below is a blockquote element which contains a character inside a `data-*` attribute which CSS can use (e.g. inside the `::before` and `::after` pseudo-element) using this function.

```
<blockquote data-mark=''''></blockquote>
```

In the following CSS block, the character is appended before and after the text inside the element:

```
blockquote[data-mark]::before,
blockquote[data-mark]::after {
    content: attr(data-mark);
}
```

Section 36.3: var() function

The var() function allows CSS variables to be accessed.

```
/* set a variable */
:root {
```

```
    --primary-color: blue;
}

/* access variable */
selector {
    color: var(--primary-color);
}
```

This feature is currently under development. Check caniuse.com for the latest browser support.

Section 36.4: radial-gradient() function

Creates an image representing a gradient of colors radiating from the center of the gradient

```
radial-gradient(red, orange, yellow) /*A gradient coming out from the middle of the
gradient, red at the center, then orange, until it is finally yellow at the edges*/
```

Section 36.5: linear-gradient() function

Creates a image representing a linear gradient of colors.

```
linear-gradient( 0deg, red, yellow 50%, blue);
```

This creates a gradient going from bottom to top, with colors starting at red, then yellow at 50%, and finishing in blue.

Chapter 37: Custom Properties (Variables)

CSS Variables allow authors to create reusable values which can be used throughout a CSS document.

For example, it's common in CSS to reuse a single color throughout a document. Prior to CSS Variables this would mean reusing the same color value many times throughout a document. With CSS Variables the color value can be assigned to a variable and referenced in multiple places. This makes changing values easier and is more semantic than using traditional CSS values.

Section 37.1: Variable Color

```
:root {
  --red: #b00;
  --blue: #4679bd;
  --grey: #ddd;
}
.Bx1 {
  color: var(--red);
  background: var(--grey);
  border: 1px solid var(--red);
}
```

Section 37.2: Variable Dimensions

```
:root {
  --W200: 200px;
  --W10: 10px;
}
.Bx2 {
  width: var(--W200);
  height: var(--W200);
  margin: var(--W10);
}
```

Section 37.3: Variable Cascading

CSS variables cascade in much the same way as other properties, and can be restated safely.

You can define variables multiple times and only the definition with the highest specificity will apply to the element selected.

Assuming this HTML:

```
<a class="button">Button Green</a>
<a class="button button_red">Button Red</a>
<a class="button">Button Hovered On</a>
```

We can write this CSS:

```
.button {
  --color: green;
  padding: .5rem;
  border: 1px solid var(--color);
  color: var(--color);
}
```

```
.button:hover {
  --color: blue;
}

.button_red {
  --color: red;
}
```

And get this result:

Section 37.4: Valid/Invalids

Naming When naming CSS variables, it contains only letters and dashes just like other CSS properties (eg: line-height, -moz-box-sizing) but it should start with double dashes (--)

```
//These are Invalids variable names
--123color: blue;
--#color: red;
--bg_color: yellow
--$width: 100px;

//Valid variable names
--color: red;
--bg-color: yellow
--width: 100px;
```

CSS Variables are case sensitive.

```
/* The variable names below are all different variables */
--pcolor: ;
--Pcolor: ;
--pColor: ;
```

Empty Vs Space

```
/* Invalid */
    --color:;

/* Valid */
  --color: ; /* space is assigned */
```

Concatenations

```
    /* Invalid - CSS doesn't support concatenation*/
    .logo{
        --logo-url: 'logo';
        background: url('assets/img/' var(--logo-url) '.png');
    }

    /* Invalid - CSS bug */
    .logo{
        --logo-url: 'assets/img/logo.png';
        background: url(var(--logo-url));
    }
```

```
    /* Valid */
    .logo{
        --logo-url: url('assets/img/logo.png');
        background: var(--logo-url);
    }
```

Careful when using Units

```
    /* Invalid */
    --width: 10;
    width: var(--width)px;

    /* Valid */
    --width: 10px;
    width: var(--width);

    /* Valid */
    --width: 10;
    width: calc(1px * var(--width)); /* multiply by 1 unit to convert */
    width: calc(1em * var(--width));
```

Section 37.5: With media queries

You can re-set variables within media queries and have those new values cascade wherever they are used, something that isn't possible with pre-processor variables.

Here, a media query changes the variables used to set up a very simple grid:

HTML

```
<div></div>
<div></div>
<div></div>
<div></div>
```

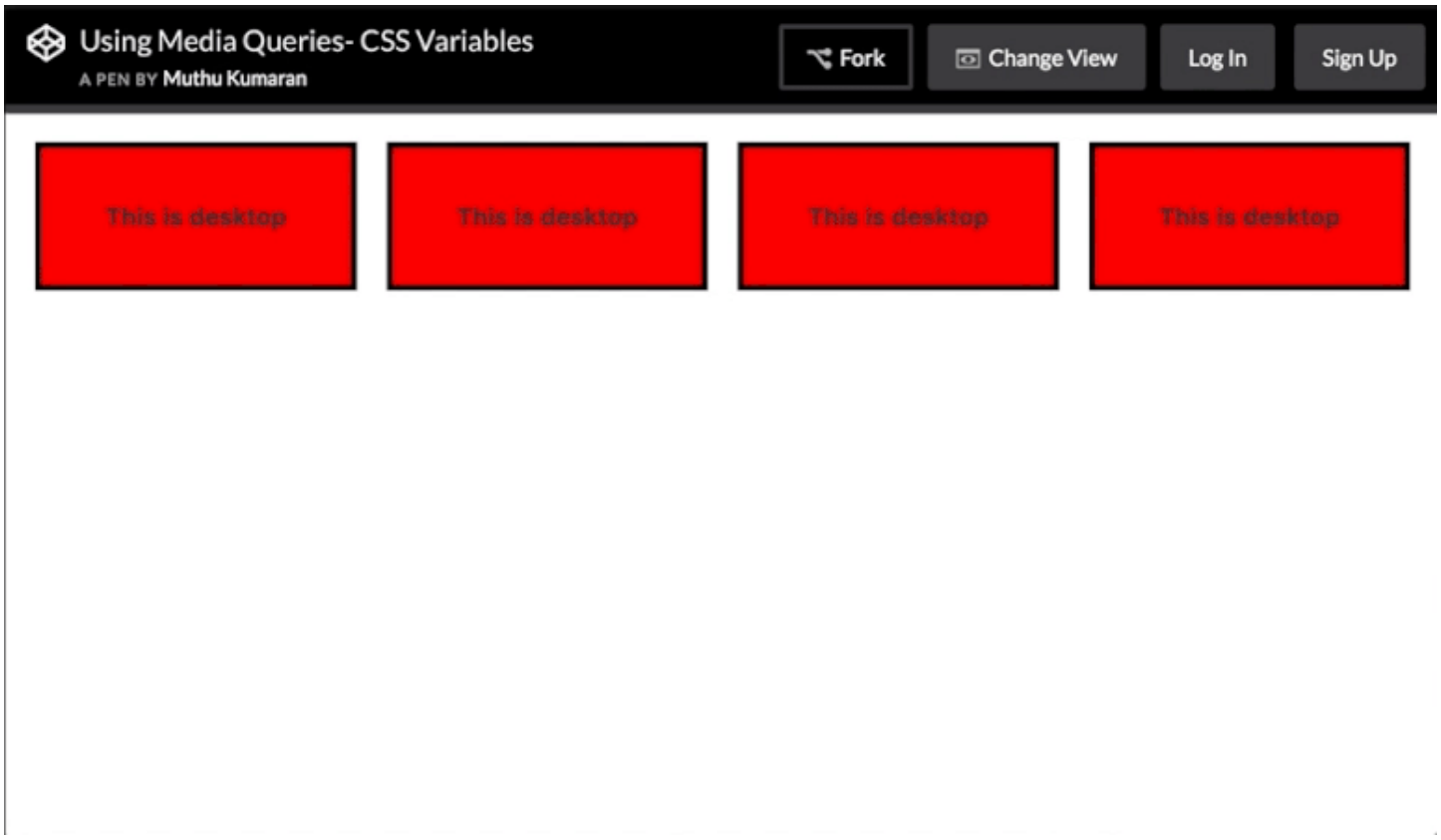
CSS

```
:root{
    --width: 25%;
    --content: 'This is desktop';
}
@media only screen and (max-width: 767px){
    :root{
        --width:50%;
        --content: 'This is mobile';
    }
}
@media only screen and (max-width: 480px){
    :root{
        --width:100%;
    }
}

div{
    width: calc(var(--width) - 20px);
    height: 100px;
}
div:before{
    content: var(--content);
```

```
}

/* Other Styles */
body {
    padding: 10px;
}

div{
    display: flex;
    align-items: center;
    justify-content: center;
    font-weight:bold;
    float:left;
    margin: 10px;
    border: 4px solid black;
    background: red;
}
```

You can try resizing the window in this CodePen Demo

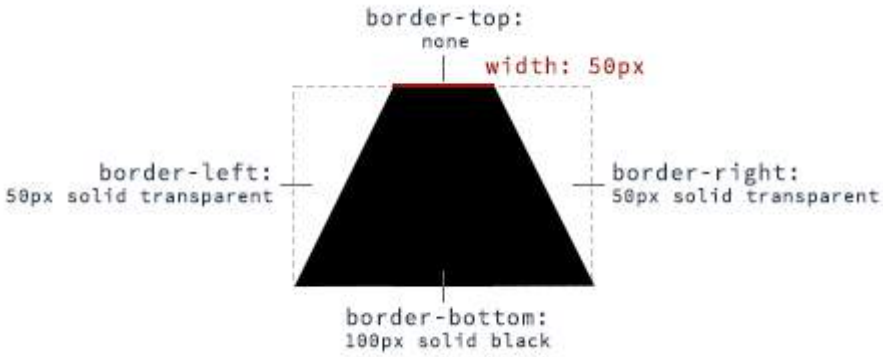
Here's an animated screenshot of the resizing in action:



Chapter 38: Single Element Shapes

Section 38.1: Trapezoid

A trapezoid can be made by a block element with zero height (height of `0px`), a width greater than zero and a border, that is transparent except for one side:



HTML:

```
<div class="trapezoid"></div>
```

CSS:

```
.trapezoid {
    width: 50px;
    height: 0;
    border-left: 50px solid transparent;
    border-right: 50px solid transparent;
    border-bottom: 100px solid black;
}
```

With changing the border sides, the orientation of the trapezoid can be adjusted.

Section 38.2: Triangles

To create a CSS triangle define an element with a width and height of 0 pixels. The triangle shape will be formed using border properties. For an element with 0 height and width the 4 borders (top, right, bottom, left) each form a triangle. Here's an element with 0 height/width and 4 different colored borders.

By setting some borders to transparent, and others to a color we can create various triangles. For example, in the Up triangle, we set the bottom border to the desired color, then set the left and right borders to transparent. Here's an image with the left and right borders shaded slightly to show how the triangle is being formed.

The dimensions of the triangle can be altered by changing the different border widths - taller, shorter, lopsided, etc. The examples below all show a 50x50 pixel triangle.

Triangle - Pointing Up

```
<div class="triangle-up"></div>
```

```
.triangle-up {
  width: 0;
  height: 0;
  border-left: 25px solid transparent;
  border-right: 25px solid transparent;
  border-bottom: 50px solid rgb(246, 156, 85);
}
```

Triangle - Pointing Down

```
<div class="triangle-down"></div>
```

```
.triangle-down {
  width: 0;
  height: 0;
  border-left: 25px solid transparent;
  border-right: 25px solid transparent;
  border-top: 50px solid rgb(246, 156, 85);
}
```

Triangle - Pointing Right

```
<div class="triangle-right"></div>
```

```
.triangle-right {
  width: 0;
  height: 0;
  border-top: 25px solid transparent;
  border-bottom: 25px solid transparent;
  border-left: 50px solid rgb(246, 156, 85);
}
```

Triangle - Pointing Left

```
<div class="triangle-left"></div>
```

```
.triangle-left {
  width: 0;
  height: 0;
  border-top: 25px solid transparent;
  border-bottom: 25px solid transparent;
  border-right: 50px solid rgb(246, 156, 85);
}
```

Triangle - Pointing Up/Right

```
<div class="triangle-up-right"></div>
```

```
.triangle-up-right {
  width: 0;
  height: 0;
  border-top: 50px solid rgb(246, 156, 85);
  border-left: 50px solid transparent;
}
```

Triangle - Pointing Up/Left

```
<div class="triangle-up-left"></div>
```

```
.triangle-up-left {
```

```
  width: 0;
  height: 0;
  border-top: 50px solid rgb(246, 156, 85);
  border-right: 50px solid transparent;
}
```

Triangle - Pointing Down/Right

```
<div class="triangle-down-right"></div>
```

```
.triangle-down-right {
  width: 0;
  height: 0;
  border-bottom: 50px solid rgb(246, 156, 85);
  border-left: 50px solid transparent;
}
```

Triangle - Pointing Down/Left

```
<div class="triangle-down-left"></div>
```

```
.triangle-down-left {
  width: 0;
  height: 0;
  border-bottom: 50px solid rgb(246, 156, 85);
  border-right: 50px solid transparent;
}
```

Section 38.3: Circles and Ellipses

Circle

To create a **circle**, define an element with an equal `width` and `height` (a *square*) and then set the `border-radius` property of this element to `50%`.

HTML

```
<div class="circle"></div>
```

CSS

```
.circle {
   width: 50px;
   height: 50px;
   background: rgb(246, 156, 85);
   border-radius: 50%;
}
```

Ellipse

An **ellipse** is similar to a circle, but with different values for `width` and `height`.

HTML

```
<div class="oval"></div>
```

CSS

```
.oval {
  width: 50px;
  height: 80px;
  background: rgb(246, 156, 85);
  border-radius: 50%;
}
```

Section 38.4: Bursts

A burst is similar to a star but with the points extending less distance from the body. Think of a burst shape as a square with additional, slightly rotated, squares layered on top.

The additional squares are created using the `::before` and `::after` psuedo-elements.

8 Point Burst

An 8 point burst are 2 layered squares. The bottom square is the element itself, the additional square is created using the `:before` pseudo-element. The bottom is rotated 20°, the top square is rotated 135°.

```
<div class="burst-8"></div>

.burst-8 {
  background: rgb(246, 156, 85);
  width: 40px;
```

```
  height: 40px;
  position: relative;
  text-align: center;
  -ms-transform: rotate(20deg);
  transform: rotate(20eg);
}

.burst-8::before {
  content: "";
  position: absolute;
  top: 0;
  left: 0;
  height: 40px;
  width: 40px;
  background: rgb(246, 156, 85);
  -ms-transform: rotate(135deg);
  transform: rotate(135deg);
}
```

12 Point Burst

An 12 point burst are 3 layered squares. The bottom square is the element itself, the additional squares are created using the `:before` and `:after` pseudo-elements. The bottom is rotated 0°, the next square is rotated 30°, and the top is rotated 60°.

```
<div class="burst-12"></div>

.burst-12 {
  width: 40px;
  height: 40px;
  position: relative;
  text-align: center;
  background: rgb(246, 156, 85);
}

.burst-12::before, .burst-12::after {
  content: "";
  position: absolute;
  top: 0;
  left: 0;
  height: 40px;
  width: 40px;
  background: rgb(246, 156, 85);
}

.burst-12::before {
  -ms-transform: rotate(30deg);
  transform: rotate(30deg);
}

.burst-12::after {
  -ms-transform: rotate(60deg);
  transform: rotate(60deg);
```

```
}
```

Section 38.5: Square

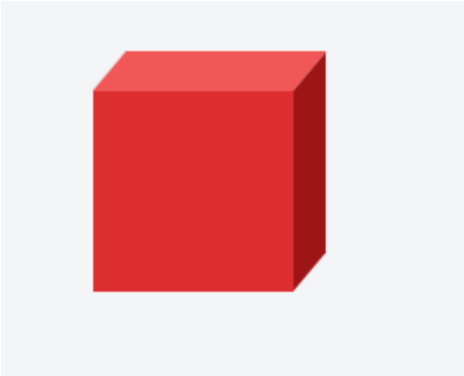
To create a square, define an element with both a width and height. In the example below, we have an element with a `width` and `height` of 100 pixels each.

```
<div class="square"></div>
```

```
.square {
    width: 100px;
    height: 100px;
    background: rgb(246, 156, 85);
}
```

Section 38.6: Cube

This example shows how to create a cube using 2D transformation methods `skewX()` and `skewY()` on pseudo elements.



HTML:

```
<div class="cube"></div>
```

CSS:

```
.cube {
  background: #dc2e2e;
  width: 100px;
  height: 100px;
  position: relative;
  margin: 50px;
}

.cube::before {
  content: '';
```

```
  display: inline-block;
  background: #f15757;
  width: 100px;
  height: 20px;
  transform: skewX(-40deg);
  position: absolute;
  top: -20px;
  left: 8px;
}

.cube::after {
  content: '';
  display: inline-block;
  background: #9e1515;
  width: 16px;
  height: 100px;
  transform: skewY(-50deg);
  position: absolute;
  top: -10px;
  left: 100%;
}
```

See demo

Section 38.7: Pyramid

This example shows how to create a **pyramid** using borders and 2D transformation methods `skewY()` and `rotate()` on pseudo elements.

HTML:

```
<div class="pyramid"></div>
```

CSS:

```
.pyramid {
  width: 100px;
  height: 200px;
  position: relative;
  margin: 50px;
}

.pyramid::before, .pyramid::after {
  content: '';
  display: inline-block;
  width: 0;
  height: 0;
  border: 50px solid;
```

```
  position: absolute;
}

.pyramid::before {
  border-color: transparent transparent #ff5656 transparent;
  transform: scaleY(2) skewY(-40deg) rotate(45deg);
}

.pyramid::after {
  border-color: transparent transparent #d64444 transparent;
  transform: scaleY(2) skewY(40deg) rotate(-45deg);
}
```

Chapter 39: Columns

Section 39.1: Simple Example (column-count)

The CSS multi-column layout makes it easy to create multiple columns of text.

Code

```
<div id="multi-columns">Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod
tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud
exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in
reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint
occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est
laborum</div>
```

```
.multi-columns {
  -moz-column-count: 2;
  -webkit-column-count: 2;
  column-count: 2;
}
```

Result

Stack Overflow is a privately held website, the flagship site of the Stack Exchange Network,[4][5][6] created in 2008 by Jeff Atwood and Joel Spolsky.[7][8] It was created to be a more open alternative to earlier Q&A sites such as Experts-Exchange. The name for the website was chosen by voting in April 2008 by readers of Coding Horror, Atwood's popular programming blog.

The website serves as a platform for users to ask and answer questions, and, through membership and active participation, to vote questions and answers up or down and edit questions and answers in a fashion similar to a wiki or Digg.[13] Users of Stack Overflow can earn reputation points and "badges"; for example, a person is awarded 10 reputation points for receiving an "up" vote on an answer given to a question, and can receive badges for their valued contributions,[14] which represents a kind of gamification of the traditional Q&A site or forum. All user-generated content is licensed under a Creative Commons Attribute-ShareAlike license.

Section 39.2: Column Width

The `column-width` property sets the minimum column width. If `column-count` is not defined the browser will make as many columns as fit in the available width.

Code:

```
<div id="multi-columns">
    Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut
labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris
nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit
esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt
in culpa qui officia deserunt mollit anim id est laborum
```

```
</div>
```

```
.multi-columns {
  -moz-column-width: 100px;
  -webkit-column-width: 100px;
  column-width: 100px;
}
```

Result

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Chapter 40: Multiple columns

CSS allows to define that element contents wrap into multiple columns with gaps and rules between them.

Section 40.1: Create Multiple Columns

```
<div class="content">
Lorem ipsum dolor sit amet, consectetuer adipiscing elit, sed diam nonummy nibh
euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim
ad minim veniam, quis nostrud exerci tation ullamcorper suscipit lobortis nisl
ut aliquip ex ea commodo consequat. Duis autem vel eum iriure dolor in
hendrerit in vulputate velit esse molestie consequat, vel illum dolore eu
feugiat nulla facilisis at vero eros et accumsan et iusto odio dignissim qui
blandit praesent luptatum zzril delenit augue duis dolore te feugait nulla
facilisi. Nam liber tempor cum soluta nobis eleifend option congue nihil
imperdiet doming id quod mazim placerat facer possim assum.
</div>
```

CSS

```
.content {
-webkit-column-count: 3; /* Chrome, Safari, Opera */
-moz-column-count: 3; /* Firefox */
column-count: 3;
}
```

Section 40.2: Basic example

Consider the following HTML markup:

```
<section>
  <p>Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor
invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et
justo duo dolores et ea rebum.</p>
  <p> Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem
ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore
et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea
rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.</p>
  <p>Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor
invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et
justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum
dolor sit amet.</p>
</section>
```

With the following CSS applied the content is split into three columns separated by a gray column rule of two pixels.

```
section {
  columns: 3;
  column-gap: 40px;
  column-rule: 2px solid gray;
}
```

See a live sample of this on JSFiddle.

Chapter 41: Inline-Block Layout

Section 41.1: Justified navigation bar

The horizontally justified navigation (menu) bar has some number of items that should be justified. The first (left) item has no left margin within the container, the last (right) item has no right margin within the container. The distance between items is equal, independent on the individual item width.

HTML

```
<nav>
    <ul>
        <li>abc</li>
        <li>abcdefghijkl</li>
        <li>abcdef</li>
    </ul>
</nav>
```

CSS

```
nav {
    width: 100%;
    line-height: 1.4em;
}
ul {
    list-style: none;
    display: block;
    width: 100%;
    margin: 0;
    padding: 0;
    text-align: justify;
    margin-bottom: -1.4em;
}
ul:after {
    content: "";
    display: inline-block;
    width: 100%;
}
li {
    display: inline-block;
}
```

Notes

- The `nav`, `ul` and `li` tags were chosen for their semantic meaning of 'a list of navigation (menu) items'. Other tags may also be used of course.
- The `:after` pseudo-element causes an extra 'line' in the `ul` and thus an extra, empty height of this block, pushing other content down. This is solved by the negative `margin-bottom`, which has to have the same magnitude as the `line-height` (but negative).
- If the page becomes too narrow for all the items to fit, the items will break to a new line (starting from the right) and be justified on this line. The total height of the menu will grow as needed.

Chapter 42: Inheritance

Section 42.1: Automatic inheritance

Inheritance the a fundamental mechanism of CSS by which the computed values of some properties of an element are applied to its' children. This is particularly useful when you want to set a global style to your elements rather than having to set said properties to each and every element in your markup.

Common properties that are automatically inherited are: `font`, `color`, `text-align`, `line-height`.

Assume the following stylesheet:

```
#myContainer {
  color: red;
  padding: 5px;
}
```

This will apply `color: red` not only to the `<div>` element but also to the `<h3>` and `<p>` elements. However, due to the nature of `padding` its value will ***not*** be inherited to those elements.

```
<div id="myContainer">
  <h3>Some header</h3>
  <p>Some paragraph</p>
</div>
```

Section 42.2: Enforced inheritance

Some properties are not automatically inherited from an element down to its' children. This is because those properties are typically desired to be unique to the element (or selection of elements) to which the property is applied to. Common such properties are `margin`, `padding`, `background`, `display`, etc.

However, sometimes inheritance is desired anyway. To achieve this, we can apply the `inherit` value to the property that should be inherited. The `inherit` value can be appied to *any* CSS property and *any* HTML element.

Assume the following stylesheet:

```
#myContainer {
  color: red;
  padding: 5px;
}
#myContainer p {
  padding: inherit;
}
```

This will apply `color: red` to both the `<h3>` and `<p>` elements due to the inheritance nature of the `color` property. However, the `<p>` element will also inherit the `padding` value from its' parent because this was specified.

```
<div id="myContainer">
  <h3>Some header</h3>
  <p>Some paragraph</p>
</div>
```

Chapter 43: CSS Image Sprites

Section 43.1: A Basic Implementation

What's an image sprite?

An image sprite is a single asset located within an image sprite sheet. An image sprite sheet is an image file that contains more than one asset that can be extracted from it.

For example:

The image above is an image sprite sheet, and each one of those stars is a sprite within the sprite sheet. These sprite sheets are useful because they improve performance by reducing the number of HTTP requests a browser might have to make.

So how do you implement one? Here's some example code.

HTML

```
<div class="icon icon1"></div>
<div class="icon icon2"></div>
<div class="icon icon3"></div>
```

CSS

```
.icon {
    background: url(“icons-sprite.png”);
    display: inline-block;
    height: 20px;
    width: 20px;
}
.icon1 {
      background-position: 0px 0px;
}
.icon2 {
      background-position: -20px 0px;
}
.icon3 {
      background-position: -40px 0px;
}
```

By using setting the sprite's width and height and by using the background-position property in CSS (with an x and y value) you can easily extract sprites from a sprite sheet using CSS.

Chapter 44: Clipping and Masking

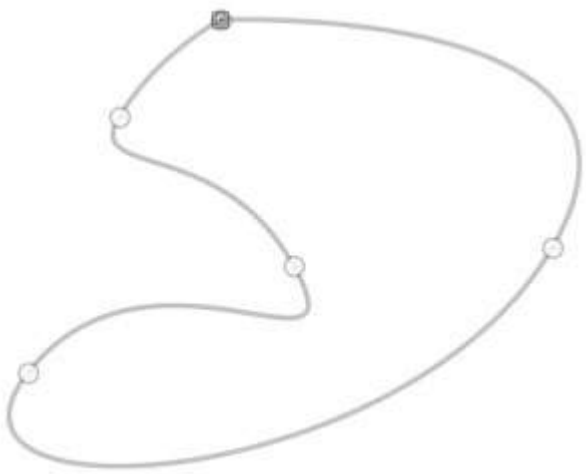
Parameter	Details
clip-source	A URL which can point to an inline SVG element (or) an SVG element in an external file that contains the clip path's definition.
basic-shape	Refers to one among `inset()`, `circle()`, `ellipse()` or `polygon()`. Using one of these functions the clipping path is defined. These shape functions work exactly the same way as they do in Shapes for Floats
clip-geometry-box	This can have one among `content-box`, `padding-box`, `border-box`, `margin-box`, `fill-box`, `stroke-box`, `view-box` as values. When this is provided without any value for <basic-shape>, the edges of the corresponding box is used as the path for clipping. When used with a <basic-shape>, this acts as the reference box for the shape.
mask-reference	This can be `none` or an image or a reference URL to a mask image source.
repeat-style	This specifies how the mask should be repeated or tiled in the X and Y axes. The supported values are `repeat-x`, `repeat-y`, `repeat`, `space`, `round`, `no-repeat`.
mask-mode	Can be `alpha` or `luminance` or `auto` and indicates whether the mask should be treated as a alpha mask or a luminance mask. If no value is provided and the mask-reference is a direct image then it would be considered as alpha mask (or) if the mask-reference is a URL then it would be considered as luminance mask.
position	This specifies the position of each mask layer and is similar in behavior to the `background-position` property. The value can be provided in 1 value syntax (like `top`, `10%`) or in 2 value syntax (like `top right`, `50% 50%`).
geometry-box	This specifies the box to which the mask should be clipped (*mask painting area*) or the box which should be used as reference for the mask's origin (*mask positioning area*) depending on the property. The list of possible values are `content-box`, `padding-box`, `border-box`, `margin-box`, `fill-box`, `stroke-box`, `view-box`. Detailed explanation of how each of those values work is available in the W3C Spec.
bg-size	This represents the size of each mask-image layer and has the same syntax as `background-size`. The value can be length or percentage or auto or cover or contain. Length, percentage and auto can either be provided as a single value or as one for each axis.
compositing-operator	This can be any one among `add`, `subtract`, `exclude`, `multiply` per layer and defines the type of compositing operation that should be used for this layer with those below it. Detailed explanation about each value is available in the W3C Specs.

Section 44.1: Clipping and Masking: Overview and Difference

With **Clipping** and **Masking** you can make some specified parts of elements transparent or opaque. Both can be applied to any HTML element.

Clipping

Clips are vector paths. Outside of this path the element will be transparent, inside it's opaque. Therefore you can define a `clip-path` property on elements. Every graphical element that also exists in SVG you can use here as a function to define the path. Examples are `circle()`, `polygon()` or `ellipse()`.



Example

```
clip-path: circle(100px at center);
```

The element will be only visible inside of this circle, which is positioned at the center of the element and has a radius of 100px.

Masking

Masks are similar to Clips, but instead of defining a path you define a mask what layers over the element. You can imagine this mask as an image what consist of mainly two colors: black and white.

Luminance Mask: Black means the region is opaque, and white that it's transparent, but there is also a grey area which is semi-transparent, so you are able to make smooth transitions.

Alpha Mask: Only on the transparent areas of the mask the element will be opaque.

This image for example can be used as a luminance mask to make for an element a very smooth transition from right to left and from opaque to transparent.

The `mask` property let you specify the the mask type and an image to be used as layer.

Example

```
mask: url(masks.svg#rectangle) luminance;
```

An element called `rectangle` defined in `masks.svg` will be used as an **luminance mask** on the element.

Section 44.2: Simple mask that fades an image from solid to transparent

CSS

```
div {
  height: 200px;
  width: 200px;
  background: url(http://lorempixel.com/200/200/nature/1);
  mask-image: linear-gradient(to right, white, transparent);
}
```

HTML

```
<div></div>
```

In the above example there is an element with an image as its background. The mask that is applied on the image (using CSS) makes it look as though it is fading out from left to right.

The masking is achieved by using a `linear-gradient` that goes from white (on the left) to transparent (on the right) as the mask. As it is an alpha mask, image becomes transparent where the mask is transparent.

Output without the mask:

Output with the mask:

Note: As mentioned in remarks, the above example would work in Chrome, Safari and Opera only when used with the `-webkit` prefix. This example (with a `linear-gradient` as mask image) is not yet supported in Firefox.

Section 44.3: Clipping (Circle)

CSS:

```
div{
```

```
  width: 200px;
  height: 200px;
  background: teal;
  clip-path: circle(30% at 50% 50%); /* refer remarks before usage */
}
```

HTML

```
<div></div>
```

This example shows how to clip a div to a circle. The element is clipped into a circle whose radius is 30% based on the dimensions of the reference box with its center point at the center of the reference box. Here since no <clip-geometry-box> (in other words, reference box) is provided, the `border-box` of the element will be used as the reference box.

The circle shape needs to have a radius and a center with `(x,y)` coordinates:

```
circle(radius at x y)
```

View Example

Output:

Section 44.4: Clipping (Polygon)

CSS:

```
div{
  width:200px;
  height:200px;
  background:teal;
  clip-path: polygon(0 0, 0 100%, 100% 50%); /* refer remarks before usage */
}
```

HTML:

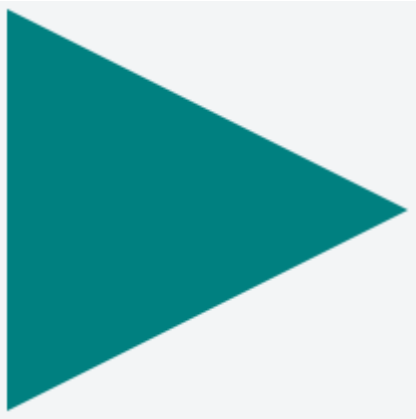
```
<div></div>
```

In the above example, a **polygonal** clipping path is used to clip the square (200 x 200) element into a triangle shape. The output shape is a triangle because the path starts at (that is, first coordinates are at) `0 0` - which is the top-left corner of the box, then goes to `0 100%` - which is bottom-left corner of the box and then finally to `100% 50%` which is nothing but the right-middle point of the box. These paths are self closing (that is, the starting point will be the ending point) and so the final shape is that of a triangle.

This can also be used on an element with an image or a gradient as background.

View Example

Output:



Section 44.5: Using masks to cut a hole in the middle of an image

CSS

```
div {
  width: 200px;
  height: 200px;
  background: url(http://lorempixel.com/200/200/abstract/6);
  mask-image: radial-gradient(circle farthest-side at center, transparent 49%, white 50%); /* check
remarks before using */
}
```

HTML

In the above example, a transparent circle is created at the center using `radial-gradient` and this is then used as a mask to produce the effect of a circle being cut out from the center of an image.

Image without mask:

Image with mask:

Section 44.6: Using masks to create images with irregular shapes

CSS

```
div { /* check remarks before usage */
  height: 200px;
  width: 400px;
  background-image: url(http://lorempixel.com/400/200/nature/4);
  mask-image: linear-gradient(to top right, transparent 49.5%, white 50.5%), linear-gradient(to top
left, transparent 49.5%, white 50.5%), linear-gradient(white, white);
  mask-size: 75% 25%, 25% 25%, 100% 75%;
  mask-position: bottom left, bottom right, top left;
  mask-repeat: no-repeat;
}
```

HTML

```
<div></div>
```

In the above example, three `linear-gradient` images (which when placed in their appropriate positions would cover 100% x 100% of the container's size) are used as masks to produce a transparent triangular shaped cut at the bottom of the image.

Image without the mask:

Image with the mask:

Chapter 45: Fragmentation

Value	Description
auto	Default. Automatic page breaks
always	Always insert a page break
avoid	Avoid page break (if possible)
left	Insert page breaks so that the next page is formatted as a left page
right	Insert page breaks so that the next page is formatted as a right page
initial	Sets this property to its default value.
inherit	Inherits this property from its parent element.

Section 45.1: Media print page-break

```
@media print {
  p {
    page-break-inside: avoid;
  }
  h1 {
    page-break-before: always;
  }
  h2 {
    page-break-after: avoid;
  }
}
```

This code does 3 things:

- it prevents a page break inside any p tags, meaning a paragraph will never be broken in two pages, if possible.
- it forces a page-break-before in all h1 headings, meaning that before every h1 occurrence, there will be a page break.
- it prevents page-breaks right after any h2

Chapter 46: CSS Object Model (CSSOM)

Section 46.1: Adding a background-image rule via the CSSOM

To add a background-image rule via the CSSOM, first get a reference to the rules of the first stylesheet:

```
var stylesheet = document.styleSheets[0].cssRules;
```

Then, get a reference to the end of the stylesheet:

```
var end = stylesheet.length - 1;
```

Finally, insert a background-image rule for the body element at the end of the stylesheet:

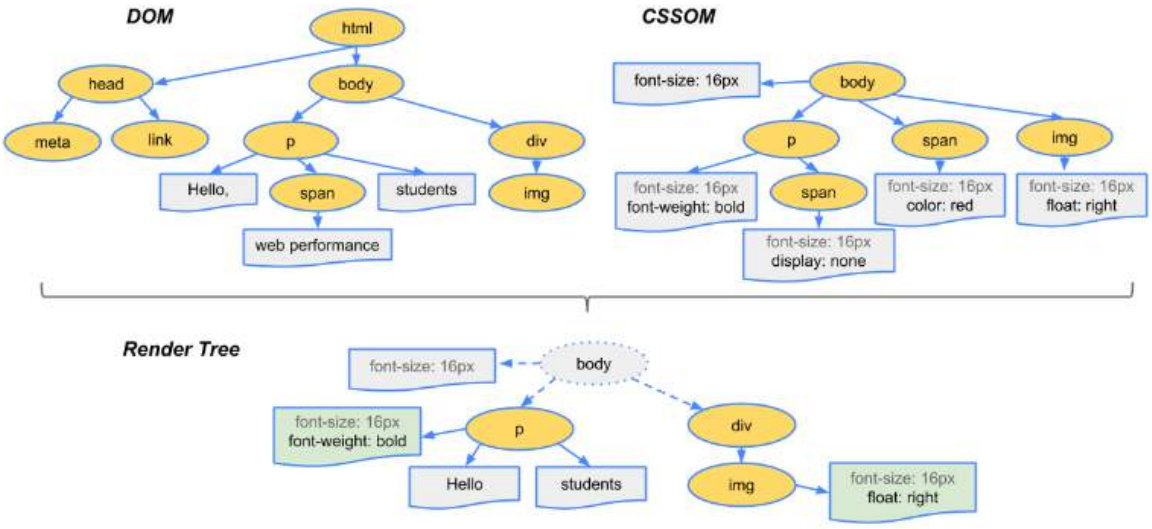
```
stylesheet.insertRule("body { background-image:
url('http://cdn.sstatic.net/Sites/stackoverflow/img/favicon.ico'); }", end);
```

Section 46.2: Introduction

The browser identifies tokens from stylesheet and coverts them into nodes which are linked into a tree structure. The entire map of all the nodes with their associated styles of a page would be the CSS Object Model.

To display the webpage, a web browser takes following steps.

1. The web browser examines your HTML and builds the DOM (Document Object Model).
2. The web browser examines your CSS and builds the CSSOM (CSS Object Model).
3. The web browser combines the DOM and the CSSOM to create a render tree. The web browser displays your webpage.



Chapter 47: Feature Queries

Parameter	Details
`(property: value)`	Evaluates true if the browser can handle the CSS rule. The parenthesis around the rule are required.
`and`	Returns true only if both the previous and next conditions are true.
`not`	Negates the next condition
`or`	Returns true if either the previous or next condition is true.
`(...)`	Groups conditions

Section 47.1: Basic @supports usage

```
@supports (display: flex) {
  /* Flexbox is available, so use it */
  .my-container {
    display: flex;
  }
}
```

In terms of syntax, `@supports` is very similar to `@media`, but instead of detecting screen size and orientation, `@supports` will detect whether the browser can handle a given CSS rule.

Rather than doing something like `@supports (flex)`, notice that the rule is `@supports (display: flex)`.

Section 47.2: Chaining feature detections

To detect multiple features at once, use the `and` operator.

```
@supports (transform: translateZ(1px)) and (transform-style: preserve-3d) and (perspective: 1px) {
  /* Probably do some fancy 3d stuff here */
}
```

There is also an `or` operator and a `not` operator:

```
@supports (display: flex) or (display: table-cell) {
  /* Will be used if the browser supports flexbox or display: table-cell */
}
@supports not (-webkit-transform: translate(0, 0, 0)) {
  /* Will *not* be used if the browser supports -webkit-transform: translate(...) */
}
```

For the ultimate `@supports` experience, try grouping logical expressions with parenthesis:

```
@supports ((display: block) and (zoom: 1)) or ((display: flex) and (not (display: table-cell))) or
(transform: translateX(1px)) {
  /* ... */
}
```

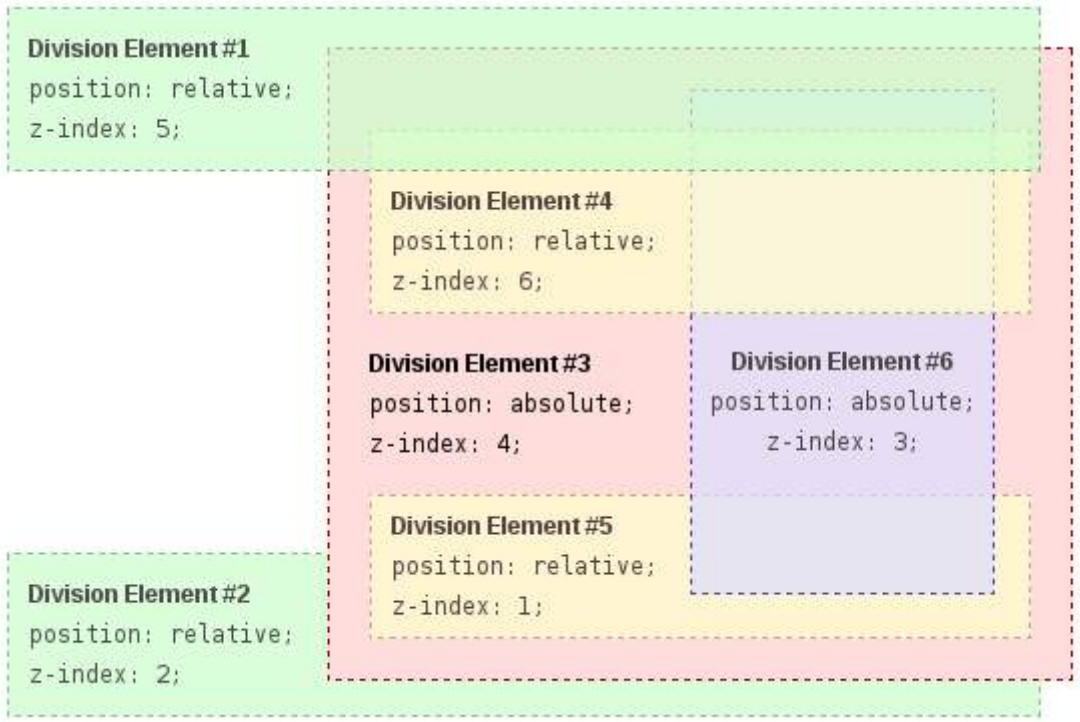
This will work if the browser

1. Supports `display: block` AND `zoom: 1`, or
2. Supports `display: flex` AND NOT `display: table-cell`, or
3. Supports `transform: translateX(1px)`.

Chapter 48: Stacking Context

Section 48.1: Stacking Context

In this example every positioned element creates its own stacking context, because of their positioning and z-index values. The hierarchy of stacking contexts is organized as follows:



- Root
  - DIV #1
  - DIV #2
  - DIV #3
  - DIV #4
  - DIV #5
  - DIV #6

It is important to note that DIV #4, DIV #5 and DIV #6 are children of DIV #3, so stacking of those elements is completely resolved within DIV#3. Once stacking and rendering within DIV #3 is completed, the whole DIV #3 element is passed for stacking in the root element with respect to its sibling's DIV.

HTML:

```
<div id="div1">
  <h1>Division Element #1</h1>
  <code>position: relative;<br/>
  z-index: 5;</code>
</div>
<div id="div2">
  <h1>Division Element #2</h1>
  <code>position: relative;<br/>
  z-index: 2;</code>
</div>
<div id="div3">
  <div id="div4">
```

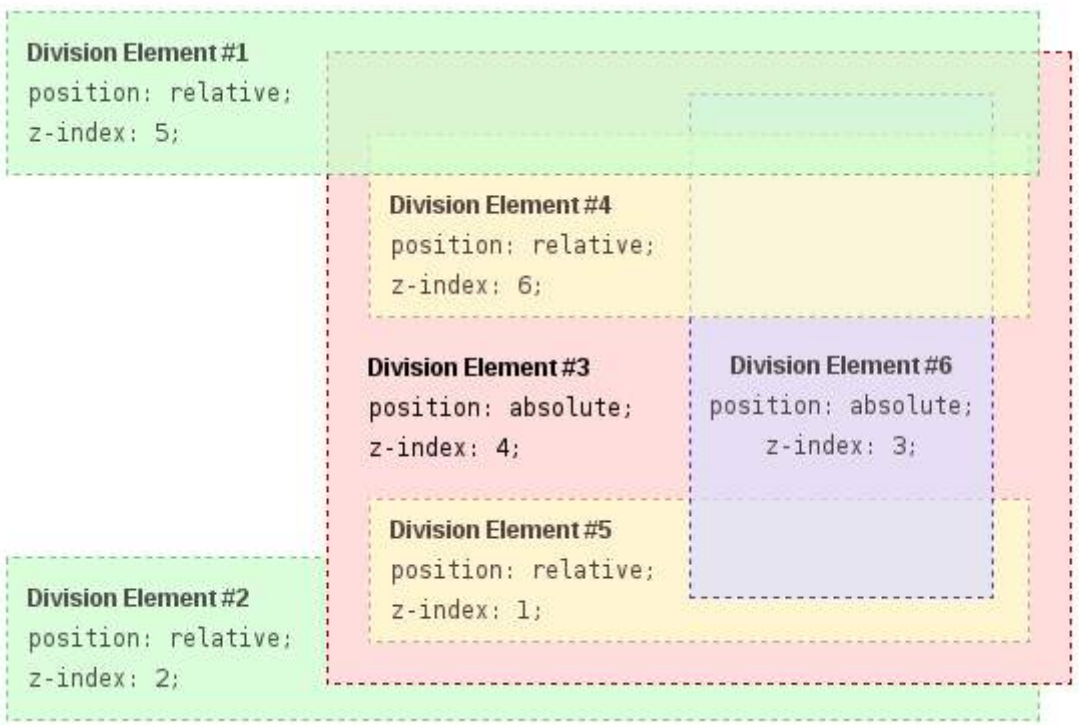
```
    <h1>Division Element #4</h1>
    <code>position: relative;<br/>
    z-index: 6;</code>
  </div>
  <h1>Division Element #3</h1>
  <code>position: absolute;<br/>
  z-index: 4;</code>
  <div id="div5">
    <h1>Division Element #5</h1>
    <code>position: relative;<br/>
    z-index: 1;</code>
  </div>
  <div id="div6">
    <h1>Division Element #6</h1>
    <code>position: absolute;<br/>
    z-index: 3;</code>
  </div>
</div>
```

CSS:

```
* {
    margin: 0;
}
html {
    padding: 20px;
    font: 12px/20px Arial, sans-serif;
}
div {
    opacity: 0.7;
    position: relative;
}
h1 {
    font: inherit;
    font-weight: bold;
}
#div1,
#div2 {
    border: 1px dashed #696;
    padding: 10px;
    background-color: #cfc;
}
#div1 {
    z-index: 5;
    margin-bottom: 190px;
}
#div2 {
    z-index: 2;
}
#div3 {
    z-index: 4;
    opacity: 1;
    position: absolute;
    top: 40px;
    left: 180px;
    width: 330px;
    border: 1px dashed #900;
    background-color: #fdd;
    padding: 40px 20px 20px;
}
#div4,
```

```
#div5 {
    border: 1px dashed #996;
    background-color: #ffc;
}
#div4 {
    z-index: 6;
    margin-bottom: 15px;
    padding: 25px 10px 5px;
}
#div5 {
    z-index: 1;
    margin-top: 15px;
    padding: 5px 10px;
}
#div6 {
    z-index: 3;
    position: absolute;
    top: 20px;
    left: 180px;
    width: 150px;
    height: 125px;
    border: 1px dashed #009;
    padding-top: 125px;
    background-color: #ddf;
    text-align: center;
}
```

Result:

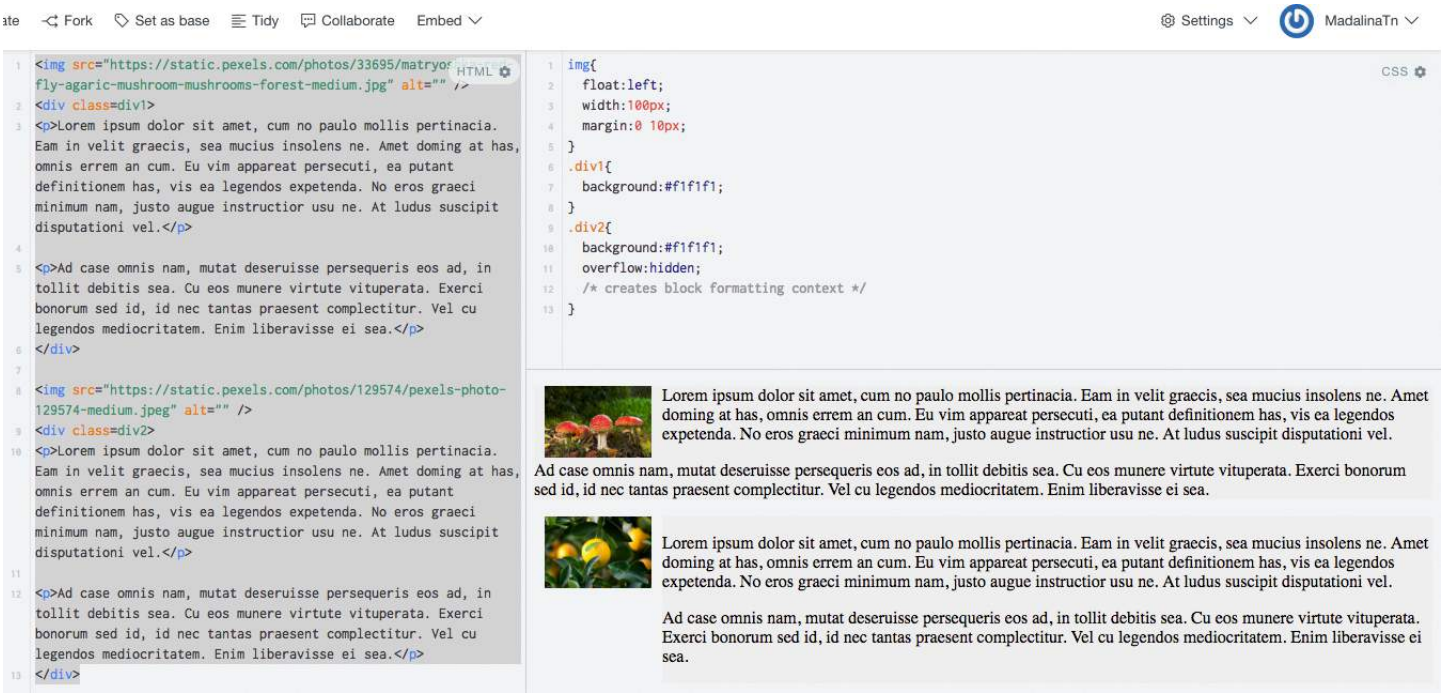


Source:
https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_Positioning/Understanding_z_index/The_stacking_context.

Chapter 49: Block Formatting Contexts

Section 49.1: Using the overflow property with a value different to visible

```
img{
  float:left;
  width:100px;
  margin:0 10px;
}
.div1{
  background:#f1f1f1;
  /* does not create block formatting context */
}
.div2{
  background:#f1f1f1;
  overflow:hidden;
  /* creates block formatting context */
}
```



https://jsfiddle.net/MadalinaTn/qkwwmu6m/2/

> Using the overflow property with a value different to visible (its default) will create a new block formatting context. This is technically necessary — if a float intersected with the scrolling element it would forcibly rewrap the content.

This example that show how a number of paragraphs will interact with a floated image is similar to this example, on css-tricks.com.

2: https://developer.mozilla.org/en-US/docs/Web/CSS/overflow MDN

Chapter 50: Vertical Centering

Section 50.1: Centering with display: table

HTML:

```
<div class="wrapper">
    <div class="outer">
        <div class="inner">
            centered
        </div>
    </div>
</div>
```

CSS:

```
.wrapper {
  height: 600px;
  text-align: center;
}
.outer {
  display: table;
  height: 100%;
  width: 100%;
}
.outer .inner {
  display: table-cell;
  text-align: center;
  vertical-align: middle;
}
```

Section 50.2: Centering with Flexbox

HTML:

```
<div class="container">
    <div class="child"></div>
</div>
```

CSS:

```
.container {
  height: 500px;
  width: 500px;
  display: flex;              // Use Flexbox
  align-items: center;        // This centers children vertically in the parent.
  justify-content: center;    // This centers children horizontally.
  background: white;
}

.child {
  width: 100px;
  height: 100px;
  background: blue;
}
```

Section 50.3: Centering with Transform

HTML:

```
<div class="wrapper">
    <div class="centered">
        centered
    </div>
</div>
```

CSS:

```
.wrapper {
  position: relative;
  height: 600px;
}
.centered {
  position: absolute;
  z-index: 999;
  transform: translate(-50%, -50%);
  top: 50%;
  left: 50%;
}
```

Section 50.4: Centering Text with Line Height

HTML:

```
<div class="container">
    <span>vertically centered</span>
</div>
```

CSS:

```
.container{
    height: 50px;           /* set height */
    line-height: 50px;      /* set line-height equal to the height */
    vertical-align: middle; /* works without this rule, but it is good having it explicitly set */
}
```

Note: This method will only vertically center a *single line of text*. It will not center block elements correctly and if the text breaks onto a new line, you will have two very tall lines of text.

Section 50.5: Centering with Position: absolute

HTML:

```
<div class="wrapper">
  <img
src="http://cdn.sstatic.net/Sites/stackoverflow/company/img/logos/so/so-icon.png?v=c78bd457575a">
</div>
```

CSS:

```
.wrapper{
   position:relative;
```

```
   height: 600px;
}
.wrapper img {
  position: absolute;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  margin: auto;
}
```

If you want to center other then images, then you must give height and width to that element.

HTML:

```
<div class="wrapper">
  <div class="child">
     make me center
  </div>
</div>
```

CSS:

```
.wrapper{
   position:relative;
   height: 600px;
}
.wrapper .child {
  position: absolute;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  margin: auto;
  width: 200px;
  height: 30px;
  border: 1px solid #f00;
}
```

Section 50.6: Centering with pseudo element

HTML:

```
<div class="wrapper">
  <div class="content"></div>
</div>
```

CSS:

```
.wrapper{
   min-height: 600px;
}

.wrapper:before{
  content: "";
  display: inline-block;
  height: 100%;
  vertical-align: middle;
}
```

```
.content {
  display: inline-block;
  height: 80px;
  vertical-align: middle;
}
```

This method is best used in cases where you have a varied-height `.content` centered inside `.wrapper`; and you want `.wrapper`'s height to expand when `.content`'s height exceed `.wrapper`'s min-height.

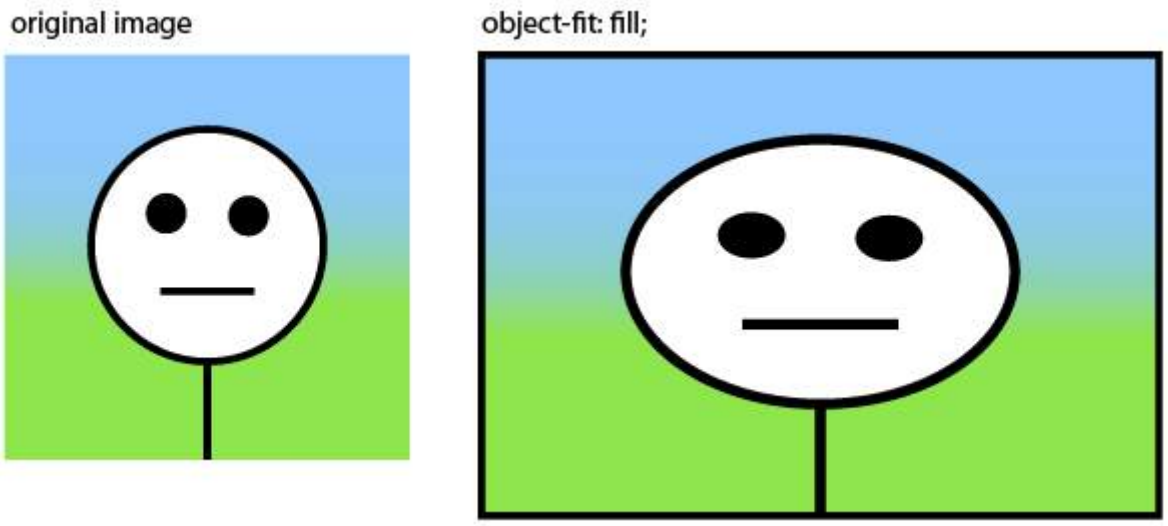
Chapter 51: Object Fit and Placement

Section 51.1: object-fit

The **object-fit** property will defines how an element will fit into a box with an established height and width. Usually applied to an image or video, Object-fit accepts the following five values:

FILL

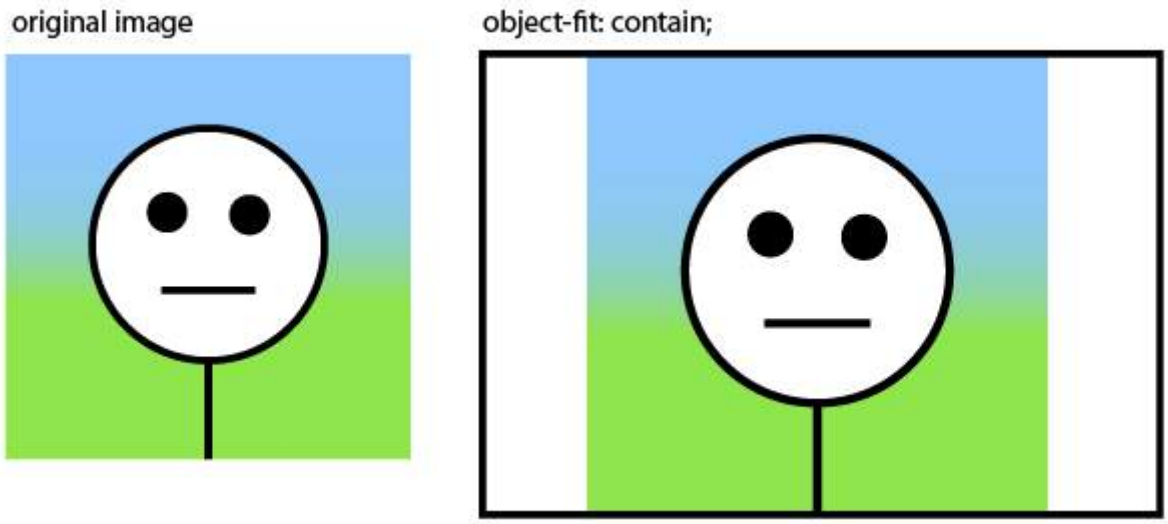
```
object-fit:fill;
```



Fill stretches the image to fit the content box without regard to the image's original aspect ratio.

CONTAIN

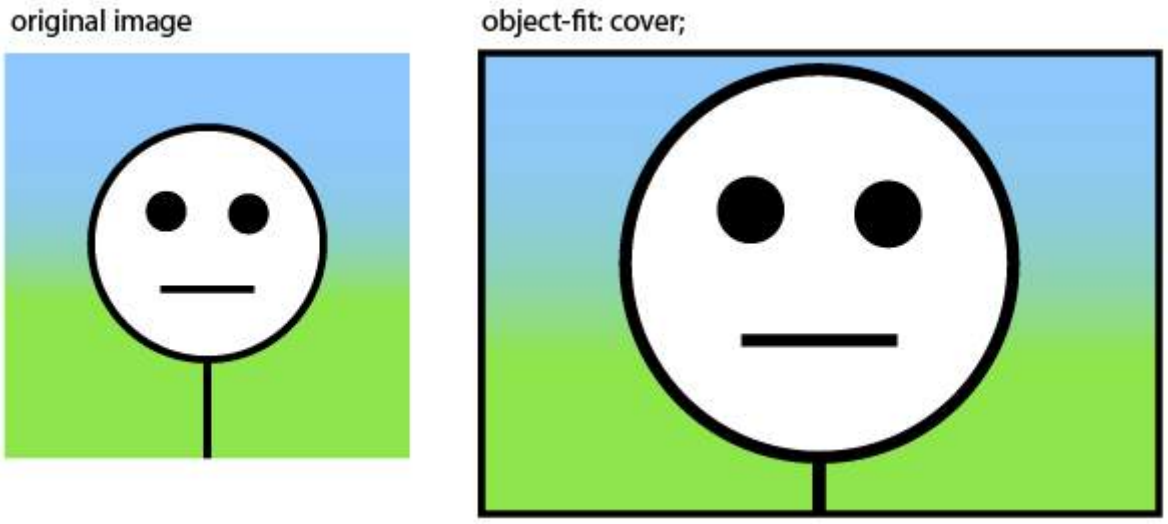
```
object-fit:contain;
```



Contain fits the image in the box's height or width while maintaining the image's aspect ratio.

COVER

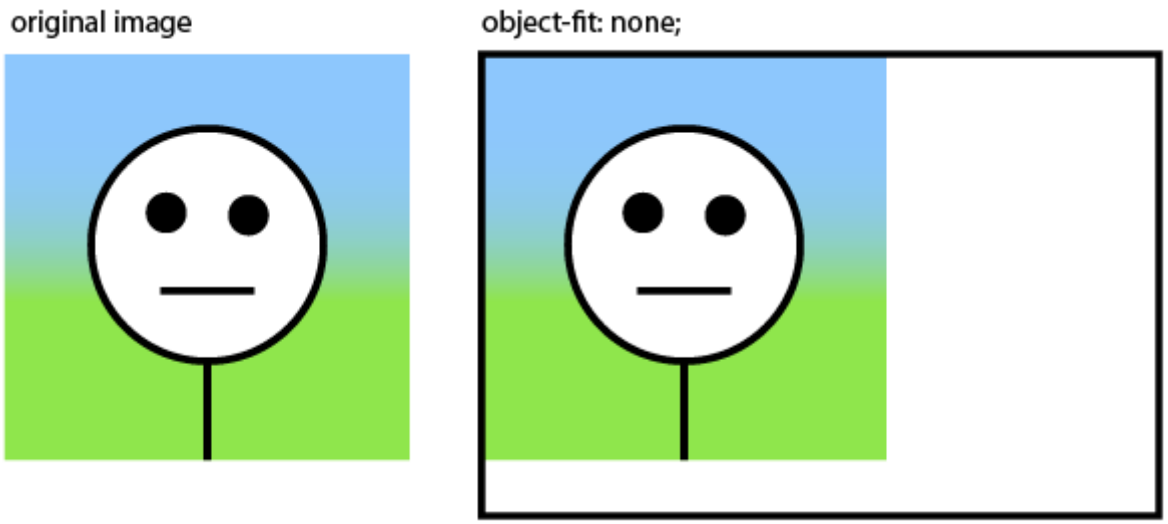
```
object-fit:cover;
```



Cover fills the entire box with the image. The image aspect ratio is preserved, but the image is cropped to the dimensions of the box.

NONE

```
object-fit:none;
```



None ignores the size of the box and is not resized.

SCALE-DOWN

```
object-fit:scale-down;
```

Scale-down either sizes the object as `none` or as `contain`. It displays whichever option results in a smaller image size.

original image

object-fit: scale-down;

Chapter 52: CSS design patterns

These examples are for documenting CSS-specific design patterns like BEM, OOCSS and SMACSS.

These examples are NOT for documenting CSS frameworks like Bootstrap or Foundation.

Section 52.1: BEM

BEM stands for `Blocks, Elements and Modifiers`. It's a methodology initially conceived by Russian tech company Yandex, but which gained quite some traction among American & Western-European web developers as well.

As the same implies, BEM metholology is all about componentization of your HTML and CSS code into three types of components:

- **Blocks:** standalone entities that are meaningful on their own

  Examples are `header`, `container`, `menu`, `checkbox` & `textbox`

- **Elements:** Part of blocks that have no standalone meaning and are semantically tied to their blocks.

  Examples are `menu item`, `list item`, `checkbox caption` & `header title`

- **Modifiers:** Flags on a block or element, used to change appearance or behavior

  Examples are `disabled`, `highlighted`, `checked`, `fixed`, `size big` & `color yellow`

The goal of BEM is to keep optimize the readability, maintainability and flexibility of your CSS code. The way to achieve this, is to apply the following rules.

- Block styles are never dependent on other elements on a page
- Blocks should have a simple, short name and avoid `_` or `-` characters
- When styling elements, use selectors of format `blockname__elementname`
- When styling modifiers, use selectors of format `blockname--modifiername` and `blockname__elementname--modifiername`
- Elements or blocks that have modifiers should inherit everything from the block or element it is modifying except the properties the modifier is supposed to modify

Code example

If you apply BEM to your form elements, your CSS selectors should look something like this:

```
.form { }                       // Block
.form--theme-xmas { }           // Block + modifier
.form--simple { }               // Block + modifier
.form__input { }                // Block > element
.form__submit { }               // Block > element
.form__submit--disabled { }     // Block > element + modifier
```

The corresponding HTML should look something like this:

```html
<form class="form form--theme-xmas form--simple">
  <input class="form__input" type="text" />
  <input class="form__submit form__submit--disabled" type="submit" />
</form>
```

Chapter 53: Browser Support & Prefixes

Prefix	Browser(s)
`-webkit-`	Google Chrome, Safari, newer versions of Opera 12 and up, Android, Blackberry and UC browsers
`-moz-`	Mozilla Firefox
`-ms-`	Internet Explorer, Edge
`-o-`, `-xv-`	Opera until version 12
`-khtml-`	Konquerer

Section 53.1: Transitions

```
div {
    -webkit-transition: all 4s ease;
       -moz-transition: all 4s ease;
         -o-transition: all 4s ease;
            transition: all 4s ease;
}
```

Section 53.2: Transform

```
div {
    -webkit-transform: rotate(45deg);
       -moz-transform: rotate(45deg);
        -ms-transform: rotate(45deg);
         -o-transform: rotate(45deg);
            transform: rotate(45deg);
}
```

Chapter 54: Normalizing Browser Styles

Every browser has a default set of CSS styles that it uses for rendering elements. These default styles may not be consistent across browsers because: the language specifications are unclear so base styles are up for interpretation, browsers may not follow specifications that are given, or browsers may not have default styles for newer HTML elements. As a result, people may want to normalize default styles across as many browsers as possible.

Section 54.1: normalize.css

Browsers have a default set of CSS styles they use for rendering elements. Some of these styles can even be customised using the browser's settings to change default font face and size definitions, for example. The styles contain the definition of which elements are supposed to be block-level or inline, among other things.

Because these default styles are given some leeway by the language specifications and because browsers may not follow the specs properly they can differ from browser to browser.

This is where normalize.css comes into play. It overrides the most common inconsistencies and fixes known bugs.

What does it do

> - Preserves useful defaults, unlike many CSS resets.
> - Normalizes styles for a wide range of elements.
> - Corrects bugs and common browser inconsistencies.
> - Improves usability with subtle modifications.
> - Explains what code does using detailed comments.

So, by including normalize.css in your project your design will look more alike and consistent across different browsers.

Difference to reset.css

You may have heard of `reset.css`. What's the difference between the two?

While normalize.css provides consistency by setting different properties to unified defaults, reset.css achieves consistency by **removing** all basic styling that a browser may apply. While this might sound like a good idea at first, this actually means you have to write **all** rules yourself, which goes against having a solid standard.

Section 54.2: Approaches and Examples

CSS resets take separate approaches to browser defaults. Eric Meyer's Reset CSS has been around for a while. His approach nullifies many of the browser elements that have been known to cause problems right off the back. The following is from his version (v2.0 | 20110126) CSS Reset.

```
html, body, div, span, applet, object, iframe,
h1, h2, h3, h4, h5, h6, p, blockquote, pre,
a, abbr, acronym, address, big, cite, code,
del, dfn, em, img, ins, kbd, q, s, samp,
small, strike, strong, sub, sup, tt, var,
b, u, i, center,
dl, dt, dd, ol, ul, li,
fieldset, form, label, legend,
table, caption, tbody, tfoot, thead, tr, th, td,
```

```
article, aside, canvas, details, embed,
figure, figcaption, footer, header, hgroup,
menu, nav, output, ruby, section, summary,
time, mark, audio, video {
    margin: 0;
    padding: 0;
    border: 0;
    font-size: 100%;
    font: inherit;
    vertical-align: baseline;
}
```

Eric Meyer's Reset CSS

Normalize CSS on the other and deals with many of these separately. The following is a sample from the version (v4.2.0) of the code.

```
/**
 * 1. Change the default font family in all browsers (opinionated).
 * 2. Correct the line height in all browsers.
 * 3. Prevent adjustments of font size after orientation changes in IE and iOS.
 */

/* Document
   ========================================================================== */

html {
  font-family: sans-serif; /* 1 */
  line-height: 1.15; /* 2 */
  -ms-text-size-adjust: 100%; /* 3 */
  -webkit-text-size-adjust: 100%; /* 3 */
}

/* Sections
   ========================================================================== */

/**
 * Remove the margin in all browsers (opinionated).
 */

body {
  margin: 0;
}

/**
 * Add the correct display in IE 9-.
 */

article,
aside,
footer,
header,
nav,
section {
  display: block;
}

/**
 * Correct the font size and margin on `h1` elements within `section` and
 * `article` contexts in Chrome, Firefox, and Safari.
 */
```

```
h1 {
  font-size: 2em;
  margin: 0.67em 0;
}
```

Normalize CSS

Chapter 55: Internet Explorer Hacks

Section 55.1: Adding Inline Block support to IE6 and IE7

```
display: inline-block;
```

The `display` property with the value of `inline-block` is not supported by Internet Explorer 6 and 7. A work-around for this is:

```
zoom: 1;
*display: inline;
```

The `zoom` property triggers the `hasLayout` feature of elements, and it is available only in Internet Explorer. The `*display` makes sure that the invalid property executes only on the affected browsers. Other browsers will simply ignore the rule.

Section 55.2: High Contrast Mode in Internet Explorer 10 and greater

In Internet Explorer 10+ and Edge, Microsoft provides the `-ms-high-contrast` media selector to expose the "High Contrast" setting from the browser, which allows the programmer to adjust their site's styles accordingly.

The `-ms-high-contrast` selector has 3 states: `active`, `black-on-white`, and `white-on-black`. In IE10+ it also had a `none` state, but that is no longer supported in Edge going forward.

Examples

```
@media screen and (-ms-high-contrast: active), (-ms-high-contrast: black-on-white) {
   .header{
      background: #fff;
      color: #000;
   }
}
```

This will change the header background to white and the text color to black when high contrast mode is active *and* it is in `black-on-white` mode.

```
@media screen and (-ms-high-contrast: white-on-black) {
   .header{
      background: #000;
      color: #fff;
   }
}
```

Similar to the first example, but this specifically selects the `white-on-black` state only, and inverts the header colors to a black background with white text.

More Information:

Microsoft Documentation on `-ms-high-contrast`

Section 55.3: Internet Explorer 6 & Internet Explorer 7 only

To target Internet Explorer 6 and Internet Explorer 7, start your properties with `*`:

```
.hide-on-ie6-and-ie7 {
    *display : none; // This line is processed only on IE6 and IE7
}
```

Non-alphanumeric prefixes (other than hyphens and underscores) are ignored in IE6 and IE7, so this hack works for any unprefixed `property: value` pair.

Section 55.4: Internet Explorer 8 only

To target Internet Explorer 8, wrap your selectors inside `@media \0 screen { }`:

```
@media \0 screen {
    .hide-on-ie8 {
        display : none;
    }
}
```

Everything between `@media \0 screen { }` is processed only by I

Chapter 56: Performance

Section 56.1: Use transform and opacity to avoid trigger layout

Changing some CSS attribute will trigger the browser to synchronously calculate the style and layout, which is a bad thing when you need to animate at 60fps.

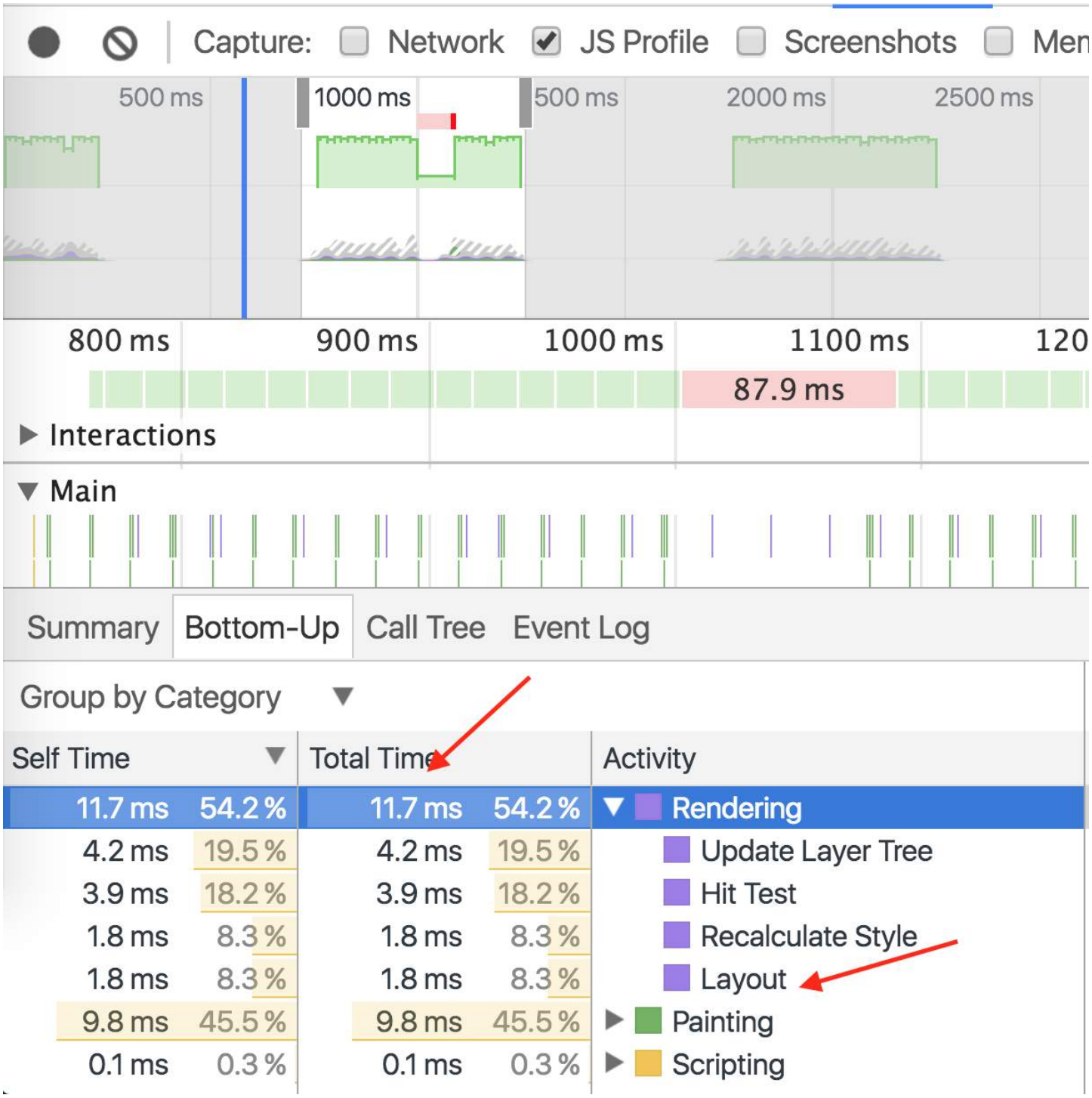
DON'T

Animate with `left` and `top` trigger layout.

```
#box {
  left: 0;
  top: 0;
  transition: left 0.5s, top 0.5s;
  position: absolute;
  width: 50px;
  height: 50px;
  background-color: gray;
}

#box.active {
  left: 100px;
  top: 100px;
}
```

Demo took **11.7ms** for rendering, **9.8ms** for painting



DO

Animate with `transform` with the same animation.

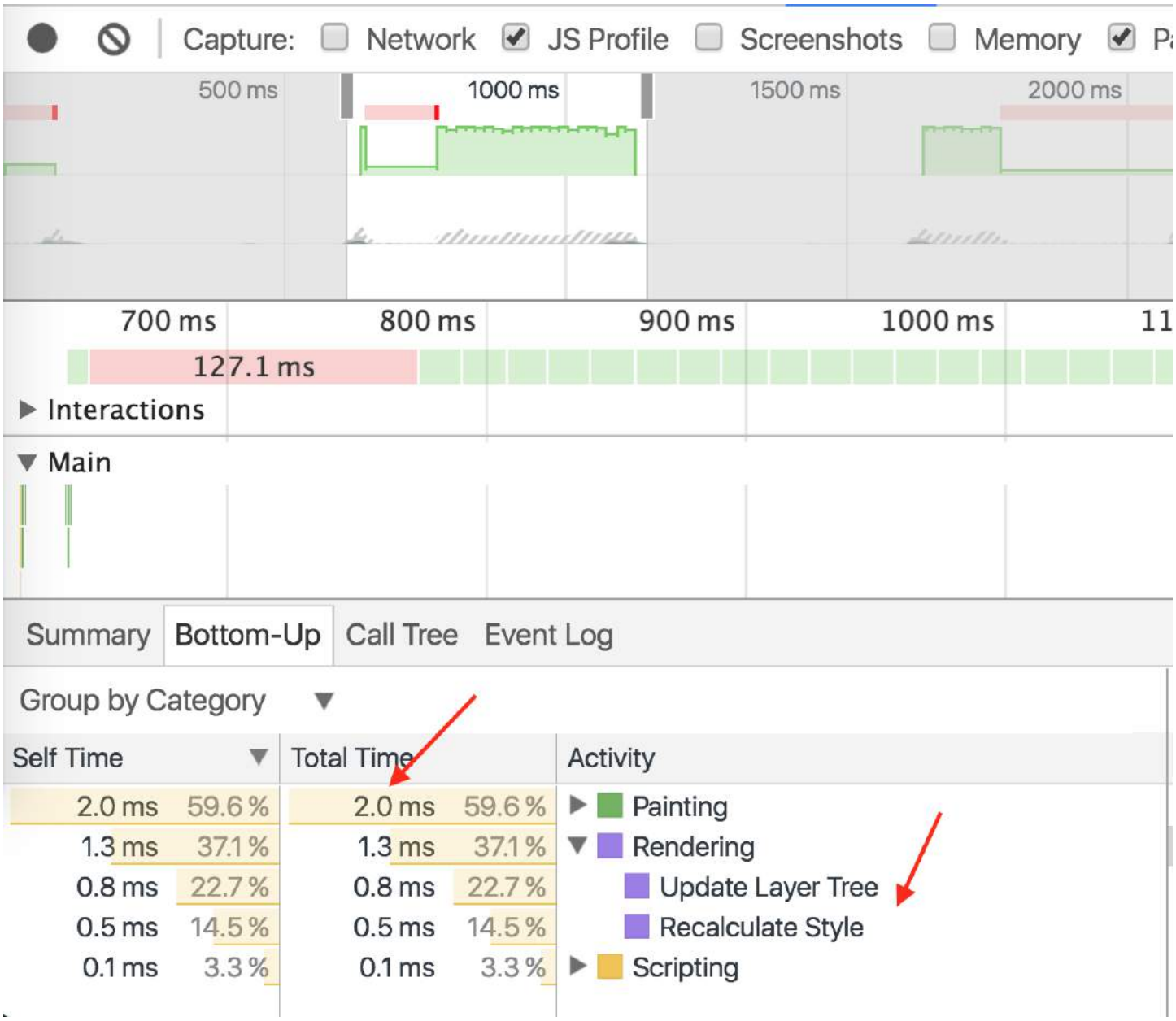
```
#box {
  left: 0;
  top: 0;
  position: absolute;
  width: 50px;
  height: 50px;
  background-color: gray;

  transition: transform 0.5s;
  transform: translate3d(0, 0, 0);
}

#box.active {
```

```
  transform: translate3d(100px, 100px, 0);
}
```

Demo same animation, took **1.3ms** for rendering, **2.0ms** for painting.



Credits

Thank you greatly to all the people from Stack Overflow Documentation who helped provide this content, more changes can be sent to web@petercv.com for new content to be published or updated

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Andrew Myers	Chapter 47
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animuson	Chapters 4, 50 and 53
apaul	Chapters 6 and 27
Araknid	Chapter 4
Arif	Chapter 11
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Asim K T	Chapters 5 and 16
AVAVT	Chapter 50
awe	Chapter 1
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Bipon	Chapter 40
BiscuitBaker	Chapter 7
Boris	Chapter 5
Boysenb3rry	Chapter 1
brandaemon	Chapter 17
Brett DeWoody	Chapters 18, 38 and 39
CalvT	Chapters 5 and 9
Casey	Chapter 11
Cassidy Williams	Chapters 10 and 22
cdm	Chapters 5 and 8
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Chiller	Chapter 38
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CocoaBean	Chapter 5
coderfin	Chapter 3
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CPHPython	Chapter 4

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darrylyeo	Chapters 2, 13 and 18
Darthstroke	Chapter 5
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demonofthemist	Chapter 14
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Dex Star	Chapter 27
Diego V	Chapter 6
Dinidu Hewage	Chapter 4
dippas	Chapters 4, 17 and 21
doctorsherlock	Chapter 10
dodopok	Chapters 13, 36 and 45
Elegant.Scripting	Chapter 43
Eliran Malka	Chapter 6
Emanuele Parisio	Chapter 6
Evgeny	Chapter 15
Farzad YZ	Chapter 6
fcalderan	Chapter 5
feeela	Chapters 46 and 55
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fzzylogic	Chapter 16
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gandreadis	Chapter 4
geek1011	Chapter 21
geeksal	Chapter 17
Gerardas	Chapter 1
Gnietschow	Chapter 10
GoatsWearHats	Chapter 1
Gofilord	Chapter 21
Grant Palin	Chapter 54
H. Pauwelyn	Chapters 4, 18 and 36
HansCz	Chapter 4
Harish Gyanani	Chapter 1
Harry	Chapters 10, 26, 28, 29, 33, 35 and 44
henry	Chapter 4
Horst Jahns	Chapter 5
Hristo	Chapter 32
Hugo Buff	Chapter 4

Hynes	Chapters 4, 5 and 15
insertusernamehere	Chapter 15
J Atkin	Chapters 1 and 4
J F	Chapters 4 and 20
Jacob Gray	Chapters 4, 5 and 22
James Donnelly	Chapters 7 and 17
James Taylor	Chapter 5
jaredsk	Chapters 10, 36 and 50
JedaiCoder	Chapter 6
Jef	Chapter 16
Jeffery Tang	Chapter 30
jehna1	Chapter 6
jgh	Chapter 12
JHS	Chapter 25
Jmh2013	Chapters 13, 23 and 43
joejoe31b	Chapters 4 and 13
JoelBonetR	Chapter 4
joe_young	Chapters 1 and 15
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Kuhan	Chapter 18
Kyle Ratliff	Chapter 6
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LiLacTac	Chapter 55
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Matas Vaitkevicius	Chapters 4 and 13
Mattia Astorino	Chapter 22
Maximillian Laumeister	Chapters 5 and 13
Maxouhell	Chapter 6
Michael Moriarty	Chapters 5, 15 and 18
Michael_B	Chapters 4 and 6
Mifeet	Chapter 6
Mike McCaughan	Chapter 24
Miles	Chapters 12 and 51
Miro	Chapter 18
MMachinegun	Chapter 54
mmativ	Chapter 50
Mod Proxy	Chapter 6
Mr. Alien	Chapter 5

Mr. Meeseeks	Chapter 29
Mr_Green	Chapter 8
Muthu Kumaran	Chapter 37
Naeem Shaikh	Chapter 4
Nate	Chapter 5
Nathan Arthur	Chapters 1, 4, 6, 8, 13, 14, 15 and 16
Nemanja Trifunovic	Chapter 48
Niek Brouwer	Chapter 23
niyasc	Chapter 18
Nobal Mohan	Chapter 10
o.v.	Chapter 6
Obsidian	Chapters 37 and 53
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Phil	Chapter 50
pixelbandito	Chapter 9
Praveen Kumar	Chapters 4, 6, 13, 15, 26, 28, 50 and 55
Qaz	Chapter 12
Rahul Nanwani	Chapter 22
RamenChef	Chapter 43
rdans	Chapter 4
RedRiderX	Chapter 37
rejnev	Chapter 8
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Sergey Denisov	Chapters 5 and 26
Shaggy	Chapters 5, 21 and 53
Siavas	Chapter 6
Someone	Chapter 6
Sourav Ghosh	Chapters 5 and 22
Squazz	Chapters 16 and 31
srikarg	Chapter 13
StefanBob	Chapter 9
Stewartside	Chapters 4, 5, 6, 18, 20 and 21
Stratboy	Chapter 5
sudo bangbang	Chapter 4
Sumner Evans	Chapter 4
Sun Qingyao	Chapter 8
Sunnyok	Chapters 4, 6 and 8
Sverri M. Olsen	Chapter 1

takeradi	Chapter 16
Taylor	Chapter 6
Ted Goas	Chapters 12, 15, 34 and 43
Teo Dragovic	Chapters 1 and 13
ThatWeirdo	Chapter 4
TheGenie OfTruth	Chapter 36
Theodore K.	Chapter 22
think123	Chapter 5
Timothy Miller	Chapter 55
Toby	Chapters 15 and 20
Todd	Chapter 1
ToniB	Chapter 15
Tot Zam	Chapter 8
Trevor Clarke	Chapters 5, 8, 10 and 15
TrungDQ	Chapter 56
TylerH	Chapters 1, 4, 5, 36 and 53
Ulrich Schwarz	Chapter 43
user007	Chapter 18
user2622348	Chapter 20
vishak	Chapter 14
vkopio	Chapter 7
Vlusion	Chapter 15
Volker E.	Chapter 15
web	Chapters 6, 26, 28, 29 and 44
Will DiFruscio	Chapter 9
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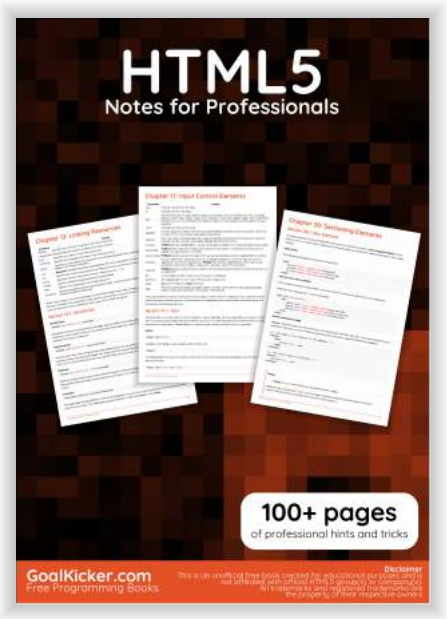
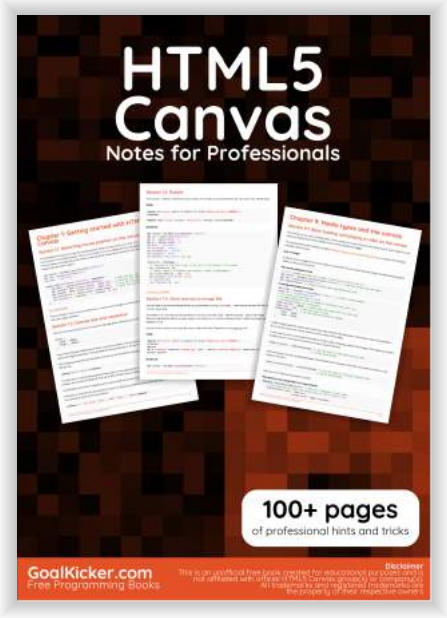
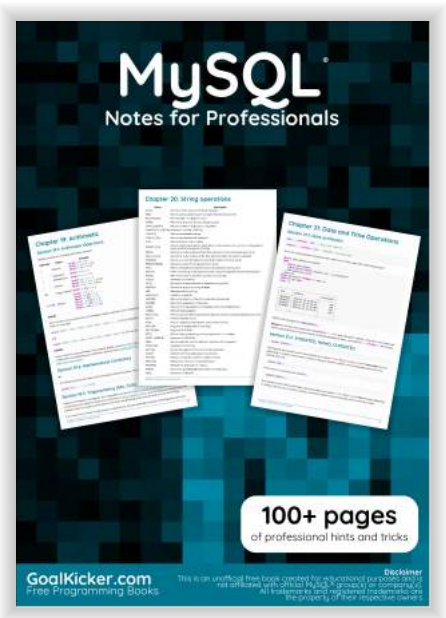
X	Chapter 18
Xinyang Li	Chapter 1
xpy	Chapter 4
Yury Fedorov	Chapter 4
Zac	Chapters 5 and 12
Zaffy	Chapter 4
Zakaria Acharki	Chapter 20
Zaz	Chapter 4
Ze Rubeus	Chapter 4
zeel	Chapter 6
zer00ne	Chapter 20
Zeta	Chapter 5
Zze	Chapter 5

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